# From Human Markets to Machine Markets

Feedback Stability, Auditable Memory, and Financial Constraints

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## Abstract

When trading algorithms learn from prices they help set, price formation becomes a closed loop. We ask what a surveillance authority can learn about such feedback from market data. In a linear market–learning model with heterogeneous adaptive cohorts, aggregate price responses identify only merged rate–weight pairs, so individual cohorts are not identified (an exact realization-theoretic equivalence class). Separating two cohorts whose adaptation rates differ by $\delta$ requires a probe budget of order $\delta^{-4}$. Any stability classifier with bounded error must abstain near the boundary $S = 1$, and with finite observation windows it must abstain at every budget. Local price stability does not imply dealer safety. The exact worst-case safe shock radius for a leveraged dealer is set by $\sup_t \|CJ^t\|$, which the Kreiss constant bounds and the spectral radius does not. In a funded simulation with genuine feedback, halt rates rise steeply inside the stable region because feedback inflates the zero-frequency gain of order flow by $(1 - S)^{-1}$; non-adaptive flow with the same persistence produces the same halts. We give a size-valid test of the completely monotone (positive-memory) class, correcting a common Hankel-only criterion. On high-frequency crypto order books (spanning a 12-day high-resolution Coinbase baseline, an 11-day cross-venue check, and a systematic 768 asset-day multi-year panel on Binance Spot across 2021–2024), a model-derived reduced-form estimate of the flow-feedback component is consistently small and positive ($\hat{Q} = 0.0082$, loop ratio $\hat{\mathcal{L}} = 0.099 \ll 1$). It survives long-memory controls, structural SMM estimation ($Q_{\mathrm{SMM}} = 0.0086$), and out-of-sample holdout validation, while Rigobon heteroskedasticity reveals contemporaneous shock buffering ($Q_{\mathrm{Rig}} = -0.01913$). These estimates are not causal and do not attribute feedback to algorithms. All reported numbers are generated by released code from the stated data. **Keywords:** adaptive agents; endogenous risk; financial stability; market microstructure; performative prediction; system identification.

**JEL codes:** C63, D83, G01, G14.

# 1 Introduction, Related Work, and Information Architecture

## 1.1 Introduction

A growing share of liquidity is supplied and consumed by algorithms that update their behavior using the prices they help set. Price formation then contains a feedback loop: order flow moves prices, and prices change the order flow that adaptive strategies submit. Episodes such as the May 2010 Flash Crash have been linked to interactions among automated strategies [7], but whether such feedback is present, how strong it is, and who generates it are difficult to determine from market data.

The question we study is:

> *Can a market surveillance authority detect whether interacting algorithms are amplifying systemic volatility using only observable transaction data? And if so, can public surveillance identify which specific group of participants is responsible?*

We give partial answers, organized around four results.

1. **Limits of passive identification.** Cohorts that share adaptation rates can exchange feedback allocations without changing any aggregate price response. The identifiable object is the merged rate–weight representation, and the minimal realization order is one plus the number of distinct active rates (Theorem 3.4). Detecting heterogeneity between rates $\delta$ apart needs a probe budget of order $\delta^{-4}$ (Theorem 3.5). Any error-controlled classifier must abstain near $S = 1$ (Theorem 3.6).
2. **Price stability versus dealer safety.** Because the joint Jacobian is generally nonnormal, the exact

worst-case safe shock radius for a leveraged dealer is $\gamma L E_0 / \sup_t \|CJ^t\|$. The Kreiss constant bounds this radius and the spectral radius does not (Theorem 3.8). In a funded simulation in which $S$ actually drives prices, halts rise steeply inside the stable region. Matched benchmarks show the mechanism is the low-frequency content of flow, which feedback inflates by $(1-S)^{-2}$ (theorem 9.2), rather than anything specific to learning agents.

3. **A testable memory class.** The positive heterogeneous-memory class is exactly the completely monotone cone. Hankel positivity alone does not characterize it. We give a size-valid finite-sample test (Theorem 3.7).
4. **What public data show.** We derive a reduced-form feedback estimand from the model and evaluate it on high-frequency crypto books across a 12-day Coinbase Pro baseline, an 11-day cross-venue replication, and a pre-registered 768 asset-day multi-year panel on Binance Spot (2021–2024). We validate it by Monte Carlo, including against spurious feedback from order splitting. The flow-feedback component is consistently small and positive ($\hat{Q} = 0.0082$), closed-loop persistence is strictly subcritical ($\hat{S} = 0.921 < 1$), the loop is far from the instability boundary, and the purely contrarian memory class is rejected. Structural minimum-distance estimation corroborates the feedback level ($Q_{\mathrm{SMM}} = 0.0086$), while Rigobon heteroskedasticity identifies instantaneous liquidity-shock absorption ($Q_{\mathrm{Rig}} = -0.01913$; section 7).

The paper is organized as follows:

- **Section 2:** the closed-loop market–learning model and its positive and signed stability boundaries (Theorems 2.5 and 2.6);
- **Section 3:** identification limits (Theorems M1–M5);
- **Section 4:** observable identification with public probes;
- **Section 5:** group identification;
- **Section 6:** the funded market mechanism and a learned-agent transfer study;
- **Section 7:** the public-data evidence;
- **Section 8:** model extensions;
- **Section 9:** discussion, the rebuilt solvency simulation, and conclusions.

Technical proofs, replication lineage, and research frontiers are provided in Appendices A–C and the accompanying Mathematical Supplement.

## 1.2 Related work and the precise research gap

### 1.2.1 Adaptive expectations and endogenous financial risk

Arthur et al. [1] model endogenous expectations in an artificial stock market. Brock and Hommes [3] analyze heterogeneous beliefs and routes to complex asset-price dynamics. These are direct precedents for any claim that agents and market conditions co-evolve. A convincing extension must identify something not already explained by adaptive strategy selection: for example, a measurable feedback operator that predicts behavior across materially different decision architectures.

Funding and trading conditions provide a second foundation. Brunnermeier and Pedersen [4] connect market liquidity and funding liquidity, including mutually reinforcing liquidity deterioration under specified conditions. Consequently, losses generated by common risk controls or liquidation pressure cannot automatically be attributed to AI. A meaningful test must retain ordinary liquidity and balance-sheet feedback in the non-AI benchmark.

### 1.2.2 Algorithmic interaction and financial-market experiments

Calvano et al. [5] show that Q-learning pricing algorithms can sustain supracompetitive outcomes in a repeated oligopoly experiment without direct communication. This establishes that individually trained agents can generate consequential collective behavior, although product-market pricing is not itself a model of systemic financial loss. Colliard et al. [12] bring learning algorithms into securities-market pricing and identify consequences of limited experimentation and noisy feedback. Their mechanisms are necessary comparators for a proposed market-making application.

Anand et al. [11] connect learning and reasoning architectures to coordination and financial fragility in fund-redemption experiments. Our model does not reproduce that environment or establish transfer across their architectures. The narrower comparison is between aggregate feedback and labeled-channel identification in a common-rate linear model (Section 5.1). The exchange exercises in Section 7 are separate observational diagnostics, not validation of this equivalence for real learning algorithms.

### 1.2.3 Performative prediction and learning in games

Perdomo et al. [8] formalize situations in which a deployed predictor changes the distribution on which it is subsequently evaluated. Narang et al. [9] extend the analysis to multiple decision makers and distinguish performatively stable equilibria from Nash equilibria. These

works already provide a language for prediction–action–distribution feedback.

A recent preprint by Farina and Perdomo [10] studies online algorithms and mixed performative stability. It is important not to equate that learning-theoretic solution concept with every form of financial stability. Convergence of average decisions or equilibrium distributions does not, without an additional argument, bound interim drawdowns, market depth, collateral shortfalls, or the solvency of a leveraged institution. Our intended bridge concerns such financial outcomes, not a denial of existing convergence results.

#### 1.2.4 Dynamical systems, robustness, and market simulation

Linearization, Lyapunov analysis, and common-metric robustness are established mathematical tools [16, 22]. Rebranding a spectral radius is not a new theorem. Their use here is to make assumptions, state variables, and failure modes explicit before attempting empirical measurement. The challenge is particularly acute for time-varying agents, discrete decisions, and partially observed policy states.

For experimental infrastructure, the ABIDES paper describes a discrete-event market simulator with exchange interaction and configurable communication latency [23]. It is an appropriate candidate for a later implementation, not evidence that our proposed agent population or market mechanism has already been validated. A second implementation or an independent simulator would help distinguish a general result from a software-specific artifact.

#### 1.2.5 Identification from imperfect input–output observations

Set-membership identification already asks which models are compatible with measured data and declared uncertainty [27]. Data informativity distinguishes recovering a unique model from learning enough to establish a property such as stability or stabilizability [29]. Positive-realization and steady-state-gain information have also been incorporated into regularized identification [32, 33]. Consequently, positivity constraints, confidence sets and gain side-information are essential predecessors, not newly invented concepts in this paper.

Our finite-family confidence construction is deliberately narrow: Gaussian independent episode vectors with arbitrary within-episode covariance. Sign-perturbed sums demonstrates that exact finite-sample identification methods also exist under other assumptions [34]; we have not implemented or subsumed them. Likewise, linear-system sample-complexity lower bounds are an established literature [30]. The new price-response two-point example supplies a model-specific observation limit, not a new general testing principle. The difference between noisy hidden kernels and noisy measured prices is central to the present integration.

#### 1.2.6 What is newly established, and what is not

Table 1 distinguishes our theoretical and empirical contributions from nearest predecessors in financial microstructure, algorithmic learning, and system identification.

### 1.3 Research question, scope, and claim discipline

The operational question is:

> Can changes in the joint market–learning feedback system identify economically meaningful changes in disturbance amplification, beyond what is explained by conventional liquidity, leverage, and volatility measures?

There are three different levels of claim. A *mathematical result* follows from an explicitly stated model. A *mechanism result* requires controlled experiments demonstrating that a specified feedback channel causes an outcome. An *external-validity result* requires evidence that the mechanism and its measurement apply beyond the experimental environment. This paper supplies analytical theorems, a controlled group measurement-and-mechanism study, and exploratory public-market analysis across live cryptocurrency markets. We explicitly distinguish endogenous algorithmic adaptation from static market liquidity, while avoiding unverified causal claims regarding AI specificity versus equally adaptive non-AI algorithmic institutions.

The primary domain for the first empirical paper should be a single market with an explicit price-formation mechanism, inventories, and funding constraints. An equity-like limit-order-book environment is a reasonable experimental starting point. A fund-redemption game is a possible alternative, but these should not be combined into one vaguely specified "global market" simulation. Banking, insurance, payments, and household finance have different state variables and institutional constraints; they are later transfer tests.

We structure the analysis around six core microstructural dimensions: machine autonomy exposure, behavioral concentration, adaptation rate, leverage, liquidity, and intervention latency. These are organizing dimensions rather than sufficient statistics by assumption. A critical surface in those coordinates may be model dependent, nonmonotone, or not estimable with available observations. Failure to find a sharp boundary is an admissible research outcome.

Table 1: Nearest-predecessor comparison matrix recording scope, assumptions, and explicit contribution increments.

| Predecessor | Core Question & Setting | Established Result | Specific Increment of This Work |
|---|---|---|---|
| Zelvyte and Griffin [40] | Multiplex credit default swap network systemic risk | Hawkes branching model across similarity, solvency, and profitability transmission layers | Separates aggregate closed-loop market feedback $S = a + Q$ from group attribution; proves that aggregate price surveillance cannot disentangle layer-specific feedback without channel-specific excitation in the common-rate symmetric benchmark. |
| Colliard et al. [12] | Quote competition and market-maker learning | Pricing under selected-action reward feedback and limited experimentation | Different environment: randomized probing of a prescribed linear feedback model. No replication or generalization of the learning-market-maker equilibrium is claimed. |
| Anand et al. [11] | Architecture fragility in fund redemptions | Stylized strategic run game; Q-learning vs. LLMs differ | Different object: labeled-channel identification within a common-rate linear class, not an architecture comparison or transfer theorem. |
| Khosravi and Smith [32] | Regularized identification of positive systems | Optimization with internal positivity constraints | A price-probe observation contract and continuous-rate outer bounds. Positivity and regularized system identification are prior tools, not inventions here. |
| van Waarde et al. [29] | Data informativity for system properties | Property identification need not require full model identification | A restricted common-rate example separates aggregate feedback from labeled group weights; the linear rank argument is an application, not a new general informativity theorem. |

# 2 The Closed-Loop Market–Learning Model

## 2.1 A joint market–learning system

### 2.1.1 States, decisions, and update timing

Let $x_t \in \mathbb{R}^d$ denote market state and $\theta_t \in \mathbb{R}^p$ denote the collection of adaptive policy states. Market state can include price deviations, inventory, funding conditions, and lagged variables. Policy state can include learned value estimates, forecasts, risk estimates, memory, and version identifiers represented in an appropriate local state description. Let $u_t \in \mathbb{R}^k$ collect economically consequential decisions. We write

$$u_t = \Pi(x_t, \theta_t), \tag{1}$$

$$x_{t+1} = F(x_t, u_t, \varepsilon_{t+1}), \tag{2}$$

$$\theta_{t+1} = G(\theta_t, x_t, u_t). \tag{3}$$

The ordering is part of the model. If learning uses the newly realized price or reward, that dependency must be substituted into an effective $G$ before differentiation. If decisions are delayed, the necessary lags must be included in the state. A delay cannot be appended as an informal qualifier after calculating a delay-free Jacobian.

For local deterministic analysis, set the innovation to zero and define

$$z_t = \begin{bmatrix} x_t \\ \theta_t \end{bmatrix}, \qquad \Phi(z_t) = \begin{bmatrix} F(x_t, \Pi(x_t, \theta_t), 0) \\ G(\theta_t, x_t, \Pi(x_t, \theta_t)) \end{bmatrix}. \tag{4}$$

Assume $z^*$ is a fixed point and that the effective map is continuously differentiable in a neighborhood of $z^*$. This assumption excludes unmodeled discontinuous execution rules. For stochastic agents it may describe an explicitly defined conditional-mean map, but mean-map stability alone is not a claim of stochastic or almost-sure stability.

### 2.1.2 The complete feedback Jacobian

Applying the chain rule at $z^*$ gives

$$J^* = D\Phi(z^*) = \begin{bmatrix} F_x + F_u\Pi_x & F_u\Pi_\theta \\ G_x + G_u\Pi_x & G_\theta + G_u\Pi_\theta \end{bmatrix}_{z^*}. \tag{5}$$

The upper-left block includes the market's direct persistence and instantaneous decision feedback. The upper-right block transmits a change in learned state into market outcomes. The lower-left block transmits a market disturbance into policy adaptation. The lower-right block describes policy-state persistence and action-dependent updating.

The product $F_u\Pi_x$ is only one channel. It is generally not the full transition derivative. Omitting the learning blocks can classify the frozen-policy system correctly while classifying the actual adaptive system incorrectly. Misclassification is possible, not inevitable: decoupled learning or other special structures can leave the relevant stability conclusion unchanged.

**Proposition 2.1** (Steady-state Kalman filtering yields exponential memory)**.** *Suppose an algorithmic market maker tracks an unobservable, drifting asset fundamental $v_{t+1} = v_t + \xi_{t+1}$ with innovation variance $\sigma_\xi^2 > 0$ under noisy order book execution signals $y_t = v_t + \varepsilon_t$ with observation variance $\sigma_\varepsilon^2 > 0$. Under Gaussian innovations, the steady-state one-step predictor that minimizes mean-squared error is the Kalman recursion:*

$$\hat{v}_{t+1} = (1 - K)\hat{v}_t + Ky_t, \tag{6}$$

*where the steady-state gain is the positive root of the scalar Riccati equation:*

$$K = \frac{-\sigma_\xi^2 + \sqrt{\sigma_\xi^4 + 4\sigma_\xi^2\sigma_\varepsilon^2}}{2\sigma_\varepsilon^2} \in (0, 1). \tag{7}$$

*If, in addition, quotes follow the linear rule $u_t = q\hat{v}_t$ (an assumption, not a derived optimum), the induced impulse-response kernel is*

$$k_j = qK(1-K)^j, \quad j \geq 0. \tag{8}$$

*This is a single exponential mode and hence completely monotone. The result gives one signal-extraction rationale for exponential memory in this specific local-level model. It does not show that real algorithms filter this way, and other signal models (e.g. mean-reverting fundamentals or multiple components) induce different kernels.*

**Proposition 2.2** (Linearized regularized gradient updates yield exponential memory)**.** *Consider an automated execution algorithm adapting policy parameters $\theta_t \in \mathbb{R}^p$ via regularized Online Gradient Descent on a strictly convex loss $\mathcal{L}(x_t, \Pi(x_t, \theta_t))$ with $L_2$ weight decay $\gamma_{\text{weight}} \in (0,1)$:*

$$\theta_{t+1} = (1-\gamma_{\text{weight}})\theta_t - \alpha_t \nabla_\theta \mathcal{L}(x_t, \Pi(x_t, \theta_t)). \tag{9}$$

*For constant learning rate $\alpha > 0$ and stationary fixed point $(\theta^*, x^*)$, linearization along an invariant coordinate of the loss Hessian with eigenvalue $\lambda_H > 0$ and cross-derivative $\ell_{\theta x}$ yields the recursion*

$$\delta\theta_{t+1} = (1-\eta)\delta\theta_t + \eta q \delta x_t, \tag{10}$$

*with effective adaptation rate $\eta = \gamma_{\text{weight}} + \alpha\lambda_H$, assumed to lie in $(0,1)$, and gain $\eta q = -\alpha\ell_{\theta x}$. The memory is positive ($q \geq 0$) only if $\ell_{\theta x} \leq 0$, which is a property of the loss and not guaranteed.*

Propositions 2.1 and 2.2 show that exponential memory arises in two simple, standard learning rules. They do not show that deployed trading systems have this structure. The mapping is local: it requires a smooth policy, a fixed point, and stationary learning rates. Threshold rules, discrete retraining, nonconvex losses, replay buffers and adaptive step sizes produce state-dependent, delayed or switching responses that are outside the class analyzed here. Whether a particular algorithm's kernel lies in the positive class is an empirical question; Theorem 3.7 gives a test, and section 7.4 applies it to aggregate flow, not to any identified algorithm.

#### 2.1.3 Units and financial interpretation

Prices, inventories, and learning states have different units. Specify a diagonal scaling matrix $D$ using an ex ante reference regime, then use $\widetilde{J} = DJD^{-1}$ for norm-based comparisons. Eigenvalues are invariant under this change of coordinates, but Euclidean finite-horizon gains are not. An economically meaningful output map $C_y$ and disturbance map $B_\varepsilon$ should connect the state to such outcomes as price dislocation, funding demand, or inventory loss:

$$\delta z_{t+1} = J^* \delta z_t + B_\varepsilon \varepsilon_{t+1}, \qquad y_t = C_y \delta z_t. \tag{11}$$

The specification of $C_y$ is necessary. Instability of an unobservable or economically irrelevant internal coordinate is not automatically systemic financial instability.

### 2.2 Local stability, finite-horizon risk, and switching

#### 2.2.1 A standard local criterion

**Proposition 2.3** (Fixed-system local stability; standard result)**.** *Suppose the deterministic joint map in equation (4) is continuously differentiable near a fixed point $z^*$. If $\rho(J^*) < 1$, that fixed point is locally exponentially asymptotically stable. If $\rho(J^*) > 1$, it is unstable. The case $\rho(J^*) = 1$ is not resolved by this first-order criterion.*

A proof of the stable case and the scope of the unstable case are given in section A. This proposition applies an established linearization principle; it is not a claim of a new financial law [22].

To summarize the instantaneous spectral amplification of the joint system, define the descriptive diagnostic

$$\text{MMC}_{\text{local}} = \rho(J^*). \tag{12}$$

It is a local, model-conditioned diagnostic. Values above or below one do not by themselves specify crisis probability, loss magnitude, or welfare. At equality, nonlinear terms can determine the outcome: $z_{t+1} = z_t - z_t^3$ is attracting sufficiently near zero, whereas $z_{t+1} = z_t + z_t^3$ is repelling, although both derivatives at zero equal one.

#### 2.2.2 Stable eigenvalues can conceal transient amplification

For a fixed linear system, define the finite-horizon state gain

$$\mathcal{A}_H(J) = \max_{0 \leq h \leq H} \left\| J^h \right\|_2. \tag{13}$$

A nonnormal matrix can have $\rho(J) < 1$ and a large $\mathcal{A}_H(J)$. Consider

$$J_N = \begin{bmatrix} 0.8 & 4 \\ 0 & 0.8 \end{bmatrix}, \qquad J_N^h = \begin{bmatrix} 0.8^h & 4h(0.8)^{h-1} \\ 0 & 0.8^h \end{bmatrix}. \tag{14}$$

Its spectral radius is 0.8, but the executed numerical calculation gives $\mathcal{A}_{50}(J_N) \approx 8.2124$, attained at $h = 4$. Such a state amplification can matter before eventual convergence. Whether it causes financial loss depends on state scaling, shock direction, output exposure, and constraints. This example establishes a diagnostic limitation, not the existence of a particular market event.

#### 2.2.3 Frozen-time stability is not switching stability

A model update or time-varying policy changes the transition matrix. Define

$$\Phi_{t,h} = J_{t+h-1} \cdots J_t, \qquad \lambda_{t,h} = h^{-1} \log \|\Phi_{t,h}\|. \tag{15}$$

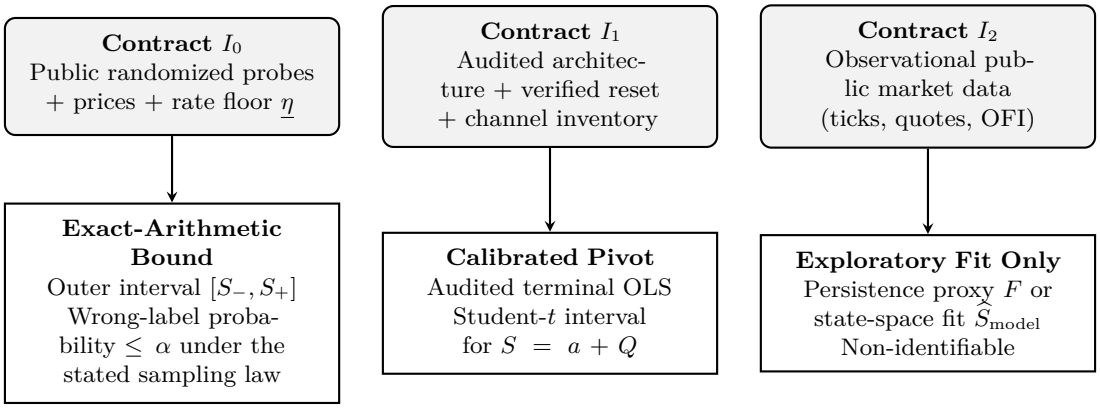



Figure 1: Information contracts and identification boundaries. Contract $I_0$ supports an exact-arithmetic outer-set theorem; the numerical conic implementation currently withholds certification. Contract $I_1$ supports a finite-sample pivot under audited disclosures and the stated sampling law, not noiseless point identification. Contract $I_2$ is exploratory and does not identify structural feedback.

The norm and time unit must be fixed before comparing finite-horizon values. Consider

$$J_1 = \begin{bmatrix} 0.5 & 2 \\ 0 & 0.5 \end{bmatrix}, \qquad J_2 = \begin{bmatrix} 0.5 & 0 \\ 2 & 0.5 \end{bmatrix}. \tag{16}$$

Both have spectral radius 0.5, yet $\rho(J_2 J_1) \approx 4.4861$. Alternation therefore amplifies disturbances despite each frozen matrix being stable. Monitoring $\rho(J_t)$ alone cannot certify stability under arbitrary updates. Common-norm conditions in section 9.1 provide one sufficient alternative.

#### 2.2.4 Calm observations do not identify weak feedback

**Proposition 2.4** (Volatility and amplification need not move together)**.** *For the stable scalar process $x_{t+1} = rx_t + \sigma\varepsilon_{t+1}$ with $0 \leq r < 1$ and independent, zero-mean, unit-variance innovations, stationary variance is $\sigma^2/(1-r^2)$ and the response sum to a unit additive state shock (rather than a unit standardized $\varepsilon$ innovation) is $1/(1-r)$. There exist parameter pairs with lower stationary variance but a higher response sum.*

*Proof.* Stationarity gives $v = r^2 v + \sigma^2$. The response sum is the convergent geometric series $\sum_{h \geq 0} r^h$. Choosing $(r, \sigma) = (0.5, 1)$ gives variance $4/3$ and response sum 2; choosing $(0.9, 0.2)$ gives variance $4/19$ and response sum 10. □

This does *not* prove that unconditional tail risk rises in the second process. For Gaussian innovations its unconditional tails contract with variance. Establishing a "machine stability paradox" involving lower ordinary volatility and higher tail losses requires a specified stress distribution, exposure mechanism, and additional evidence. The narrower result is that ordinary volatility alone is an inadequate identifier of feedback amplification.

### 2.3 A tractable positive-feedback market model

#### 2.3.1 A price-adjustment and belief-update specialization

Let $x_t \in \mathbb{R}^d$ be deviations of market prices from a local reference level, and $m_t \in \mathbb{R}^n$ be deviations of adaptive forecasts or decision states. Consider

$$x_{t+1} = ax_t + gBm_t + \varepsilon_{t+1}, \tag{17}$$
$$m_{t+1} = (1-\eta)m_t + \eta C x_t. \tag{18}$$

Here $0 \leq a < 1$, $0 < \eta \leq 1$, $g \geq 0$, and $B \in \mathbb{R}_+^{d \times n}$, $C \in \mathbb{R}_+^{n \times d}$. The matrix $B$ maps policy-state changes into price pressure; $C$ maps prices into target learning states. The nonnegativity assumption imposes reinforcing responses in the chosen coordinates. It is not a realistic description of all financial strategies.

The market component can be interpreted as a local reduced form of price adjustment under order imbalance. The memory component is an exponential update, not a derivation of optimal behavior or of a specific foundation model. Human adaptive expectations can satisfy the same equations. Machine-specific interpretation requires an empirically validated link from actual agents to $B$, $C$, $g$, and $\eta$.

The joint transition matrix is

$$J_+ = \begin{bmatrix} aI_d & gB \\ \eta C & (1-\eta)I_n \end{bmatrix}. \tag{19}$$

A genuinely frozen-policy baseline holds the relevant $m$ deviations at zero and has market transition $aI_d$. Merely setting $\eta = 0$ in the full augmented system leaves neutral memory eigenvalues at one; that is not the same stability claim as conditioning on a fixed policy.

#### 2.3.2 An exact stability boundary

**Theorem 2.5** (Positive-feedback boundary; model specialization)**.** *Under the assumptions of equations* (17) *and* (18)*, the origin of the deterministic linear joint system is asymptotically stable if and only if*

$$g\rho(BC) < 1 - a. \tag{20}$$

*Writing* $\nu = \rho(BC)$*, its spectral radius is*

$$\rho(J_+) = \frac{a + 1 - \eta + \sqrt{(a - 1 + \eta)^2 + 4g\eta\nu}}{2}. \tag{21}$$

*Equality in equation* (20) *produces an eigenvalue at one. A strict reversal produces an eigenvalue larger than one.*

**Theorem 2.6** (Signed-feedback stability boundary)**.** *Let* $a \in [0, 1)$*,* $\eta \in (0, 1]$ *and* $g\kappa \in \mathbb{R}$ *of either sign, so that both momentum feedback* ($g\kappa > 0$) *and contrarian feedback* ($g\kappa < 0$) *are allowed. The scalar closed-loop Jacobian* $J = \left(\begin{smallmatrix} a & g \\ \eta\kappa & 1-\eta \end{smallmatrix}\right)$ *is Schur stable if and only if*

$$-\frac{1 - a(1 - \eta)}{\eta} \;<\; g\kappa \;<\; 1 - a. \tag{22}$$

*The upper inequality is the momentum boundary. It is equivalent to* $S = a + g\kappa < 1$ *and to* $\mathcal{L} = g\kappa/(1 - a) < 1$*, since* $S - 1 = (1 - a)(\mathcal{L} - 1)$*. The lower inequality is an oscillatory (flip) boundary that has no counterpart in the positive model.*

The proof of theorem 2.5 in section A.2 is a positive-systems argument specialized to this model; the proof of theorem 2.6 (Jury conditions) is in section A.6. The elementary spectral result should not be presented as an original general theorem without a literature-priority review. Its role is to produce explicit, falsifiable consequences and to check a proposed empirical measurement strategy.

Define the model-specific loop ratio

$$\mathcal{L} = \frac{g\rho(BC)}{1 - a}. \tag{23}$$

Then $\mathcal{L} < 1$ characterizes stability in this restricted model. The surface $g\rho(BC) = 1 - a$ depends on response structure and market persistence, not merely the fraction of AI participants.

**Corollary 2.7** (No universal participation cutoff in this model)**.** *Suppose, as an additional model parameterization,* $g = \alpha\ell/q$*, where* $\alpha \in [0, 1]$ *is economically weighted machine participation,* $\ell > 0$ *is a specified exposure multiplier, and* $q > 0$ *is normalized depth. If* $\nu > 0$*, the formal boundary is*

$$\alpha^* = \frac{(1 - a)q}{\ell\nu}. \tag{24}$$

*If* $\alpha^* > 1$*, no crossing occurs on the feasible participation interval. The formula is not an empirical identity and does not establish that these factors combine multiplicatively in real markets.*

#### 2.3.3 Learning speed: a useful negative result

For every strictly positive $\eta$ in the assumed interval, the boundary in equation (20) is unchanged. Update speed changes eigenvalues and adjustment times but does not shift this particular boundary. A claim that faster learning necessarily lowers the instability threshold is therefore false even within a simple reinforcing model. Heterogeneous update rates, delays, nonlinear risk limits, signed feedback, and endogenous depth can change that conclusion; those extensions require fresh analysis.

For the scalar case $B = 1$, $C = \kappa \geq 0$,

$$J_s = \begin{bmatrix} a & g \\ \eta\kappa & 1 - \eta \end{bmatrix}, \qquad g\kappa < 1 - a. \tag{25}$$

At $a = 0.4$, $\kappa = 1$, the boundary is $g = 0.6$. Figure 2 plots the analytic eigenvalue formula, not a fitted empirical relationship.

# 3 Fundamental Identification Boundaries

## 3.1 Measurement and causal identification

#### 3.1.1 Machine autonomy exposure without arbitrary category scores

Rather than assign unvalidated values such as 0.1 to advisory software and 0.7 to execution software, define a decision event and audit who had executable authority. For event $e$, let $v_e \geq 0$ be a predeclared exposure weight and $d_e \in \{0, 1\}$ indicate machine execution without a contemporaneous discretionary human approval. For a window $W$,

$$\mathrm{MAE}_W = \frac{\sum_{e \in W} v_e d_e}{\sum_{e \in W} v_e}. \tag{26}$$

The denominator must be positive, and weights should represent comparable exposure, not arbitrary numbers of messages. Advisory use, order generation, execution authority, and online learning should be reported separately. Net exposure, gross executed exposure, and assets under management answer different questions and should not be mixed.

If authority is unobserved for some events, report an interval: the lower bound treats unknown events as nonautonomous and the upper bound treats them as autonomous. This makes missingness visible rather than assigning undocumented scores. It does not solve measurement error in $v_e$.

#### 3.1.2 Behavioral concentration is not provider concentration

Provider concentration and common-cloud dependence are operational quantities. Behavioral concentration in-

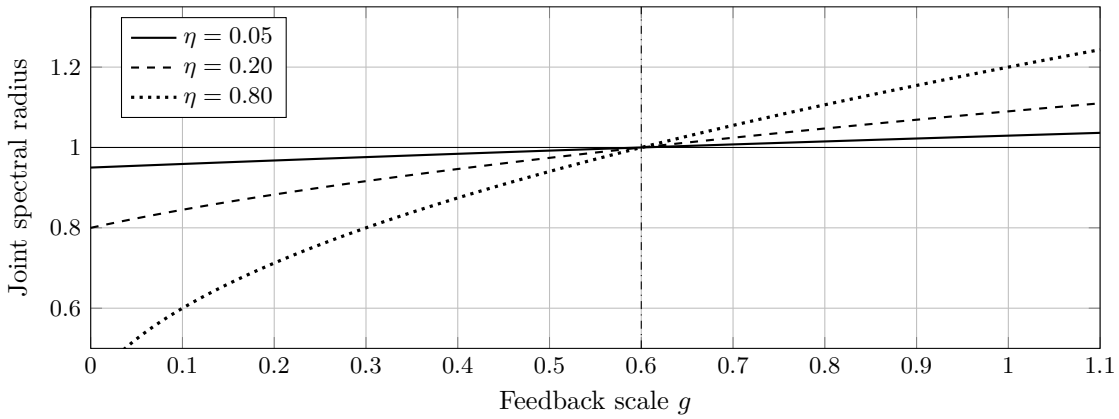



Figure 2: Analytic illustration for $a = 0.4$, $\kappa = 1$. The three update rates share the boundary $g = 0.6$ but yield different growth or decay rates. No AI models were trained to produce this figure.

stead asks whether agents react along similar directions. Let $r_i \in \mathbb{R}^K$ be a nonzero response vector for agent $i$ on an independently designed panel of controlled perturbations, normalized to unit length after reporting its original response magnitude. With $w_i \geq 0$ and $\sum_i w_i = 1$, define

$$S = \sum_i w_i r_i r_i^\top, \quad \text{ADC} = \lambda_{\max}(S), \quad D_{\text{eff}} = \frac{1}{\text{tr}(S^2)}. \tag{27}$$

Here $\text{tr}(S) = 1$, so ADC is a directional-concentration descriptor. Zero-response agents require a separate reported mass; silently normalizing them is invalid. The perturbation panel, units, weights, and sampling variability all affect the estimate.

There is an important limitation. Two equal-weight agents with responses $r, r$ and two with responses $r, -r$ both yield the same $S$ and ADC $= 1$. Yet their aggregate signed responses are $r$ and zero, respectively. Thus equation (27) cannot substitute for the signed aggregate response, price-impact direction, or the product $BC$. Diversity can remove reinforcing feedback, but it can also remove stabilizing behavior. There is no unconditional "more diversity is safer" theorem here.

#### 3.1.3 An identification impossibility without provenance

**Proposition 3.1** (Observational equivalence of decision–maker labels)**.** *Consider two economies with the same initial market-state distribution, the same market transition kernel conditional on aggregate actions, and the same conditional law of aggregate actions given every observable history. The economies can have different assignments of human and machine decision-maker labels. They nevertheless induce the same law for all observable market histories. No estimator using only those histories can consistently distinguish the two label assignments or identify different autonomy shares attached solely to those assignments.*

*Proof.* The initial observable distributions coincide. If observable histories coincide in distribution through $t$, the assumed conditional action law and common market transition kernel imply the same distribution for the next observation. Induction establishes equality of all finite-dimensional observable distributions. Any statistic of those observations therefore has the same law in both economies; it cannot converge in probability to two distinct label-dependent constants. □

This is an application of ordinary observational equivalence, not a claim that financial attribution is impossible with richer data. Account-level decision provenance, randomly assigned architecture, authorized policy-update experiments, or credible external variation can break the equivalence. Historical prices alone do not reveal whether a self-exciting pattern was produced by humans, fixed algorithms, adaptive agents, or a mixture.

#### 3.1.4 Estimating the response system

The general operator-estimation problem below is broader than the completed scalar experiment. Sections 4.1–4.3 provide one restricted solution using signed filled probes and prices; they neither reconstruct every private policy state nor identify AI participation from prices alone.

In a simulator with full state access, estimate $J$ by perturbing each normalized coordinate around a specified operating point and measuring the resulting joint next-state response. Use multiple perturbation sizes to test local linearity and independent random streams to quantify agent stochasticity. Finite differences of a mean response estimate a mean-map derivative, not a pathwise derivative of a discrete sampled action.

In observational data, fitting $z_{t+1} = Jz_t + e_{t+1}$ requires sufficient excitation, a defensible state representation, and restrictions controlling confounding and feedback-induced endogeneity. Hidden policy states typically require a state-space or input–output model. A reduced-

form estimate may predict outcomes without identifying architecture-specific mechanisms; the paper must report which claim its design supports.

An uncertainty set $\mathcal{J}$ should reflect estimation uncertainty and relevant model misspecification. There is no distribution-free license to assign a narrow confidence interval to $\rho(J)$ by ignoring dependence, near-defective matrices, hidden states, or selected operating points. A claim of pre-crisis warning additionally requires lead time, false-alarm rates, and a prospectively fixed event definition.

## 3.2 Hidden slow learners and finite-horizon identification limits

This section establishes the fundamental mathematical limits of identifying latent adaptation from finite-horizon observations. Its purpose is to distinguish a well-defined structural stability target from the empirical information available to estimate it under passive surveillance. The finite-window construction, its limitations and its adverse controls are analyzed below.

### 3.2.1 Heterogeneous memory and the scalar stability target

Consider the fixed positive linear specialization

$$x_{t+1} = ax_t + \sum_{i=1}^{n} y_{i,t} + \kappa u_t + e_{t+1}, \tag{28}$$

$$y_{i,t+1} = (1-\eta_i)y_{i,t} + \eta_i q_i x_t, \tag{29}$$

where $0 \le a < 1$, $q_i \ge 0$, $0 < \eta_i \le 1$, and $\kappa > 0$. Positivity refers to transition coefficients, not to signs of realized price deviations or positions. Define $Q = \sum_i q_i$ and $G = Q/(1-a)$. Throughout the new experiments, $S = a + Q$ is the unnormalized stability target and $G$ is used only to index generating configurations. Confusing $G$ with $Q$ changes the threshold.

**Proposition 3.2** (Positive heterogeneous-memory boundary)**.** *The joint transition matrix of* (29) *is Schur stable exactly when $S = a + Q < 1$. Equality gives an eigenvalue at one; $S > 1$ gives spectral radius greater than one.*

*Proof.* The matrix is $J = \begin{bmatrix} a & \mathbf{1}^\top \\ \operatorname{diag}(\eta) q & \operatorname{diag}(1-\eta) \end{bmatrix}$. The lower diagonal block of $I - J$ is positive. Its Schur complement is $1 - a - \sum_i q_i$. The nonsingular M-matrix characterization of a nonnegative transition matrix establishes the stable case. At equality, $(1, q^\top)^\top$ is an eigenvector. Above the boundary, the function

$$f(\lambda) = \lambda - a - \sum_i \frac{\eta_i q_i}{\lambda - 1 + \eta_i}$$

is negative at one, strictly increasing on $[1, \infty)$, and tends to infinity. It therefore has a root greater than one. This is a specialization of standard positive-system analysis, not a new general stability principle [26]. □

With zero initial adaptive states, the loop-open kernel and unit closed-loop response are

$$k_j = \sum_i q_i \eta_i (1-\eta_i)^j, \quad \sum_{j=0}^{\infty} k_j = Q, \tag{30}$$

$$h_0 = 1, \quad h_1 = a, \quad h_{t+1} = a h_t + \sum_{j=0}^{t-1} k_j h_{t-1-j}. \tag{31}$$

The kernel sum exists even when the closed loop is unstable. Summing a diverging closed-loop response and calling it $Q$ would be incorrect.

### 3.2.2 Exact finite-window ambiguity

**Theorem 3.3** (Retained exact construction)**.** *For every positive integer $K$ and $a \in [0, 1)$, there are a stable and an unstable model of* (29) *with the same $a$, the same impact normalization, and the same number of active memory states, whose kernels agree for $j = 0, \dots, K-1$ and whose price responses agree for $t = 0, \dots, K+1$. Rational nodes and $a$ give rational coefficients.*

*Proof.* Choose distinct $0 \le r_0 < \cdots < r_K < 1$ and let

$$w_i = \left[ \prod_{\ell \ne i} (r_i - r_\ell) \right]^{-1}.$$

Comparing the leading coefficients in polynomial interpolation gives $\sum_i w_i r_i^j = 0$ for $j < K$ and $\sum_i w_i r_i^K = 1$. Partial fractions evaluated at one give

$$\sum_i \frac{w_i}{1 - r_i} = \frac{1}{\prod_i (1 - r_i)} > 0.$$

Set $B_+ = \sum_i \max(w_i, 0)/(1 - r_i)$, $B_- = \sum_i \max(-w_i, 0)/(1 - r_i)$ and $s = 2(1-a)/(B_+ + B_-)$. Alternating signs imply $B_+ > B_- > 0$. Choose $\eta_i = 1 - r_i$ and $q_i^{\pm} = s \max(\pm w_i, 0)/(1 - r_i)$. The totals have midpoint $1 - a$ and lie on opposite sides of it. Their kernel difference is $s \sum_i w_i r_i^j$, so the stated terms coincide. Equation (31) propagates this equality to the price responses, with first unmatched difference $h^+_{K+2} - h^-_{K+2} = s$. Delete zero weights and split positive weights among identical-rate copies until each model has $K + 1$ active states; splitting changes neither kernel nor stability. Rationality is immediate. □

Equal deterministic observations imply balanced forced-classification error 1/2 on this pair, even after adding identically distributed observation noise. This is a finite-window, increasing-order construction, not a result that every known low-order market remains unidentifiable indefinitely. Under a fixed valid learning-rate floor

below one the construction remains possible, but the stability separation can shrink. A floor is useful for bounds, not a guarantee of finite-sample decisiveness arbitrarily near the boundary.

**Theorem 3.4** (Equivalence class and minimal realization)**.** *Consider two economies $\mathcal{E}_A = \{(\eta_i^A, q_i^A)\}_{i=1}^{n_A}$ and $\mathcal{E}_B = \{(\eta_j^B, q_j^B)\}_{j=1}^{n_B}$ evolving according to (29) with the same persistence $a$ and impact normalization $\kappa$, with $q_i > 0$ and $\eta_i \in (0,1]$. Let $\mathcal{M}(\mathcal{E}) \equiv \{(\eta, \sum_{i:\eta_i=\eta} q_i) : \eta \in \operatorname{supp}(\eta)\}$ be the reduced representation obtained by merging identical adaptation rates. Then:*

1. ***Impulse-response equivalence.*** *$\mathcal{E}_A$ and $\mathcal{E}_B$ generate identical price impulse responses $h_t$ at all horizons $t \geq 0$ if and only if $\mathcal{M}(\mathcal{E}_A) = \mathcal{M}(\mathcal{E}_B)$.*
2. ***Minimal order.*** *The transfer function from $u$ to $x$ is coprime of degree*

$$n_{\min} = 1 + |\operatorname{supp}(\mathcal{M}(\mathcal{E}))|, \tag{32}$$

*which is therefore the minimal state dimension of any realization (Ho–Kalman; Kalman–Gilbert). Variations of the individual $q_i$ that keep each merged total fixed (an $(n - |\operatorname{supp}\mathcal{M}|)$-dimensional set of directions) leave the price response unchanged.*

**Theorem 3.5** (Local minimax risk and a $\delta^{-4}$ separation bound)**.** *Suppose each of $R$ independent probe episodes of amplitude $A$ yields $y_{r,t} = A\lambda k_t(\theta) + \varepsilon_{r,t}$, $t = 1, \ldots, H$, with $\varepsilon_{r,t} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$, where $\lambda > 0$ is the depth impact and $k_t(\theta)$ the loop-open kernel.*

1. ***Single rate.*** *Let $k_t(\eta) = (1-\eta)^t$ and $H_{\text{eff}}(\eta) = \sum_{t=1}^{H} t^2(1-\eta)^{2(t-1)}$. For every $\eta_0$ and $\delta_R = \sigma/(A\lambda\sqrt{R\bar{H}})$, where $\bar{H} = \sup_{|\eta-\eta_0|\leq\delta_R} H_{\text{eff}}(\eta)$,*

$$\inf_{\hat{\eta}} \max_{\eta\in\{\eta_0,\eta_0+\delta_R\}} \mathbb{E}_\eta\left[(\hat{\eta}-\eta)^2\right] \;\geq\; \frac{\sigma^2}{16\,R\,A^2\lambda^2\,\bar{H}}. \tag{33}$$

*Under standard regularity the maximum-likelihood estimator attains the matching Cramér–Rao rate $\sigma^2/(RA^2\lambda^2 H_{\text{eff}}(\eta))$ asymptotically as $R \to \infty$; no finite-sample matching upper bound is claimed.*

2. ***Separating near-coincident cohorts.*** *Let $H_0$ be a single cohort, $k_t = wr^t$, and $H_1$ a symmetric split, $k_t = \frac{w}{2}[(r+\delta)^t + (r-\delta)^t]$, with $0 < \delta < r < r+\delta < 1$. Any test whose two error probabilities are both at most $\beta < 1/2$ requires*

$$R \;\geq\; \frac{16\,(1-2\beta)^2\,\sigma^2}{A^2\lambda^2 w^2\,C(r+\delta)}\,\delta^{-4}, \qquad C(\rho) = \sum_{t\geq 2} t^2(t-1)^2\rho^{2(t-2)} < \infty. \tag{34}$$

**Theorem 3.6** (Mandatory abstention near the boundary)**.** *Let $\mathcal{T}$ map data to $\{\text{Stable}, \text{Unstable}, \text{Abstain}\}$ and satisfy, uniformly over the model class, $\mathbb{P}_\theta(\mathcal{T} = \text{Unstable}) \leq \alpha$ whenever $S(\theta) < 1$ and $\mathbb{P}_\theta(\mathcal{T} = \text{Stable}) \leq \alpha$ whenever $S(\theta) > 1$.*

1. ***Two-point bound.*** *For any $\theta_0, \theta_1$ with $S(\theta_0) < 1 < S(\theta_1)$ and $i \in \{0,1\}$,*

$$\mathbb{P}_{\theta_i}(\mathcal{T} = \text{Abstain}) \;\geq\; 1-2\alpha-\text{TV}(\mathbb{P}_{\theta_0}, \mathbb{P}_{\theta_1}) \;\geq\; 1-2\alpha-\sqrt{\text{KL}(\mathbb{P}_{\theta_0}| \tag{35}$$

2. ***Indifference zone under Gaussian probes.*** *In the probe model of Theorem 3.5, fix the rates and write $k_t = Q\phi_t$ with $\phi \geq 0$ and $\sum_t \phi_t = 1$. For the two models with $S = 1 \mp \varepsilon$, the abstention probability is at least $1 - 2\alpha - \varepsilon A\lambda \|\phi_{1:H}\|_2 \sqrt{R}/\sigma$. Hence it is at least $\gamma$ for all $\varepsilon \leq \varepsilon^* = (1 - 2\alpha - \gamma)\,\sigma/(A\lambda \|\phi_{1:H}\|_2 \sqrt{R})$.*
3. ***Finite windows.*** *If $H \leq K+1$, the pair of Theorem 3.3 has $\text{TV} = 0$, so the abstention probability at both members is at least $1 - 2\alpha$ for every $R$.*

**Theorem 3.7** (Completely monotone memory and a size-valid finit

***Characterization.*** *Every kernel $k_j = \sum_i q_i\eta_i(1-\eta_i)^j$ of (29) is a Hausdorff moment sequence $k_j = \int_{[0,1]} r^j\, d\mu(r)$ with $\mu \geq 0$, equivalently completely monotone: $(-1)^m\Delta^m k_j \geq 0$ for all $m, j \geq 0$ [17]. Conversely, every completely monotone sequence is a limit of such kernels (finitely atomic measures are weak-$*$ dense).*

2. ***Hankel positivity alone is not sufficient.*** *$k_j = 2^j$ has positive semidefinite Hankel matrices $[k_{i+j}]$ of every order but is not completely monotone. For a finite vector $(k_0, \ldots, k_{2m})$, membership in the truncated moment cone on $[0,1]$ holds if and only if both $[k_{i+j}]_{i,j=0}^{m} \succeq 0$ and $[k_{i+j+1} - k_{i+j+2}]_{i,j=0}^{m-1} \succeq 0$ (Markov–Lukács; 19).*
3. ***Test.*** *Let $\hat{k} \in \mathbb{R}^K$ be approximately $\mathcal{N}(k, V)$ with $V$ known or consistently estimated, let $\mathcal{C}$ be the cone generated by $\{(r^j)_{j<K} : r \in \mathcal{G}\}$ for a grid $\mathcal{G} \subset [0,1]$, and let $T(\hat{k}) = \min_{c\in\mathcal{C}}(\hat{k}-c)^\top V^{-1}(\hat{k}-c)$. Rejecting when $T$ exceeds the $1-\alpha$ quantile of $T(Z)$, $Z \sim \mathcal{N}(0, V)$, gives a test of size at most $\alpha$ at every $k \in \mathcal{C}$.*

**Theorem 3.8** (Exact safe radius and the Kreiss bound)**.** *Let $z_t = J^t z_0$ with $\rho(J) < 1$, let $C$ map the state into dealer exposure, and say a breach occurs if $\|CJ^t z_0\|_2 > B := \gamma L E_0$ for some $t \geq 0$. Let $M_C = \sup_{t\geq 0} \|CJ^t\|_2$ and $r^* = B/M_C$.*

1. ***Exact worst-case safe radius.*** *Every $z_0$ with $\|z_0\|_2 < r^*$ is safe. For every $\epsilon > 0$ some $z_0$ with $\|z_0\|_2 \leq (1+\epsilon)r^*$ breaches.*
2. ***Kreiss bound.*** *With $C = I$,*

$$\sup_{t\geq 0} \|J^t\|_2 \;\geq\; \mathcal{K}(J) \equiv \sup_{|z|>1} (|z|-1)\|(zI-J)^{-1}\|_2, \tag{36}$$

*so $r^* \leq B/\mathcal{K}(J)$: for every $\epsilon > 0$ there is a breaching shock of norm at most*

$$(1+\epsilon)\,\frac{\gamma L E_0}{\mathcal{K}(J)}. \tag{37}$$

*Conversely,* $\sup_t \|J^t\|_2 \le e\,n\,\mathcal{K}(J)$ *for* $J \in \mathbb{R}^{n\times n}$ *[21], so* $\mathcal{K}$ *determines* $r^*$ *up to the factor* $en$.

3. ***Spectral radius does not bound the safe radius.*** *For* $J = \left[\begin{smallmatrix} \lambda & m \\ 0 & \lambda \end{smallmatrix}\right]$ *with* $0 < \lambda < 1$, $\rho(J) = \lambda$ *for all* $m$, *while* $\sup_t \|J^t\|_2 \ge m$, *so* $r^* \le B/m \to 0$ *as* $m \to \infty$.

*The breach statement in item 2 holds for* some *shock direction, not for every shock of that size. The theorem concerns the deterministic linearization; it says nothing about the frequency of such shocks.*

## 3.3 Mathematical Lineage and Microstructural Increments

Table 2 separates the classical result behind each theorem from what is added here. In every case the mathematics is standard. The contribution is the specialization to the market-learning model and the resulting statements about surveillance.

Before equal-count splitting, the witness uses rates $(1, 1/2, 1/20)$ with weights $(0, 12/65, 0)$ and $(54/1235, 0, 240/247)$. Displayed decimals are rounded; the equality is rational, not an approximate visual match. A one-rate extrapolation from two exact taps yields

$$\widehat{Q}_{\text{one}} = \frac{k_0^2}{k_0 - k_1} \le Q.$$

Indeed, with $p_i = q_i\eta_i/k_0$, Jensen's inequality for $r \mapsto 1/(1-r)$ gives the bound. The inequality is strict for heterogeneous active rates. Thus a low one-rate estimate can be false reassurance, not a stability certificate.

### 3.3.1 What the first measurement experiment actually assumed

If $\eta_i \ge \underline{\eta} > 0$, $r = 1 - \underline{\eta}$, and $T_H = \sum_{j<H} k_j$, positivity gives

$$T_H \le Q \le \min\left\{\frac{T_H}{1-r^H},\ T_H + \frac{r}{\underline{\eta}}k_{H-1}\right\}. \qquad (38)$$

These inequalities follow directly by summing each geometric component and bounding its omitted tail. They are not optimal use of the entire moment sequence.

Under ideal loop-open impulse measurements with known noise variance, a three-way decision rule evaluated across 21,600 noisy measurement datasets (108 configurations) achieved decision coverage of 82.69% and 66.20% in lower- and higher-noise conditions at 256 taps without false confident calls. However, those measurements were *loop-open impulse responses with known noise standard error*; they were not ordinary price records. Deconvolving noisy prices does not preserve that clean noise model. Moreover, a falsely specified learning rate floor can label an unstable model stable. These limitations demonstrate why passive impulse measurement is insufficient for market-wide stability certification.

## 3.4 Full-horizon identification audit and fixed-order limits

We next benchmark the passive diagnostic against full-horizon moment-based estimators that receive identical information. This audit compares the diagnostic with non-parametric alternatives and documents where identification breaks down under high noise. A difficult identification problem does not establish superiority of an arbitrary ad hoc estimator.

### 3.4.1 A conventional full-information comparison

All methods received the same noisy normalized kernel vector $\widehat{g}_j$, known common standard error $v$, and a valid $\underline{\eta}$, where $g_j = k_j/(1-a)$. Neither generating weights, rates, component count nor class label was supplied. A nonnegative least-squares mixture fit used all observed taps. A second comparator bounded $G$ over positive mixtures consistent with simultaneous confidence intervals, using continuous rate bins to protect off-grid modes. For a feature $f$ and mass $w_l$ in bin $B_l$, its outer inequalities were

$$\sum_l \min_{\eta\in B_l} f(\eta) w_l \le U_f, \qquad \sum_l \max_{\eta\in B_l} f(\eta) w_l \ge L_f.$$

This belongs to the established set-membership and positive-measure optimization tradition [27, 28]. Stable-model fitting alone would not be an acceptable comparator for detecting instability; stability was not imposed in these fits.

The bounds resolved 578 and 1,128 additional vectors respectively, but did not dominate on every realization. Some earlier decisions became model conflicts, and four high-noise decisions were wrong. All four occurred outside the simultaneous confidence event. Across the 82,364 rows on that event, the reported interval contained the true gain and made no wrong confident call. The full audit comprised 86,400 rows over four horizons. Its strongest interpretation is a measurement requirement and benchmark, not an improved original summary diagnostic.

### 3.4.2 A fixed-order lower bound in the earlier observation law

For two common-rate models with rates $(1/2, \eta)$ and normalized weights $(0.7, 0.2)$ and $(0.7, 0.4)$, the gains are 0.9 and 1.1. With $\Delta = 0.2$, their kernel means differ by $\Delta\eta(1-\eta)^j$. Under *independent kernel noise* of variance $v^2$, their squared Mahalanobis distance is

$$d^2_{H,\text{kernel}} = \frac{\Delta^2\eta\{1-(1-\eta)^{2H}\}}{v^2(2-\eta)}. \qquad (39)$$

The equal-prior optimal forced error is $\Phi(-d/2)$, obtained by projecting onto the standardized mean difference. If

Table 2: **Lineage of Theorems M1–M5.** Classical foundations and the specific increment. Proofs are in section A.6.

| Result | Classical foundation | Increment in this paper |
|---|---|---|
| **M1** | Kalman (1960), Gilbert (1963), Ho & Kalman (1966): minimal realization of rational transfer functions. | Exact equivalence classes of heterogeneous-memory economies: merged rate/weight pairs are identified, individual cohorts are not; $n_{\min} = 1 + \lvert \operatorname{supp} \mathcal{M} \rvert$. |
| **M2** | Le Cam (1973); Cramér–Rao; resolution limits for exponential/sinusoid mixtures (e.g. Smith 2005). | Probe-budget lower bounds in terms of depth $\lambda$, amplitude $A$ and memory horizon; a $\delta^{-4}$ sample-size lower bound for detecting cohort heterogeneity. |
| **M3** | Chow (1970), Herbei & Wegkamp (2006): classification with a reject option; two-point testing bounds. | Any error-controlled stability classifier must abstain on an explicit zone around $S = 1$; with finite windows, abstention is unavoidable at any budget. |
| **M4** | Hausdorff (1921), Markov–Lukács, Krein & Nudelman (1977); cone-restricted testing (Perlman 1969). | Corrected characterization of the positive-memory class and a size-valid test, applied to real order-flow kernels in section 7.4. |
| **M5** | Kreiss (1962), Spijker (1991), Trefethen & Embree (2005). | Exact worst-case solvency radius for a linearized dealer exposure; Kreiss constant bounds it up to $en$; spectral radius cannot. |

Table 3: Retained readable witness. Four initial price coefficients match exactly.

| Quantity | Stable member | Unstable member |
|---|---|---|
| $a$ | 2/5 | 2/5 |
| $Q$ | 12/65 | 66/65 |
| $S = a + Q$ | 0.584615 | 1.415385 |
| $\rho(J)$ | 0.757909 | 1.032345 |
| $h_0, h_1$ | 1, 0.4 | 1, 0.4 |
| $h_2, h_3$ | 0.252308, 0.184 | 0.252308, 0.184 |
| $h_4$ | 0.138428 | 0.159198 |

a rule may abstain, its balanced decision coverage $c$ and unconditional wrong-label probability $e$ obey

$$c \le \min\{1,\, 2e + 2\Phi(d/2) - 1\}. \tag{40}$$

For disjoint stable and unstable decision regions, subtract twice the error from coverage and bound the difference of probabilities by total variation. The Gaussian total variation is $2\Phi(d/2) - 1$. This is a direct specialization of classical two-point testing, not a new general lower-bound method; identification sample-complexity precedents must also be acknowledged [30].

At $\eta = 10^{-5}$, $H = 256$, $v = 0.02/\sqrt{128}$, the oracle error is 49.6394%. An unconditional error target of 5% permits at most 10.7212% coverage on that pair. Because this particular kernel experiment adds equally noisy late taps, its information can saturate. **These numbers and the saturation conclusion must not be transferred unchanged to a different price-measurement protocol.** The price experiment below has a different observation law and a separately calculated lower bound.

# 4 Observable Identification under Public Probes: Contract $I_0$

## 4.1 An observable intervention protocol without hidden-state access

The new research step replaces assumed kernel observations with recorded probe fills and quoted prices. It does not identify the full latent Jacobian or the fraction of trades performed by AI. It estimates an input–output response and asks what that response supports about $S$ under the declared positive-memory class. This change is compatible with established data informativity: a property can sometimes be determined without uniquely recovering every parameter [29].

### 4.1.1 Information available to the estimator

In each of $R$ independent episodes, assign one signed pulse $u_r \in \{-\delta, +\delta\}$, balanced across signs and randomized independently of the initial latent states and subsequent disturbances. Record the *filled quantity*, not merely the

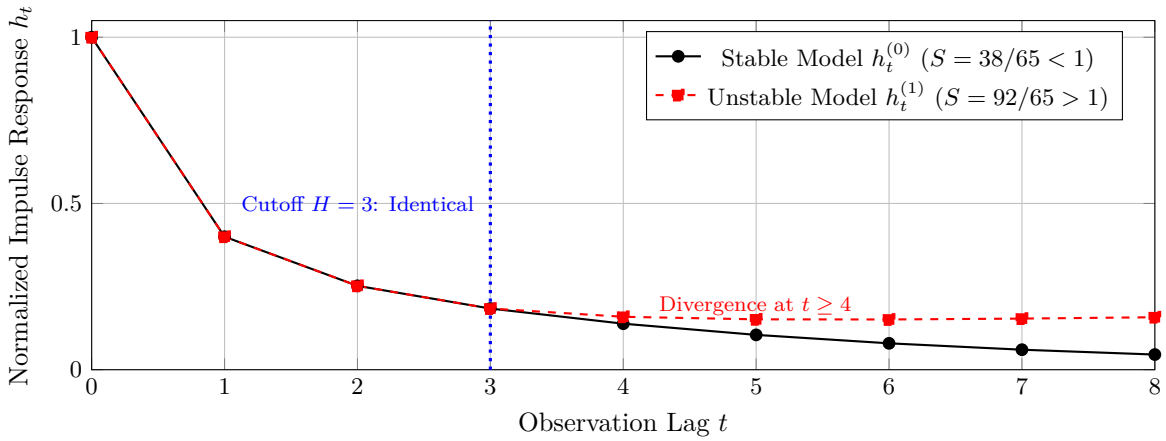



Figure 3: Exact finite-window non-identification witness (Theorem 3.3 and Table 3). Both stable ($S = 38/65$) and unstable ($S = 92/65$) economies produce mathematically identical public price responses through lag $t = 3$, proving that no universal finite-horizon response formula can classify stability.

Table 4: Retained full-horizon audit at 256 taps. Each noise condition contains 10,800 vectors. These are the earlier kernel measurements, not the new price experiment.

| | Lower noise | | Higher noise | |
|---|---|---|---|---|
| Method | Decisive (%) | Wrong | Decisive (%) | Wrong |
| Original summary bound | 82.69 | 0 | 66.20 | 0 |
| Full-horizon positive point fit | 100.00 | 24 | 100.00 | 306 |
| Full-moment positive bounds | 88.04 | 0 | 76.65 | 4 |

intended order, and the subsequent quote changes $Y_{r,j} = P_{r,j+1} - P_{r,0}$, $j = 0, \ldots, M-1$. Within a linear time-invariant episode,

$$Y_{r,j} = m_j + u_r \beta_j + \varepsilon_{r,j}, \qquad \beta_j = \kappa h_j. \tag{41}$$

The arbitrary common intercept path $m_j$ accommodates the mean no-probe trajectory. The estimator receives only $(u_r, Y_{r,\cdot})$ and a valid externally declared $\underline{\eta}$. It receives neither private states, private aggregate pressures, exact rates, weights, $a$, $\kappa$, nor the disturbance variance. Initial adaptive states need not be zero or observable: in the experiment they are independent Gaussian draws, hidden from the estimator.

For the exact confidence result, episodes have independent identically distributed Gaussian residual vectors $\varepsilon_r \sim N(0, \Sigma)$, independent of assignment, where $\Sigma$ is unknown and may have arbitrary temporal covariance. This is a finite-window condition; it does not require stationary prices or closed-loop stability. Gaussian initial states, additive flow disturbances and quote errors in the fixed linear simulator imply it. It is not guaranteed by a successful regression on a real market.

#### 4.1.2 Recovering the response and its uncertainty

Let $v_r = u_r - \bar{u}$ and $D = \sum_r v_r^2 > 0$. Ordinary regression with an intercept gives

$$\widehat{\beta}_j = D^{-1} \sum_r v_r Y_{r,j}, \qquad \widehat{m}_j = \bar{Y}_j - \bar{u}\widehat{\beta}_j, \tag{42}$$

$$\widehat{\Sigma} = \frac{1}{R-2} \sum_r (Y_r - \widehat{m} - u_r \widehat{\beta})(Y_r - \widehat{m} - u_r \widehat{\beta})^\top, \tag{43}$$

$$\widehat{\mathrm{Cov}}(\widehat{\beta}) = \widehat{\Sigma}/D. \tag{44}$$

The covariance is estimated from public price residuals; no oracle covariance is inserted. For balanced pulses, $D = R\delta^2$. Repeating episodes and increasing pulse size change precision, but larger pulses can invalidate linearity and create additional economic exposure.

**Theorem 4.1** (Finite-family response uncertainty)**.** *Under* (41) *and the stated independent Gaussian episode contract, require $R > 2$ and $D > 0$ (an even $R \geq 4$ for the implemented balanced design). Then let*

$$z = t_{R-2}^{-1}\Big(1 - \frac{\alpha}{2M}\Big), \qquad L_j = \widehat{\beta}_j - z\sqrt{\widehat{\Sigma}_{jj}/D}, \quad U_j = \widehat{\beta}_j + z\sqrt{\widehat{\Sigma}_{jj}/D}$$

*Then $\mathbb{P}(\beta_j \in [L_j, U_j] \text{ for all } j < M \mid u) \geq 1 - \alpha$. No temporal independence between price observations or between estimated response coefficients is required.*

*Proof.* Conditional on the nonconstant design, each response coordinate is a Gaussian regression with two fitted coefficients. Its standardized slope error has a Student-$t$ distribution with $R-2$ degrees of freedom; the residual variance is independent of that coordinate's slope estimate. Each two-sided interval fails with probability $\alpha/M$. A union bound over the $M$ coordinates yields the result even though these statistics are dependent. In a noiseless coordinate the interval degenerates to its exact coefficient. This is an application of ordinary Gaussian regression inference, not a claim to a new distribution-free method. Exact finite-sample methods under other assumptions already exist, including sign-perturbed sums [34]. □

The event covers all predeclared horizons that use these same coefficients. It is not anytime-valid for an unlimited search over new windows, budgets or operating points. A 95% guarantee applies separately to a declared measurement panel, not simultaneously to the entire collection of thousands of simulation panels. It is also an unconditional wrong-label bound, not a 5% conditional error guarantee among the few cases that receive labels.

#### 4.1.3 The four-metric evaluation standard and conditional error rates

In partial-identification settings, an essential econometric distinction exists between unconditional error and conditional decision error. Across an evaluation suite of $N$ panels, let $N_{\text{dec}}$ be the number of panels where the estimator issues a definitive classification (stable or unstable), and let $N_{\text{err}}$ be the number of incorrect classifications. We track four distinct metrics:

1. **Decisiveness:** $\text{Dec} = N_{\text{dec}}/N$;
2. **Coverage:** $\text{Cov} = 1 - N_{\text{param_uncovered}}/N$;
3. **Unconditional Error Rate:** $\text{Err}_{\text{uncond}} = N_{\text{err}}/N$;
4. **Conditional Error Rate:** $\text{Err}_{\text{cond}} = N_{\text{err}}/N_{\text{dec}}$ (defined for $N_{\text{dec}} > 0$).

Near the boundary $S = 1$, an estimator that is forced to decide can report a moderate unconditional error while being wrong on a large share of the calls it actually makes. The output-error row of table 6 is an example: it issues labels on 438 of 600 panels and is wrong on 204, so

$$\text{Err}_{\text{uncond}} = \frac{204}{600} = 34.00\%, \qquad \text{Err}_{\text{cond}} = \frac{204}{438} = 46.58\%. \tag{45}$$

The set-membership intervals instead abstain when their confidence set spans the boundary $S = 1$ (Dec = 0%, $\text{Err}_{\text{uncond}} = 0\%$), avoiding confident false calls on near-boundary parameter profiles.

#### 4.1.4 Exact deconvolution is possible; independent noise is not preserved

With exact response coefficients and $\beta_0 > 0$, the unknown impact and direct persistence are recoverable:

$$\kappa = \beta_0, \quad h_j = \beta_j/\beta_0, \quad a = h_1, \quad k_n = h_{n+2} - a h_{n+1} - \sum_{j=0}^{n-1} k_j h_{n-j}. \tag{46}$$

Thus finite loop-open responses can be obtained without observing the adaptive states. The exact claim is algebraic; noisy point estimates require uncertainty propagation. For example,

$$k_0 = \frac{\beta_2}{\beta_0} - \left(\frac{\beta_1}{\beta_0}\right)^2,$$

whose gradient with respect to $(\beta_0, \beta_1, \beta_2)$ is

$$\left(-\frac{\beta_2}{\beta_0^2} + \frac{2\beta_1^2}{\beta_0^3}, -\frac{2\beta_1}{\beta_0^2}, \frac{1}{\beta_0}\right).$$

Even a first-order variance approximation uses the *full* response covariance. Recursive products, division by a weak initial response and dependence across times can amplify uncertainty. Reusing the previous iid-kernel Gaussian endpoints on these estimates would be unjustified. The new pipeline therefore avoids requiring a precise point estimate of every $k_n$.

### 4.2 Positive recurrence bounds directly from uncertain prices

This section constructs an outer feasible set directly from the response intervals, following conventional set-membership logic. The relevant methodological increment in this artifact is the explicit link from randomized price measurements to that set, including unknown persistence and impact. Neither broad novelty nor optimality over all input–output estimators is asserted.

#### 4.2.1 A response-only convex outer relaxation

The response satisfies

$$\beta_{t+1} = a\beta_t + \int_{\underline{\eta}}^{1} \left[\sum_{j=0}^{t-1} \eta(1-\eta)^j \beta_{t-1-j}\right] d\mu(\eta), \quad \mu = \sum_i q_i \delta_{\eta_i}, \tag{47}$$

with an empty sum at $t = 0$. This equation is homogeneous in the price-impact scale; there is no need to divide a noisy record by an assumed known $\kappa$.

Intersect response lower bounds with zero, since the theoretical impulse response is nonnegative. An upper bound below zero is a model conflict. If $L_0 \le 0$, the present implementation remains inconclusive rather than normalizing by a poorly supported impact. Partition

$[\underline{\eta}, 1]$ into bins $B_l$ and let $w_l = \mu(B_l) \geq 0$. For $\psi_j(\eta) = \eta(1-\eta)^j$ define exact endpoint/peak bounds

$$\psi_{jl}^- = \min_{\eta \in B_l} \psi_j(\eta), \qquad \psi_{jl}^+ = \max_{\eta \in B_l} \psi_j(\eta).$$

The minimum is at an endpoint; the maximum is at the projection of $1/(j+1)$ onto the bin. Set

$$C_{tl}^- = \sum_{j=0}^{t-1} \psi_{jl}^- L_{t-1-j}, \qquad C_{tl}^+ = \sum_{j=0}^{t-1} \psi_{jl}^+ U_{t-1-j}. \quad (48)$$

Every compatible true model then satisfies the *linear* inequalities

$$aL_t + \sum_l C_{tl}^- w_l \leq U_{t+1}, \quad (49)$$

$$aU_t + \sum_l C_{tl}^+ w_l \geq L_{t+1}, \qquad 0 \leq a \leq 1, \quad w_l \geq 0. \quad (50)$$

Minimize and maximize $a + \sum_l w_l$ over this set to obtain $[S_-, S_+]$. Different terms may attain their bin extrema at different rates and different response endpoints; the relaxation is conservative, not an exact description of all realizations. Refining bins does not by itself remove the looseness of rectangular response uncertainty.

**Theorem 4.2** (Observable-data stability interval)**.** *Under the model, a valid rate floor, and the response event in Theorem 4.1, the true $S$ belongs to $[S_-, S_+]$ in exact arithmetic at every declared horizon. Declaring stable if $S_+ < 1$, unstable if $S_- > 1$, and otherwise inconclusive therefore has probability at most $\alpha$ of any wrong confident label within the declared family.*

*Proof.* On the response event, all true $\beta_t$ lie in the clipped nonnegative intervals. Positivity of $a$, the kernel and bin masses makes each lower convolution no greater than its true value and each upper convolution no smaller. Thus the true $(a, w)$ satisfies (50). Its objective is exactly $a + \mu([\underline{\eta}, 1]) = S$, so the LP extremal values enclose it. Proposition 3.2 gives the label implication; the probability statement follows from the simultaneous event. Unknown hidden states affect the observational residual distribution, not the feasibility proof, provided they satisfy the independent episode contract. □

A useful numerical anchor follows from $\beta_2 \geq \underline{\eta}\,\beta_0 Q$:

$$S \leq 1 + \frac{U_2}{\underline{\eta} L_0}, \qquad L_0 > 0. \quad (51)$$

The implementation includes this redundant true-model bound, outward feature padding, dual residual repair, primal/dual checks and a $10^{-7}$ classification margin. While the exact-arithmetic LP bounds are fully verified, the floating-point conic implementation details remain an open numerical challenge rather than a core theoretical contribution. Linear-programming bounds are guarded numerically, not formally verified with interval arithmetic. Solver failure and infeasibility are not converted into a favorable label.

## 4.3 Observable-data experiments

The following analyses evaluate the observable-data experimental protocol. The inputs begin with the post-probe price, $x_1$, and end at $x_{H+1}$. Failed fits and invalid endpoints count in the attempted-panel denominator rather than being treated as covered intervals. This ensures a strictly conservative evaluation of the interval coverage rates.

The corrected grid contains 600 independent panels within 120 fixed cells. The production design crosses $S \in \{.70, .82, 1, 1.03, 1.18\}$, slow rates $.002, .02, .10$, episode budgets $R = 128, 1024$, probe sizes $1, 4$, and horizons $H = 24, 64$, with five panels per cell. The aggregate rates are descriptive averages over this fixed, heterogeneous design, not universal performance guarantees or evidence that every cell is adequately powered.

In the grid, 472/600 panels have a cut-limit stop at one or both conic endpoints. A finite-cut outer candidate is not a converged second-order-cone optimum.

Output-error inference is unavailable on 31/600 panels. The conic routine records optimizer status, cut counts, residuals and both objective corrections. Full outward rounding of all assembled constraints is not established, so `certified=False` is returned even when a numerical candidate separates $S$ from one. The exact-arithmetic outer-set theorem is not a proof that floating-point endpoints have been certified.

**Output-error benchmark and interval interpretation.** The single-rate model fits $\theta = (a, \eta, Q, \kappa)$ by

$$\min_\theta (\hat{\beta} - \beta(\theta))^\top W (\hat{\beta} - \beta(\theta)) + \lambda(a^2 + Q^2), \qquad \lambda = 0. \quad (52)$$

The constraints are $0 \leq a \leq .995$, $\underline{\eta} \leq \eta \leq 1$, $0 \leq Q \leq 4$, with $\kappa$ initialized at $\max(.001, \hat{\beta}_0)$ and bounded between .1 and 5 times that initialization. Thus impact is estimated, not supplied by the simulator oracle. The covariance inverse uses the disclosed trace-scaled ridge $10^{-6}\,\mathrm{tr}(\widehat{\Sigma})/(H+1)$. With response Jacobian $J$, the reported covariance is the local Gauss–Newton propagation

$$BJ^\top W \widehat{\Sigma} W J B, \qquad B = (J^\top W J)^{-1}.$$

This is not a general misspecification-robust variance formula for arbitrary nonlinear models. Profile intervals use a local objective-increase threshold and numerical nuisance minimization. The primary interval is the profile interval when available; sandwich intervals are recorded separately. Neither is a global profile-set certificate or a finite-sample guarantee. Failed optimization, rank deficiency or an unavailable profile is retained as such, without inventing a standard error or counting a fallback as valid inference.

**Known versus estimated covariance in the budget sweep.** These experiments draw Gaussian response

Table 5: Methodological specifications and empirical linkages.

| Method & Contract & Empirical Specification | | |
|---|---|---|
| Recurrence LP | Theorem 4.2; Gaussian episode regression | `grid/results.json` |
| Conic relaxation | Known covariance: chi-square; estimated covariance: F. Numerical certification withheld. | `grid/`, `sweep_known/`, `sweep_estimated/` |
| Output-error | Single-rate local nonlinear fit; profile and sandwich intervals separately recorded | `grid/`, `diagnostic/` |
| Point/deconvolution | Constrained point heuristic or interval propagation | `grid/results.json` |
| Response-region pivot | Gaussian errors and independent Wishart covariance where estimated | `calibration/results.json` |

Paths are relative to `statistical_repair_20260926/verified/`. Source hashes, full per-panel observations, failure statuses and the repair protocol accompany these outputs. The separate audited-disclosure results in Section 5.3 are not a matched estimator comparison on this grid.

Table 6: Comparator grid: containment and issued labels. All attempted panels are retained.

| Method | Covered/$N$ | Issued/$N$ | Wrong | Wrong/$N$ (%) | Wrong/issued (%) | Invalid |
|---|---|---|---|---|---|---|
| Always abstain | 600/600 | 0/600 | 0 | 0.00 | – | 0 |
| Recurrence LP | 597/600 | 0/600 | 0 | 0.00 | – | 3 |
| Conic (numerical) | 600/600 | 0/600 | 0 | 0.00 | – | 0 |
| Output-error | 193/600 | 438/600 | 204 | 34.00 | 46.58 | 31 |
| Interval deconvolution | 597/600 | 0/600 | 0 | 0.00 | – | 3 |
| Constrained point fit | 228/600 | 483/600 | 165 | 27.50 | 34.16 | 0 |

Always abstain returns $[0, \infty)$ under the nonnegative model. Conic issued labels here are numerical candidates only: every certified label is withheld. Invalid or missing intervals are not covered. A zero wrong-label count with no issued labels is abstention, not successful discrimination. The point-fit interval has no nominal confidence guarantee.

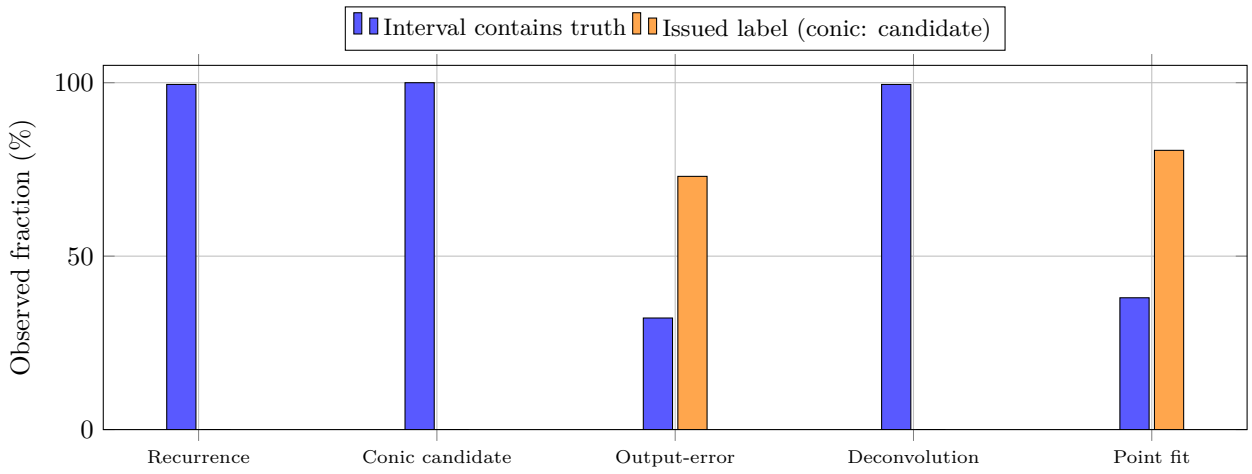



Figure 4: Empirical containment and labeling on the corrected grid. These are descriptive panel frequencies, not confidence bounds on performance. Conic certification remains withheld; a candidate label is not a certified decision. Audited-disclosure experiments use a different information contract and are therefore not placed on this plot as a matched comparison.

Table 7: Interval availability and finite-width distributions on the comparator grid.

| Method | Finite | Unbounded | Invalid | Finite mean | Finite $p_{90}$ |
|---|---|---|---|---|---|
| Always abstain | 0 | 600 | 0 | – | – |
| Recurrence LP | 596 | 1 | 3 | 51.190 | 105.066 |
| Conic (numerical) | 472 | 128 | 0 | 125.639 | 245.685 |
| Output-error | 569 | 0 | 31 | 0.332 | 0.774 |
| Interval deconvolution | 596 | 1 | 3 | 127.918 | 294.159 |
| Constrained point fit | 600 | 0 | 0 | 0.137 | 0.182 |

Finite-width summaries condition on the finite count in that row. They are not unconditional mean widths when intervals are unbounded or invalid.

Table 8: Response-region calibration, separate from structural interval optimization.

| Covariance | Pivot | Covered/draws | Coverage (%) | Exact 99% interval (%) |
|---|---|---|---|---|
| Known | $\chi^2$ | 18975/20000 | 94.875 | [94.460, 95.268] |
| Estimated | $F$ | 19000/20000 | 95.000 | [94.590, 95.389] |

Dimension 25, episode-budget parameter 128; nominal coverage 95%. Estimated mode draws an independent Wishart covariance. The fixed acceptance tolerance is one percentage point around nominal. Passing this check does not validate outward numerical constraint assembly.

coefficients directly, not noisy episode histories. Their covariance is either known exactly or estimated by an independent Wishart draw. They are useful checks of covariance calibration and numerical contraction, not evidence about fills, latency or exchange data. The two modes share coefficient-noise draws; only their covariance treatment differs. Each production cell has 20 draws and horizon $H = 24$.

Panels with a cut-limit stop at one or both endpoints are known: 300/300; estimated: 300/300. These are finite-cut outer approximations; the width curves do not demonstrate convergence to the exact conic optimum.

**Matched and misspecified output-error fits.** C1/C3 are single deterministic cases, not repeated independent observations. C2/C4 use independent Gaussian coefficient noise with known covariance, at the same fixed horizon and noise scale. C1/C2 have one true rate; C3/C4 have two rates but are fit by the same single-rate estimator. This design isolates one form of structural mismatch; it does not establish an asymptotic bias theorem or identify every reason for grid undercoverage. In particular, a profile-interval result must not be relabeled as sandwich-interval coverage.

**What the corrected studies establish.** The response-region probability calculation is a statement under its explicit Gaussian sampling law. The comparator grid evaluates implemented methods, including their numerical failures and modeling restrictions. Neither experiment proves estimator superiority across information contracts, a universal impossibility theorem, or real-market safety. The primary price-only experimental evaluation below serves as a separate benchmark.

**Primary price-only experimental evaluation.** To evaluate observable price-response identification in full market simulation, we analyze the primary 1,920-panel evaluation (generated under the double-entry market environment detailed in Section 6).

#### 4.3.1 Response recovery and confidence calibration

The simultaneous response intervals covered all 66 true coefficients in 1,895 of 1,920 panels, or 98.70%, versus a nominal lower coverage of 95%. Estimated neighboring response errors were strongly correlated: the mean absolute adjacent correlation was 0.8497. This supports the decision not to reuse an iid-kernel error model. These are measurements from the declared synthetic experiment, not estimates of correlations in a real exchange.

On all 5,685 horizon rows whose panel satisfied the response event, the output-only recurrence interval contained the true $S$. It made no wrong confident call anywhere in the 5,760 primary rows, including the 75 rows from the 25 panels outside that event. The observed absence of errors is not proof of a zero-error procedure.

The strongest primary design therefore resolves 16.67%, leaving 83.33% inconclusive. All 80 decisions in that design occur at the fastest allowed slow rate, $\eta_s = 0.10$, and the large-margin gains $G = 0.70$ and 1.30, with 40 correct decisions each. Neither near-boundary gain nor either slower class is resolved by that design. Across all three nested horizons, the recurrence method makes 256 decisions (176 stable and 80 unstable) and leaves 5,504 rows inconclusive. Nested horizon decisions are not independent replications.

The naive interval-deconvolution comparator remains inconclusive throughout the primary panel. This is evi-

Table 9: Conic numerical endpoints at the first and last budget in each sweep.

| Covariance | $S$ | First width | Last width | Last covered/$N$ | Last candidates/$N$ | Wrong |
|---|---|---|---|---|---|---|
| known | 0.60 | 7.306 | 2.343 | 20/20 | 0/20 | 0 |
| known | 0.82 | 11.203 | 4.856 | 20/20 | 0/20 | 0 |
| known | 1.25 | 17.151 | 11.198 | 20/20 | 20/20 | 0 |
| estimated | 0.60 | 7.701 | 2.344 | 20/20 | 0/20 | 0 |
| estimated | 0.82 | 12.316 | 4.857 | 20/20 | 0/20 | 0 |
| estimated | 1.25 | 18.186 | 11.199 | 20/20 | 20/20 | 0 |

Widths are unconditional means only when every interval is finite. Production budgets run from 128 to 32,768. All certified labels are withheld.

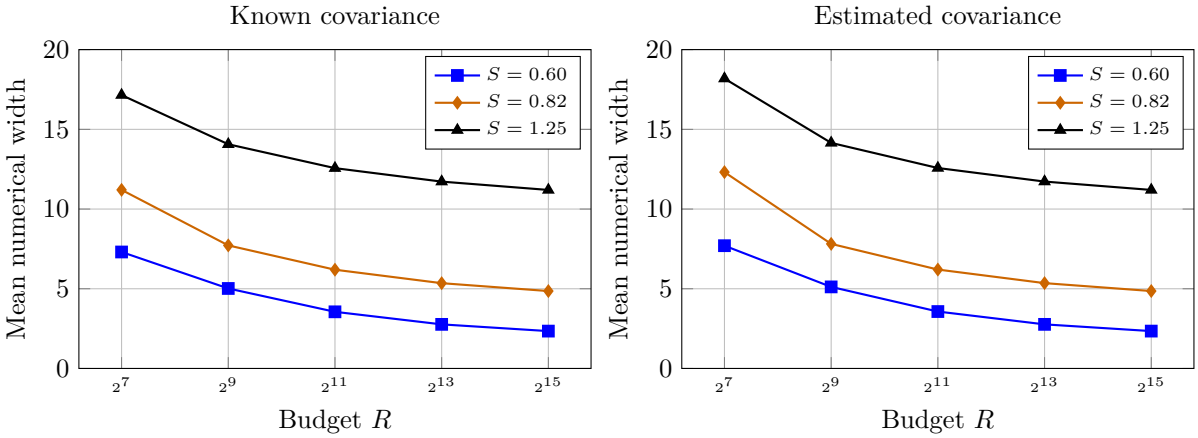



Figure 5: Descriptive numerical interval widths under the two covariance contracts, on common vertical scales. Each point averages a fixed cell, not a confidence interval. Missing means indicate an invalid or unbounded interval and are not imputed. Contraction is not certification or a universal asymptotic claim.

Table 10: Separate profile and sandwich diagnostics; true $S = .82$.

| Case | Attempts | Mean $\hat{S}$ | Bias | Profile covered/valid | Sandwich covered/valid | Unavailable |
|---|---|---|---|---|---|---|
| C1 | 1 | 0.8200 | 0.000000 | 1/1 | 1/1 | 0 |
| C2 | 100 | 0.8200 | -0.000037 | 91/100 | 93/100 | 0 |
| C3 | 1 | 0.7128 | -0.107218 | 0/1 | 0/1 | 0 |
| C4 | 100 | 0.7128 | -0.107205 | 0/100 | 0/100 | 0 |

Coverage denominators condition on an available interval of the named type; unavailable reports are counted separately. C1/C3 entries are deterministic containment only. The raw records also report optimizer failures, bound hits and interval fallbacks.

Table 11: New price-only experiment at largest fitted lag 64. Each row contains 480 panels. Parentheses show wrong labels; point fitting always gives a label but has no confidence guarantee.

| Episodes | Pulse (shares) | Recurrence bound | Interval deconvolution | Point fit |
|---|---|---|---|---|
| 128 | 1 | 0/480 (0) | 0/480 (0) | 480/480 (101) |
| 128 | 4 | 28/480 (0) | 0/480 (0) | 480/480 (56) |
| 1,024 | 1 | 0/480 (0) | 0/480 (0) | 480/480 (49) |
| 1,024 | 4 | 80/480 (0) | 0/480 (0) | 480/480 (37) |

dence of looseness in that particular interval propagation, not a theorem that all deconvolution methods fail. The point estimator makes 1,191 wrong labels across the 5,760 rows. Its apparent 100% decisiveness is therefore not a substitute for a supported stability statement. Nor is this weak point comparator a basis for claiming superiority over the strongest output-error or covariance-aware conventional methods.

Importantly, statistical reporting must distinguish unconditional error (wrong labels divided by total evaluated panels) from conditional error (wrong labels divided by issued decisions, $N_{\text{wrong}}/N_{\text{decisive}}$). For an estimator that frequently abstains, a seemingly modest unconditional error rate can conceal high conditional error among the few decisions it actually commits to. For instance, the output-error comparator in table 6 has 34.00% unconditional but 46.58% conditional error. By contrast, on the primary panel in Table 11 ($N = 1,024$, pulse 4), the recurrence interval makes 0 errors among its 80 issued decisions (0% conditional error, 16.67% decisiveness, 100% coverage), whereas the point fit issues 480 decisions with 37 wrong labels (7.71% conditional error). All four metrics—coverage, decisiveness, unconditional error, and conditional error among issued labels—must be tracked together to prevent misleading claims of reliability.

#### 4.3.2 Replication budget and the cost of a useful answer

After inspecting primary results, a separately labeled exploratory extension uses 8,192 and 32,768 episodes, four-share pulses and largest lag 64. Five panels per configuration/budget give 60 panels at each budget. There are 15/60 (25.00%) and 25/60 (41.67%) definite answers, with no wrong labels. At 32,768 episodes all 20 fastest-class panels are resolved, five moderate-rate stable large-margin panels are resolved, and all 20 slowest-class panels remain inconclusive. The extension contributes 2,457,600 simulated episodes but is not pooled with the frozen primary analysis.

These results expose an observation-budget problem. In a live market, 1,024 four-share probes would also incur slippage and fees. Natural randomizations such as tick-size pilots (section 8.4) are a possible substitute, but none is analyzed here. Low decisiveness here also reflects conservative bin/rectangle relaxation and discarded covariance geometry. They do not prove that the remaining cases are information-theoretically impossible at these budgets. Any methodological comparison must retain model mismatch, unavailable intervals and the distinction between numerical candidates and verified certificates.

#### 4.3.3 Assumption failures are not repaired by good price confidence

In the false-floor control, actual rates are $(0.5, 10^{-5})$ but inference is incorrectly told $\underline{\eta} = 0.10$. Weights $(0.42, 0.24)$ give $S = 1.06$. All price-response confidence events hold, yet the true measure is excluded from the allowed support and all 20 outputs are falsely stable. No output-only claim here certifies the rate floor itself.

The confounded control assigns signs using future background flow, deliberately breaking independence. The nonlinear control replaces $\kappa V$ by $\kappa c_0 \tanh(V/c_0)$ with $c_0 = 1$ share. Its Jacobian at the origin retains the linear reference, but finite pulses do not. In an exact noise-free scaling check, initial price response per share is 0.0195935, 0.0152319 and 0.00499665 for pulses of 0.25, 1 and 4 shares. A larger pulse improves linear-model signal-to-noise only while linearity remains valid. Passing feasibility is not a model-validation test; failing amplitude invariance can flag a problem but cannot certify the remaining assumptions.

#### 4.3.4 A new lower bound for the actual price-observation family

To separate observation limits from solver conservatism, consider a restricted subexperiment with zero initial states, an exactly observed initial price reference, no flow noise and independent subsequent quote noise $N(0, \sigma^2)$. This subexperiment is not the full noisy-dealer panel. For the two models with $a = 0.4$, rates $(0.5, \eta)$ and $q^{(0)} = 0.6(0.7, 0.2)$, $q^{(1)} = 0.6(0.7, 0.4)$, define their observable price responses $\beta^{(0)}, \beta^{(1)}$ as in (41).

**Theorem 4.3** (Fixed-budget price-response ambiguity)**.** *With $R$ balanced pulses of magnitude $\delta$, observed through lag $H$, the exact Gaussian distance is*

$$d^2_{H,\text{price}} = \frac{R\delta^2}{\sigma^2} \sum_{j=0}^{H} (\beta_j^{(1)} - \beta_j^{(0)})^2. \tag{53}$$

*The optimal balanced forced-classification error is $\Phi(-d_{H,\text{price}}/2)$, and the abstention bound* (40) *holds using this new distance. For fixed $H, R, \delta, \sigma > 0$, the distance tends to zero as $\eta \downarrow 0$, although the two gains remain $G_0 = 0.9$ and $G_1 = 1.1$ and both models have two memory states.*

*Proof.* Conditional on the known assignments, if the differenced observations $Y_{r,j}$ experience independent measurement innovations, the stacked observation covariance is $\sigma^2 I$ and each episode's mean difference is $u_r(\beta^{(1)} - \beta^{(0)})$. Summing squared standardized differences gives (53). Projection onto that direction gives the same scalar Gaussian likelihood-ratio calculation as before.

If, instead, underlying quote levels are observed with independent errors $P_{r,t} = p_{r,t} + \varepsilon_{r,t}$ with $\varepsilon_{r,t} \sim N(0, \sigma^2)$, differencing against the noisy baseline $P_{r,0}$ induces the equicorrelated covariance $\Sigma_Y = \sigma^2(I + \mathbf{1}\mathbf{1}^\top)$. By the Sherman–Morrison formula, $(I + \mathbf{1}\mathbf{1}^\top)^{-1} = I - \frac{1}{H+2}\mathbf{1}\mathbf{1}^\top$, and the Mahalanobis distance becomes

$$d^2_{H,\text{level}} = \frac{R\delta^2}{\sigma^2} (\Delta\beta)^\top \left( I - \frac{1}{H+2} \mathbf{1}\mathbf{1}^\top \right) (\Delta\beta). \tag{54}$$

Table 12: Twenty panels per control, each with 4,096 episodes and four-share pulses. Controls are not part of the primary calibration claim.

| Condition | Observed result | Interpretation |
|---|---|---|
| Valid reference | 20 stable labels; all response bands cover | Consistent with the declared class |
| False rate floor | 20 stable labels for an unstable linear system; all response bands cover | Accurate measured responses do not validate hidden-tail assumptions |
| Confounded assignment | 20 model conflicts; zero response bands cover the causal response | Randomization is part of identification, not decoration |
| Saturating price impact | 20 stable labels; zero bands cover the origin's linear response | A plausible label does not establish linearity or global safety |

For the limit, let $J_0, J_1$ be the pressure-coordinate matrices in Proposition 3.2. Their difference has a single entry $\eta\Delta q$, where $\Delta q = 0.12$. A common induced norm bound $\|J_i\| \le M$ can be chosen independently of $\eta \in (0, 1]$. The telescoping identity for powers gives

$$\|J_1^j - J_0^j\| \le jM^{j-1}\eta\Delta q.$$

Since $\beta_j^{(i)} = \kappa e_1^\top J_i^j e_1$, every finite coefficient difference tends to zero and so does the finite sum in (53) and (54). The gains do not depend on $\eta$. This proves the restricted finite-budget impossibility under both noise models, not a claim about all possible excitation policies. □

For $R = 1,024$, $\delta = 4$, $\sigma = 0.002$, $H = 64$, and $\kappa = 0.02$, under independent difference errors the exact oracle errors are 49.94%, 49.42%, 44.27%, and 7.45% at rates $10^{-7}$, $10^{-6}$, $10^{-5}$, and $10^{-4}$. Under the alternative noisy-baseline level-error model (54), the corresponding oracle errors are 49.98%, 49.76%, 47.56%, and 27.02%. At $\eta = 10^{-5}$ an unconditional 5% wrong-label target permits at most 21.46% balanced decision coverage under difference errors (and at most 14.88% under level errors, following $c \le 2e + 1 - 2p^* = 0.10 + 1 - 2(0.4756)$) on this pair. These are newly computed *price-law* examples, not the 49.64% kernel-law number in the previous audit. The limit lets the allowable slowest rate approach zero; it does not hold that rate above one fixed positive floor while simultaneously sending it to zero. Each numerical pair must be assigned a class floor no greater than its actual slow rate. A 100,000-draw likelihood-score Monte Carlo check accompanies the exact calculation. Neither this limit nor the primary low decisiveness establishes a universal crisis-prediction impossibility.

#### 4.3.5 Accounting, numerical checks and reproduction

Across all primary simulated steps, maximum absolute conservation errors were $3.37 \times 10^{-11}$ dollars in cash transfers, $3.52 \times 10^{-13}$ shares and $4.61 \times 10^{-11}$ dollars in aggregate marked wealth. The largest price displacement was about 1.010 dollars from the artificial 100-dollar reference. These are accounting and implementation checks, not claims of economic realism.

Twenty-two new tests pass, including off-grid feasible-mixture checks, unknown-parameter interface tests, exact response recovery, amplitude scaling and accounting. The previous releases' 29 and 23 tests were rerun and passed; the original 625 scalar and 250 network algebra checks were also rerun. A separately assembled group-means regression agrees to $1.95 \times 10^{-16}$ on 48 regenerated panels. Alternative interior-point optimization agrees with the main dual-simplex bounds to $1.30 \times 10^{-12}$ across 144 comparisons. Maximum primary primal residual was $2.23 \times 10^{-10}$; the guarded dual residual was zero to the reported precision. These are internal cross-checks, not independent outside review.

A second full run in a clean output directory regenerated all primary panels, response summaries and comparison rows. Both panel and comparison CSV files match byte for byte; all nine repeated NPZ datasets match array for array. Runtime is recorded separately and is not expected to match. The supplemental archive contains the versioned protocols, source, tests, all result rows, every public response summary, seed manifests, four complete public example datasets and source-paper provenance. The new study uses NumPy 2.3.5 and SciPy 1.17.0; solver options are recorded rather than relying on changing library defaults [36].

# 5 Group Identification and Audited Memory: Contract $I_1$

## 5.1 Group identification: fundamental limits and targeted separation

The preceding sections show how randomized aggregate probes recover total feedback $S = a + Q$ from market prices. However, evaluating systemic financial risk often requires identifying *which group of adaptive agents* is responsible for shock amplification. We now establish what aggregate price observations can identify about group decomposition, and prove an exact non-identifiability result.

### 5.1.1 The swapped-feedback non-identifiability theorem

Consider two economies with two adaptive agents sharing a common learning rate $\eta \in (0,1]$, price persistence $a \in [0,1)$, and impact $\kappa > 0$, initialized at zero:

$$x_{t+1} = ax_t + \sum_{i=1}^{2} y_{i,t} + \kappa u_t + e_{t+1}, \tag{55}$$

$$y_{i,t+1} = (1-\eta)y_{i,t} + \eta q_i x_t. \tag{56}$$

Let Economy A have feedback weights $(q_1, q_2) = (0.30, 0.10)$ and Economy B have $(q_1, q_2) = (0.10, 0.30)$.

**Proposition 5.1** (Swapped-feedback non-identifiability)**.** *Both economies induce identical aggregate price impulse responses at all horizons. If aggregate disturbances also have the same joint law in both economies, no estimator observing only prices and market-wide probes can distinguish them. Identical deterministic responses alone do not imply this stochastic claim when disturbance laws differ. Nevertheless, for a policy intervention that suppresses Agent 1 (setting $q_1 = 0$), the feedback reduction is $\Delta Q_1 = q_1 = 0.30$ in Economy A and $\Delta Q_1 = 0.10$ in Economy B. Defining the normalized loop gain as $G = Q/(1-a)$ (distinguished from the steady-state price response to constant input, $\kappa/(1-a-Q)$), holding $a = 0.40$ fixed the corresponding reduction in normalized loop gain is $\Delta G_1 = q_1/(1-a) = 0.50$ in Economy A versus $\Delta G_1 \approx 0.167$ in Economy B, differing by a factor of three.*

*Proof.* Summing the memory equations gives $Y_t = \sum_i y_{i,t}$, satisfying $Y_{t+1} = (1-\eta)Y_t + \eta(q_1+q_2)x_t$. The closed-loop price dynamics depend on the parameter vector $q$ solely through the scalar sum $Q = q_1 + q_2$. Since $Q = 0.40$ in both economies, the rational transfer function from $u$ to $x$ is identical at all horizons; see Section A.5 for the full resolvent matrix expansion. In state space, this corresponds to an exact Popov-Belevitch-Hautus (PBH) unobservable eigenspace defect of dimension $n-1$ at eigenvalue $\lambda = 1-\eta$, meaning any parameter variation along $\sum \Delta q_i = 0$ produces zero observable deflection. Consequently, the likelihood ratio of any aggregate price sequence under the two models is identically one. Yet counterfactual suppression removes $q_1$, which differs by 0.20. □

Hence aggregate prices carry no information about group attribution in the common-rate symmetric benchmark, even with infinite price records and zero measurement noise.

### 5.1.2 Minimal separation in the common-rate observationally symmetric model

**Theorem 5.2** (Minimal separation in the common-rate observationally symmetric model)**.** *Consider the observationally symmetric feedback model with $n \geq 2$ and common learning rate $\eta_i = \eta \in (0,1]$. Aggregate price impulse responses identify at most the total feedback, not its allocation uniformly over the nonnegative parameter class. This does not rule out degenerate pointwise cases such as a known zero total. When channel probing experiments are conducted according to an $m \times n$ experimental measurement matrix $U$, the measured lag-2 response vector satisfies the linear measurement map:*

$$\mathbf{b} = \eta U \mathbf{q}. \tag{57}$$

*Identifiability over the parameter class depends strictly on the observation contract:*

1. ***Known** $\eta$, **full-rank** $U$: Full column rank $\operatorname{rank}(U) = n$ provides unique componentwise identification of $\mathbf{q} = \eta^{-1} U^{\dagger}\mathbf{b}$.*
2. ***Known** $\eta$, **known aggregate** $Q > 0$: The condition $\operatorname{rank}\begin{bmatrix}\mathbf{1}^\top \\ U\end{bmatrix} = n$ gives uniform injectivity on the nonnegative fixed-total parameter class. Thus $n-1$ appropriately chosen channels can suffice. This rank is not necessary for uniqueness at an individual boundary point. At $Q = 0$, nonnegativity already determines $\mathbf{q} = 0$.*
3. ***Unknown** $\eta$, **known aggregate** $Q > 0$: When $U$ has full column rank and $Q = \mathbf{1}^\top \mathbf{q} > 0$ is known, let $\mathbf{v} = U^{\dagger}\mathbf{b} = \eta\mathbf{q}$. Then $\mathbf{1}^\top \mathbf{v} = \eta Q$ uniquely identifies the scalar rate $\eta = \mathbf{1}^\top \mathbf{v}/Q$, and substituting back yields unique point identification of the vector $\mathbf{q} = Q\mathbf{v}/(\mathbf{1}^\top \mathbf{v})$.*
4. ***Unknown** $\eta$, **no auxiliary scale information:** When only lag-2 channel measurements $\mathbf{b}$ are available under full-rank $U$ without aggregate $Q$ or known rate, a positive lower bound $\underline{\eta} > 0$ alone yields only partial ray identification: $\mathbf{q} \in \left\{\frac{U^{\dagger}\mathbf{b}}{\tilde{\eta}} : \tilde{\eta} \in [\underline{\eta}, 1]\right\}$, where $U^{\dagger} = (U^\top U)^{-1}U^\top$. For diagonal $U$, coordinatewise division is $q_i = b_i/(\tilde{\eta}U_{ii})$; direct division of $\mathbf{b}$ by $\tilde{\eta}$ holds only for $U = I_n$.*
5. ***Rank-deficient** $U$: For a known rate and one feasible $\mathbf{q}_0$, the identified set is $(\mathbf{q}_0 + \ker U) \cap \mathbb{R}^n_+$. Its local dimension at a strictly positive point is $n - \operatorname{rank}(U)$, which may be one. At a boundary it may collapse to a singleton. Unknown-rate identification additionally requires projection over the admissible rate interval.*

*Full column rank $\operatorname{rank}(U) = n$ is a necessary condition for uniform point identification across the interior of the parameter class in the absence of auxiliary constraints (Supplement Theorem S4.1). In contrast, when both the common rate $\eta$ and aggregate mass $Q$ are known, $\operatorname{rank}\begin{bmatrix}\mathbf{1}^\top \\ U\end{bmatrix} = n$ suffices, requiring only $n-1$ channel probes (Supplement Theorem S4.2). Boundary constraints (such as nonnegativity or known sparsity) can resolve particular isolated configurations on the boundary.*

*Proof.* By Proposition 5.1, the aggregate price trans-

fer function depends on $q$ solely through the scalar sum $Q = \sum_i q_i$; any parameter variation satisfying $\sum_i \Delta q_i = 0$ leaves aggregate price dynamics invariant at all horizons, proving unidentifiability under passive aggregate surveillance. For channel excitation, suppose an observer executes $m$ probe experiments where experiment $j$ injects pulse weights $U_{j,:}$ across the $n$ agent input channels at $t = 0$. The resulting state displacement at $t = 1$ is $\Delta \mathbf{y}_1 = \eta \,\mathrm{diag}(\mathbf{q}) U^\top \mathbf{e}_j$, which generates a price deflection at lag $t = 2$ of $b_j = \eta U_{j,:}\mathbf{q}$. Stacking all $m$ experiments yields $\mathbf{b} = \eta U \mathbf{q}$. For known $\eta > 0$, full column rank makes the unrestricted linear map injective and yields $\mathbf{q} = \eta^{-1}(U^\top U)^{-1} U^\top \mathbf{b}$. Necessity for uniform identification follows by perturbing a strictly positive vector along a null direction; it is not a pointwise claim at every nonnegative boundary. With known $Q > 0$, ambiguity directions must also have zero sum. This gives the augmented-rank condition and shows how $m = n - 1$ suitably chosen channels can suffice. When $\eta$ is unknown but $U$ has full column rank and $Q = \mathbf{1}^\top \mathbf{q} > 0$ is known, the pseudo-inverse projection $\mathbf{v} = U^\dagger \mathbf{b} = \eta \mathbf{q}$ satisfies $\mathbf{1}^\top \mathbf{v} = \eta \mathbf{1}^\top \mathbf{q} = \eta Q$. Because $Q > 0$, dividing by $Q$ recovers the true learning rate $\eta = \mathbf{1}^\top \mathbf{v}/Q$ uniquely. The parameter vector is then recovered as $\mathbf{q} = \mathbf{v}/\eta = Q\mathbf{v}/(\mathbf{1}^\top \mathbf{v})$, establishing Case 3. Conversely, if neither $\eta$ nor $Q$ is known, every $\tilde{\eta} \in [\underline{\eta}, 1]$ and $\tilde{\mathbf{q}} = (\eta/\tilde{\eta})\mathbf{q}$ produces the same measurement. For full-column-rank $U$, the ray segment is nondegenerate only if $\mathbf{b} \neq 0$ and $0 < \underline{\eta} < 1$; zero observations identify $\mathbf{q} = 0$, and a singleton rate interval fixes the scale. Rank-deficient $U$ adds its nullspace subject to nonnegativity. At an interior point sufficiently small nullspace perturbations remain feasible; at a boundary they need not. The explicit boundary example in Appendix A.5 distinguishes uniform and pointwise uniqueness. □

*Remark* 5.3 (Identification under heterogeneous dynamics). When agent cohorts possess known, distinct learning rates $\eta_1 \neq \eta_2 \neq \ldots \neq \eta_n$, the aggregate impulse kernel $k_j = \sum_{i=1}^n q_i \eta_i (1 - \eta_i)^j$ comprises distinct exponential modes. In that asymmetric setting, observing $n$ distinct impulse moments can permit Vandermonde-type separation of $(q_1, \ldots, q_n)$ from aggregate prices without channel-specific probing, provided the modes $(1 - \eta_i)$ are sufficiently separated. Theorem 5.2 characterizes the minimal separation requirements specifically in the observationally symmetric benchmark where temporal mode degeneracy prevents passive separation.

## 5.2 Auditable memory tails: what a source contract can justify

Passive or unconstrained price observation experiments leave a consequential gap: a short response record cannot validate the minimum learning rate that makes an asymptotic upper bound finite without structural side-information. We therefore separate an *implementation-conditioned tail claim*, an *observable linear-gain claim*, and a *finite-horizon operational-risk claim*. They have distinct assumptions, targets, and information requirements. A successful result for one is not silently promoted to the others. In this section, we examine what an auditable execution contract can and cannot justify.

### 5.2.1 A complete channel inventory, not a guessed rate floor

The audited experimental class has the public update rule

$$\begin{aligned} m_{i,t+1} &= (1 - \eta_i) m_{i,t} + \eta_i x_t, \\ x_{t+1} &= a x_t + \sum_i q_i m_{i,t} + \lambda u_t + w_{t+1}, \end{aligned} \tag{58}$$

with $m_{i,0} = x_0 = 0$, $0 \le a < 1$, $q_i \ge 0$, $\sum_i q_i \le Q_{\max}$, and public $\eta_i \in [\eta_0, 1]$. Here $Q_{\max} = 1$ is enforced by input validation in the released linear source. The rates, reset, update timing, and absence of other feedback channels are architectural disclosures; they are not estimated from price records. The actual $q_i$, $a$, and $\lambda$ are withheld from the estimator.

The release includes a machine-readable contract with source digest, public rates, zero-reset declaration, observation specification and allowed execution regime. Unit tests reject declared changes to the reset, source digest, observation type, or execution regime. In the experiment the contract is issued from the actual public simulation configuration, and the source is available for inspection. This justifies the restriction *within that controlled implementation*, not in an arbitrary proprietary market.

A digest establishes file identity, not completeness, truthful reporting, or secure runtime attestation. A deliberately self-issued contract containing false rates still passes a syntax-and-digest check. This failure is included in the release. An outside application needs authenticated configuration and an independently inspected inventory of every feedback-relevant channel. A manifest alone is not a safety certificate. In particular, a signal-memory reset does not reset account equity, outstanding positions, leverage rules, or other agents.

**Proposition 5.4** (Implementation-conditioned tail envelope)**.** *For the nonnegative memory kernel $k_j = \sum_i q_i \eta_i (1 - \eta_i)^j$ generated by equation* (58)*, a complete valid contract implies*

$$R_H := \sum_{j=H}^{\infty} k_j = \sum_i q_i (1 - \eta_i)^H \le Q_{\max} (1 - \eta_0)^H. \tag{59}$$

*More generally, for $k_j = CA^jB$, induced-norm bounds $\|A\| \le r < 1$, $\|B\| \le b$ and $\|C\| \le c$ imply $\sum_{j=H}^{\infty} \|k_j\| \le bc\, r^H/(1 - r)$.*

*Proof.* Sum each geometric series to obtain the equality in equation (59); nonnegativity and $1 - \eta_i \le 1 - \eta_0$ give

the bound. For the second statement, submultiplicativity gives $\|CA^jB\| \le cbr^j$; summing proves the claim. These are standard geometric-series and induced-norm arguments, not new general stability theorems. □

For $Q_{\max} = 1$ and $H = 64$, the envelopes at $\eta_0 = 0.002, 0.02, 0.1$ are respectively 0.879741, 0.274454, and 0.001179. The slowest contract therefore still permits a large unseen tail. A verified small rate is not equivalent to a useful small tail. If an additional relevant inventory coordinate has unit persistence, the premise $r < 1$ for the complete hidden subsystem is unavailable; a contract for the memory module cannot fill that gap.

#### 5.2.2 Why the earlier impossibility results still matter

The exact and noisy counterexamples in Sections 3.2–3.4 are not contradicted. They concern a less informative observation class. Disclosing the true update architecture and guaranteeing a reset changes the compatible model set. That disclosure has an institutional cost and is not available merely because prices are observed accurately. The distinction between information sufficient for a property and information sufficient for full model recovery is established in data-informativity research [29]; positive side-information is likewise an existing identification tool [32]. Our contribution here is an explicit, executable conditional contract and a documented failure when its scope is misrepresented.

### 5.3 Budgeted inference from audited public dynamics

#### 5.3.1 One terminal innovation per independent episode

Given public rates, exact midprice deviations and a verified reset, the estimator reconstructs

$$m_{i,T-1} = \eta_i \sum_{s=1}^{T-2} (1-\eta_i)^{T-2-s} x_s \tag{60}$$

from prices alone. It never reads the simulator's stored memory. Assume $u_{T-1} = 0$. Each independent episode contributes exactly one regression row,

$$\begin{aligned} Y_j &= x_{j,T_j}, \\ X_j &= (1, x_{j,T_j-1}, m_{j,1,T_j-1}, \dots, m_{j,d,T_j-1}), \\ Y_j &= X_j\beta + w_{j,T_j}, \end{aligned} \tag{61}$$

where $\beta = (\mu, a, q_1, \dots, q_d)^\top$ and the intercept accommodates a common mean innovation. The endpoint $T_j$ is fixed or randomly assigned *before* observing that episode. It is not selected after a favorable price movement.

**Proposition 5.5** (Conditional terminal-row confidence; classical regression)**.** *Suppose episodes are independent, the innovations $w_{j,T_j}$ are independent Gaussian variables with common unknown variance, and each is independent of all pre-terminal regressors and preassigned endpoints. If $X$ has full column rank $p < n$, then for a fixed contrast $v$,*

$$\frac{v^\top\widehat{\beta} - v^\top\beta}{\widehat{\sigma}\sqrt{v^\top(X^\top X)^{-1}v}} \sim t_{n-p} \quad \textit{conditional on } X. \tag{62}$$

*In particular $v = (0, 1, 1, \dots, 1)$ gives a confidence interval for $A_{\text{fb}} = a + \sum_i q_i$, whose threshold is one in the maintained positive linear class.*

*Proof.* Conditional on $X$, the last innovations are an independent spherical Gaussian vector. The least-squares projection and orthogonal residual projection are independent. The contrast is Gaussian with variance $\sigma^2 v^\top(X^\top X)^{-1}v$, and the residual sum of squares divided by $\sigma^2$ is $\chi^2_{n-p}$. Their ratio gives equation (62). No independence claim is made for multiple successive transitions from the same episode; treating those as independent rows would invalidate this proof. □

This is ordinary regression under an explicitly auditable experimental design, not an ad hoc estimator. Unknown-rate recurrence bounds and the present structural estimator do not receive the same information: the former operate under sensor noise without channel disclosure, whereas the latter utilizes disclosed architectural rates and exact midquote updates. Consequently, differences in decisiveness reflect differences in information structure rather than raw statistical efficiency alone.

#### 5.3.2 Primary budget experiment and negative results

The primary identification panel investigates twelve benchmark $(G, \eta_{\text{slow}})$ combinations: $G \in \{0.7, 0.95, 1.05, 1.3\}$, $\eta_{\text{slow}} \in \{0.002, 0.02, 0.1\}$, a fast rate of 0.5, $a = 0.4$, $\lambda = 0.02$, and $q = 0.6G(0.6, 0.4)$. Gaussian innovation standard deviation is 0.002 dollars. Gains and innovations are not inputs to inference. There are $n \in \{32, 128, 512\}$ independent episodes, lengths $T \in \{16, 64\}$, two probe designs, and twenty replicate panels per cell: 2,880 panels and 645,120 episode trajectories.

The two designs have four shares of within-window gross probe volume: an initial signed four-share impulse or eight half-share orders in four round trips. They do *not* have equal squared input energy or residual inventory. The impulse has energy 16 share-squared and holds four shares at its endpoint; the round trips have energy 2 and return flat, with peak absolute inventory at most half a share. Unwinding the impulse after observation requires another four shares of volume. These economic differences are reported rather than hidden behind a claim of "equal budget." Input design under amplitude

and power constraints is established research [37]; neither pattern is claimed optimal.

Across primary panels, 2,758 of 2,880 gain intervals cover the true gain, establishing a parameter coverage of 95.76%. Of the 2,880 panels, 745 produce definite classification decisions (a decisiveness rate of 25.87%), while 2,135 abstain. Among all 2,880 panels, twenty result in confident errors (an unconditional error rate of 0.69%). Crucially, evaluated among the 745 issued decisions, the conditional error rate is $20/745 = \mathbf{2.68}\%$. Therefore, this audited protocol does not eliminate conditional error: while abstention correctly filters out most ambiguous windows near boundary $S = 1$, issued decisions still retain a non-zero error rate of 2.68%. These are empirical frequencies under per-panel 95% intervals, not simultaneous study-wide coverage. At $T = 64, n = 128$, the impulse resolves 86/240 panels, versus 46/240 for the round trips. At $n = 512$, those counts are 158/240 and 80/240. The round-trip alternative fails as a general accuracy improvement. Its smaller position exposure is a separate property, not proof of equivalent informativeness.

#### 5.3.3 Held-out endpoint diversification: useful at small budgets, not uniformly

After the primary experiment, a small pilot suggested varying the observation endpoint instead of redistributing the order volume. A fresh-seed follow-up was specified before its execution. Both arms use the same initial four-share impulse and the same audited rates. One arm records 64 transitions per episode. The other assigns equal numbers of episodes to 8, 16, 32 and 64 transitions, shuffled before inspecting prices. Its average record is 30 transitions, with the same maximum of 64. Both arms would require the same four-share post-observation unwind, outside the measurement law.

Each paired dataset is simulated to length 64, but the diversified estimator is given only the assigned prefixes; a test alters all later prices and verifies unchanged inference. There are 720 fresh panel datasets (161,280 episodes), each evaluated in both arms, yielding 1,440 decision rows. This is post-primary held-out confirmation, not external preregistration. The contrasts, model grid and new seeds are released.

At 32 episodes, 56 previously unresolved paired cases become definite and seven go the opposite way: 49 net additional decisions. At 128, the corresponding counts are 42 and sixteen. At 512, diversification loses fifteen decisions net. Thus the result is a budget-dependent tradeoff, not a uniformly dominating detector. The release includes descriptive paired tests; pooling heterogeneous model settings is not a claim of universal statistical superiority. In the original fixed-64 primary experiment, the slowest rate resolves only 4/80 panels at 128 episodes and 15/80 at 512. Architectural disclosure does not remove low signal-to-noise limitations.

#### 5.3.4 A measured tail interval under the stronger audit

When rates are genuinely public, the unseen-tail mass in equation (59) is itself a linear contrast of $\beta$. Use $v_H = (0, 0, (1-\eta_1)^H, \ldots, (1-\eta_d)^H)$ in equation (62). Intersect its interval with $[0, Q_{\max}(1-\eta_0)^H]$. An empty intersection is a model conflict, not an interval silently clipped into feasibility. Splitting $\alpha = 0.05$ equally between the gain and tail gives at least 95% joint coverage per panel by a union bound.

This analysis reuses the primary observations; it creates no additional independent market data. For $H = 64$, the joint event holds in 2,796/2,880 panels (97.08%). There are nine model conflicts. At a fixed observation length of 64, the statistical upper bounds below are medians across eighty panels; they are not worst-case bounds for every episode or every institution.

#### 5.3.5 Assumption violations remain visible

Five eighty-panel controls evaluate deliberately ungated inference. Omitting a slow state produces eighty false stable classifications in eighty panels. Omitting a positive inventory-feedback channel produces three false stable classifications in eighty panels. Noisy regressors and a violated reset produce no wrong labels in their particular runs, but their calibration theorem is still unavailable; absence of observed errors is not proof of robustness. The valid control produces eighty correct unstable classifications. All outcomes, including unhelpful and non-failing controls, are retained. A fit cannot authenticate the architecture that made its confidence statement valid.

# 6 Funded Market Mechanism: Solvency vs. Signal Stability

## 6.1 An account-consistent market environment

Stability of the price-signal recursion ($\rho(J) < 1$) is a statement about prices, not about the balance sheets that supply liquidity. This section builds a funded simulation in which the two can be compared.

Liquidity providers face cash, inventory and leverage limits. In the model below a dealer can reach those limits while the price-feedback recursion is stable, and it then stops absorbing flow. To study this gap between signal stability and dealer capacity, we embed the learning dynamics in a funded, double-entry dealer market with explicit cash and inventory ledgers.

Table 13: Held-out endpoint comparison. Each entry uses 240 panels. Wrong calls are included in, not removed from, the definite-call count.

| Episodes | Fixed 64: definite | Wrong | Diversified: definite | Wrong |
|---|---|---|---|---|
| 32 | 26 (10.83%) | 0 | 75 (31.25%) | 2 |
| 128 | 86 (35.83%) | 0 | 112 (46.67%) | 1 |
| 512 | 159 (66.25%) | 1 | 144 (60.00%) | 0 |

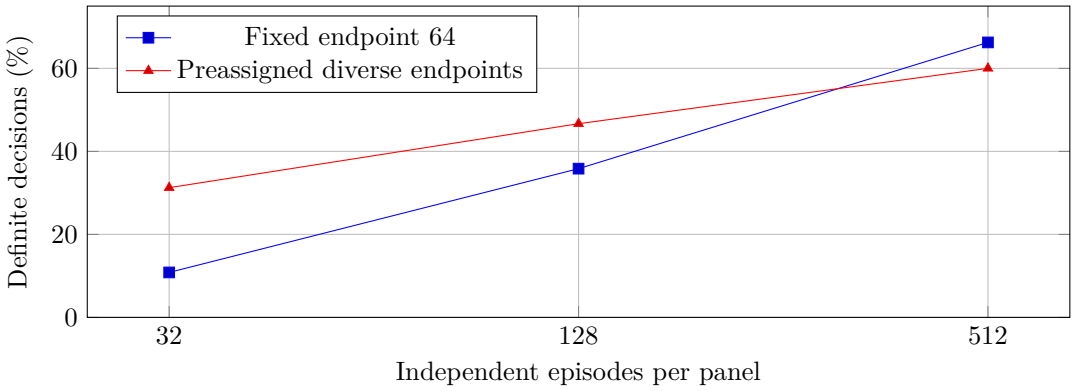



Figure 6: Observed decision rates on the held-out paired panel. The crossing is retained: a useful low-budget change does not improve every budget. Error counts appear in table 13.

Table 14: Audited tail envelope and median measured upper endpoint for the four-share impulse. Each statistical interval allocates 2.5% error to the tail.

| Minimum rate | Analytic envelope | 128 episodes | 512 episodes |
|---|---|---|---|
| 0.002 | 0.879741 | 0.788458 | 0.490520 |
| 0.020 | 0.274454 | 0.136919 | 0.103451 |
| 0.100 | 0.001179 | 0.000674 | 0.000451 |

#### 6.1.1 Orders, price formation and adaptive demand

Let $P_t = p^* + x_t$, with the constant reference value $p^* = 100$ dollars. Chartist $i$ holds a private smoothed displacement $m_{i,t}$, with demand $v_{i,t} = \chi_i m_{i,t}$. A fundamentalist submits $v_{F,t} = -\varphi x_t$, a noise trader submits $v_{N,t}$, and the experiment's trader submits $u_t$. The dealer absorbs their net quantity $V_t$. Prices and memories update as

$$V_t = \sum_i v_{i,t} + v_{F,t} + v_{N,t} + u_t, \qquad P_{t+1} = P_t + \kappa V_t, \tag{63}$$

$$m_{i,t+1} = (1 - \eta_i) m_{i,t} + \eta_i x_t. \tag{64}$$

Writing $a = 1 - \kappa\varphi$, $q_i = \kappa\chi_i$ and $y_{i,t} = \kappa\chi_i m_{i,t}$ gives (29) exactly. Updates use $x_t$, not the newly moved $x_{t+1}$; changing timing changes the response recurrence. Agents have adaptive memory, not trained neural policies or demonstrated utility-optimal behavior.

Empirical order-book research motivates considering a local relation between net pressure, price changes and depth, but does not validate this specific aggregate-demand model or its chosen parameters [35]. In particular, actual order-flow imbalance includes quote and cancellation events; the simple signed trade volume $V_t$ here is not claimed to equal that empirical measure.

#### 6.1.2 Double-entry accounting and economic scope

Trades execute at the within-step midpoint $P_t^{\mathrm{ex}} = (P_t + P_{t+1})/2$, with half-spread $c = 0.005$ dollars transferred to the dealer. Each nondealer actor with quantity $v$ has changes

$$\Delta N = v, \qquad \Delta C = -v P_t^{\mathrm{ex}} - c|v|.$$

The dealer has the opposite aggregate inventory and cash changes. Thus total shares and cash transfers sum to zero after every step; marking all inventories to one common quote also preserves aggregate marked wealth when net initial shares are zero. The cash variables are ledger changes, not evidence that all actors satisfy capital requirements. Unconstrained credit and short positions, fixed depth, no defaults and no endogenous balance-sheet limits make the primary model a local mechanism test rather than a complete financial economy.

The primary price impact is 0.02 dollars per share and $a = 0.4$, implying $\varphi = 30$ shares per dollar. Chartist elasticities are set from the declared $q_i$. Background net-flow noise has standard deviation 0.10 shares; quote observation error has standard deviation 0.002 dollars; each initial private memory has standard deviation 0.01 dollars. The estimator is not told these scales. Time is an abstract model step, not a claimed number of market seconds. No live-market orders, customer records or external financial data are used.

#### 6.1.3 Experimental design and information firewall

The frozen primary panel uses rates $(0.5, \eta_s)$ with $\eta_s \in \{0.002, 0.02, 0.10\}$ and $G \in \{0.70, 0.95, 1.05, 1.30\}$. Sixty percent of normalized feedback is fast and forty percent slow. The imposed class floor is $\underline{\eta} = \eta_s$; this is public design-side information, not an estimated individual hidden rate. The equality at the floor is a benchmark boundary choice, and off-grid support is separately tested.

For each of 12 configurations, 128 or 1,024 independent episodes use pulses of one or four shares. Forty independent panels per configuration/design give 1,920 panels. Each episode records 66 quote changes. Inference at largest lags 8, 24 and 64 gives 5,760 horizon rows. There are $H + 1$ response coefficients through lag $H$; the last collected coefficient, lag 65, is unused in these fits but included in the simultaneous confidence family, conservatively. There are 1,105,920 episodes and 72,990,720 quoted-price observations, not that many independent financial markets.

Only a public object with keys `probes` and `prices` crosses into estimation. Analytical responses, latent trajectories, true gains and configuration labels are used by the evaluation layer only. Known noise standard deviations and exact direct persistence are not estimator arguments. Full public response estimates, endpoints and standard errors are saved for every panel, together with generation seeds and raw-record hashes. Four public episode examples and their response covariances are included. All other raw episodes can be regenerated; they are not represented as external market records.

### 6.2 Binding financial constraints, balance-sheet stress, and default risk

#### 6.2.1 A funded transaction model, not just a ledger

Version 0.4 records trades but does not impose binding capital constraints. The new financial experiment contains two memory traders, a fundamentalist, a background-flow trader, a probe account and a dealer. Let $C_i$, $I_i$, $P$ and $E_i = C_i + I_i P$ denote cash, shares, midpoint and marked equity. A trade $v_i$ executes at the current midpoint plus a half-spread $s$ paid in the direction of the trade:

$$C_i' = C_i - P v_i - s|v_i|, \qquad I_i' = I_i + v_i, \quad E_i' = E_i - s|v_i|. \tag{65}$$

The dealer receives the opposite shares and all cash transfers. This leaves total cash and shares unchanged. The price then moves by $\lambda \sum_i v_i$; a declared external repricing may follow.

For previously feasible participants, requested orders

are projected onto

$$|I_i'| \le \overline{I}_i, \qquad |I_i'|P \le L_i E_i', \qquad C_i' \ge -\kappa_i E_{i,0}. \quad (66)$$

These constraints include spread costs in post-trade equity. For a fixed trade sign they are linear inequalities in the fill, so the nearest feasible fill along the requested trade direction is computed explicitly. A separate LP verifier checks 500 random cases against this projection. A breach caused by subsequent price marking triggers forced liquidation on the next step. Negative equity is recorded as default; outstanding debt is not erased or replaced with fresh capital.

The dealer also faces inventory, leverage and credit limits. When desired aggregate flow exceeds its capacity, all participant fills are proportionally reduced. If its existing account is already infeasible, the simulated venue halts rather than inventing an unconstrained outside buyer. Existing positions remain on the books. An externally imposed price shock is still recorded. The maximum leverage observed *after* a price move may exceed the execution limit; this is precisely why margin calls exist. The accounting audit distinguishes infeasible newly accepted trades from later marking losses.

Funding–liquidity feedback is established financial theory [4]. Trading under propagator and inequality constraints also has dedicated predecessors [39]. The present simulator is a simple executable validation environment for the measurement distinction. It is not a new liquidity-spiral theory, an exchange order book, or an implementation of those papers' optimal-execution algorithms.

#### 6.2.2 Primary economic settings and accounting audit

The base initial equities, in account order, are $(100, 100, 600, 300, 600, 2000)$ dollars. Leverage limits are two; credit limits equal initial equity; inventory caps are $(3, 3, 15, 10, 10, 30)$ shares. The ample regime multiplies equity and inventory caps by twenty. The tight regime uses the base values. The unlimited counterfactual ignores execution constraints and may accumulate insolvent positions; it is a diagnostic counterfactual, not a viable funding arrangement. Positions initially sum to zero.

The primary panel crosses $G \in \{0.7, 1.3\}$, slow rates 0.02/0.1, the three funding regimes, and repricing shocks of zero or minus two dollars at step 16. Each of 24 settings has 20,480 episodes of 64 transitions. The probe buys or sells two shares at step zero and requests the reverse at step eight: four gross requested shares, not a guaranteed full fill. External random tapes are shared across funding counterfactuals. Therefore 491,520 setting-trajectories use 163,840 unique exogenous episode tapes; they must not be counted as 491,520 mutually independent market shocks.

The execution engine modifies 12,388,550 order requests and records 2,240,614 account-step margin calls. These are repeated order/account-step counts, not unique crises. Maximum cash-conservation error is $2.41 \times 10^{-10}$ dollars, maximum share error $2.34 \times 10^{-12}$ shares, and maximum fresh-trade constraint residual $3.19 \times 10^{-12}$. Thus the financial limits are active, rather than decorative parameters.

#### 6.2.3 Finite-horizon risk without an unjustified linear certificate

When accounts constrain fills, the dynamics are state-dependent and nonlinear. The audited memory bound still describes that isolated signal module, but not all feedback in the financially constrained system. We therefore do not apply the linear gain interval to certify funded-system stability.

Instead define an episode event $B$ as any account default, a venue halt, or an adaptive trader's peak equity drawdown exceeding 20% of initial equity during 64 transitions. Price dislocation, margin activity, turnover and unfilled orders are reported separately. The endpoint is frozen before the primary experiment. For iid repetitions of a declared scenario distribution, $K_n = \sum_{j=1}^{n} \mathbf{1}_{B_j}$ is binomial regardless of the within-episode nonlinear dynamics. An exact Clopper–Pearson interval bounds $p_B$ [38].

**Proposition 6.1** (Planned-budget operational-risk coverage)**.** *At predeclared episode budgets $n \in \{32, 128, 512\}$, construct two-sided exact binomial intervals at error $\alpha/3$ each. Under iid episodes from the same specified scenario law, all three intervals cover $p_B$ with probability at least $1 - \alpha$. This conclusion does not require linearity, a memory-rate floor, or Gaussian observations.*

*Proof.* Each interval inverts binomial tail tests and has noncoverage probability at most $\alpha/3$. A union bound over the three planned looks gives total noncoverage at most $\alpha$, without requiring independence across the nested budgets. Arbitrary additional peeking, dependent episodes, a changed scenario distribution, or an event definition chosen after seeing losses is outside the claim. □

The illustrative acceptance threshold $p_B = 5\%$ is a research choice, not a regulatory standard or investment recommendation. At zero events, the upper endpoint is $1 - (\alpha/6)^{1/n}$. Thus 32 quiet episodes are insufficient to establish risk below 5%; 128 can be sufficient. There are 960 setting-panels evaluated at three nested budgets, yielding 2,880 rows. In the primary settings, the realized event frequencies are extreme: twenty settings have zero events and four unlimited-funding shocked settings have an event in every episode. At 128 episodes all panels consequently receive a definite probability-threshold label. This is *not* a difficult near-threshold benchmark, not proof of universal 128-episode practicality, and not a demonstration of zero population risk in the zero-event

settings. Exact coverage is separately enumerated on 3,003 $(n, p)$ grid points.

#### 6.2.4 Dealer stress: locally stable signal feedback is not system safety

The first financial panel activates participant constraints but has no dealer halts. A separately labeled post-primary extension therefore starts the tight-regime dealer with 19 shares and equity 1,000, balanced by positions of minus nine shares in the fundamentalist account and minus ten in the probe account. Cash is adjusted to preserve each initial equity and zero net shares. These positions satisfy the initial constraints. Repricing at step 16 has mean minus two or minus ten dollars and independent standard deviation one. Eight settings each contain 8,192 fresh episodes, totaling 65,536.

For $G = 0.7$, the unconstrained linear quantity is $a + \sum q_i = 0.82 < 1$, yet substantial halt frequencies occur under the stressed dealer balance sheet. This supplies a concrete model-specific counterexample to transferring the signal-stability label to operational financial safety. The difference is not attributed exclusively to artificial intelligence: ordinary adaptive demand and standard funding constraints are enough. Neither the higher-gain nor lower-gain case is uniformly safer across these outcomes. No blanket prescription for capital limits or autonomous-agent adoption follows.

**Dealer capacity-risk surface and inventory-buffer scaling.** Table 16 varies inventory capacity and initial inventory at fixed utilization. In `FundedScenarioMarket`, the argument historically named `dealer_capital_multiplier` scales the inventory limit, while initial marked equity remains \$10,000 in every arm. Thus 0.5×, 1× and 2× do not mean \$5,000, \$10,000 and \$20,000 of equity. This compares capacity/inventory regimes, not an isolated capacity effect with initial inventory held fixed or a causal effect of recapitalization. It is separate from the original funded-engine experiments with explicit credit and liquidation rules. The historical argument `a=0.4` is also not used by this scenario's price update. The effective unconstrained adaptive dynamics are

$$x_{t+1} = (1 - \kappa\phi)x_t + q_f y_{f,t} + q_s y_{s,t} + \kappa\varepsilon_t,$$
$$y_{i,t+1} = (1 - \eta_i)y_{i,t} + \eta_i x_{t+1}.$$

Here memories are unweighted and update from the new price, unlike the timing and weighted memory coordinates in the principal linear model. At the paired-control parameters, the nominal sum is 0.82 but the effective sum is 1.22 and the full transition matrix has spectral radius approximately 1.07409. The correction exposes these quantities without changing any historical trajectory. The stable-signal counterexample in Table 15 belongs to the separate original funded engine; it must not be inferred from this scenario surface.

**Event-path causal diagnostics and inventory buffer exhaustion.** At step $t = 16$, the simulator imposes $x \mathrel{+}= \mu$. Desired flow is $(q_f/\kappa)y_f + (q_s/\kappa)y_s - \phi x + \varepsilon_t$; fills are restricted by the dealer's inventory capacity. The dealer records inventory change minus filled flow and cash change equal to filled flow times the new price. When a requested inventory move exceeds the limit, the code sets a sticky `halted` flag, but does not terminate the loop or ban later trades: reverse-direction fills can still occur. The flag therefore measures a constraint-hit event. Margin checks count threshold breaches; they do not implement liquidation in this scenario class. In the paired run, mean minimum marked equity remains about \$9,999.06 in the constrained adaptive regime. Under the historically named "Unlimited Funding" arm, only the inventory constraint is disabled. Initial equity is unchanged. Time-average absolute price dislocation is 5.070 dollars and mean minimum marked equity is −\$122.03. This compares inventory-constrained with unconstrained paths in an unstable scenario; it does not establish a universal benefit of halts, a funding intervention, or failure of a valid stable-signal certificate.

#### 6.2.5 Paired counterfactual mechanism controls: active versus frozen memory updating

We compare adaptive memory with memory frozen after the pre-shock warm-up across 100 paired disturbance tapes ($\mu = -2.0$), holding the implemented scenario rules fixed (Table 17). The contrast concerns this adaptive-demand mechanism, not trained AI or an independently validated market.

Mean dislocation is 1.151 dollars with frozen memory, 1.766 with adaptive updating, and 1.127 with feedback removed. The adaptive-minus-frozen difference is about 0.6148 dollars, a 53.4% increase relative to the frozen arm. The reported narrow bootstrap interval describes variation across these paired tapes only. Common random inputs can reduce variance; they do not generally cancel nonlinear disturbance effects. Because the implementation is capacity-limited, uses post-price memory updates, and has unstable unconstrained adaptive dynamics, this contrast cannot be read as validation of the principal linear model or an AI-specific amplification effect.

### 6.3 Learned-Agent Transfer: Inventory-Avoidance and Action-Support Failure

The linear theory does not automatically describe a learned trading policy. We therefore examine a separate tabular Q-learning environment, with an explicit

Table 15: Preloaded-dealer stress extension. Event rates are synthetic scenario frequencies, not estimates of real-market crisis probabilities. All settings have binding execution limits.

| $G$ | Slow rate | Mean repricing | Event rate | Halt rate |
|---|---|---|---|---|
| 0.7 | 0.02 | −2 | 45.42% | 45.42% |
| 0.7 | 0.10 | −2 | 36.22% | 36.22% |
| 1.3 | 0.02 | −2 | 50.62% | 50.62% |
| 1.3 | 0.10 | −2 | 45.15% | 45.15% |
| 0.7 | 0.02 | −10 | 85.16% | 82.59% |
| 0.7 | 0.10 | −10 | 73.91% | 63.94% |
| 1.3 | 0.02 | −10 | 79.92% | 56.60% |
| 1.3 | 0.10 | −10 | 77.04% | 51.75% |

Table 16: Preserved scenario frequencies across inventory-capacity multipliers and price shocks ($\eta_s = 0.02$, initial utilization 20%, $N = 30$ per cell). The first column retains the historical nominal label, not the implemented feedback sum: $a_{\text{eff}} = 1 - \kappa\phi = 0.8$ implies $S_{\text{eff}} = S_{\text{nominal}} + 0.4$. These rows therefore do not establish a stable-signal counterexample. "Halt" records any inventory-constraint hit, not a persistent trading ban. The 43.3% cell is 13 hits in 30 repetitions.

| Feedback $S$ | Capacity | Shock $\mu$ | Failure Rate | Halt Rate | Mean Disloc. | Min Equity ($) |
|---|---|---|---|---|---|---|
| 0.70 | 0.5× | −0.2 | 0.0% | 0.0% | 0.005 | $9,999.38 |
| 0.70 | 0.5× | −0.6 | 100.0% | 100.0% | 0.227 | $9,998.24 |
| 0.70 | 1.0× | −0.2 | 0.0% | 0.0% | 0.005 | $9,999.40 |
| 0.70 | 1.0× | −0.6 | 100.0% | 100.0% | 0.011 | $9,994.69 |
| 0.70 | 2.0× | −0.6 | 0.0% | 0.0% | 0.011 | $9,994.63 |
| 0.70 | 2.0× | −2.0 | 100.0% | 100.0% | 0.604 | $9,984.01 |
| 0.82 | 0.5× | −0.2 | 0.0% | 0.0% | 0.006 | $9,999.39 |
| 0.82 | 0.5× | −0.4 | 100.0% | 100.0% | 0.076 | $9,999.03 |
| 0.82 | 1.0× | −0.2 | 0.0% | 0.0% | 0.006 | $9,999.39 |
| 0.82 | 1.0× | −0.4 | 0.0% | 0.0% | 0.010 | $9,997.60 |
| 0.82 | 1.0× | −0.6 | 43.3%† | 43.3%† | 0.015 | $9,994.67 |
| 0.82 | 1.0× | −0.8 | 100.0% | 100.0% | 0.152 | $9,996.10 |
| 0.82 | 2.0× | −0.6 | 0.0% | 0.0% | 0.015 | $9,994.67 |
| 0.82 | 2.0× | −1.0 | 0.0% | 0.0% | 0.024 | $9,985.24 |
| 0.82 | 2.0× | −2.0 | 100.0% | 100.0% | 0.604 | $9,982.97 |
| 1.18 | 0.5× | −0.2 | 0.0% | 0.0% | 0.025 | $9,999.54 |
| 1.18 | 0.5× | −0.4 | 100.0% | 100.0% | 0.082 | $9,999.02 |
| 1.18 | 1.0× | −0.2 | 0.0% | 0.0% | 0.024 | $9,999.48 |
| 1.18 | 1.0× | −0.6 | 0.0% | 0.0% | 0.070 | $9,996.12 |
| 1.18 | 1.0× | −0.8 | 100.0% | 100.0% | 0.161 | $9,996.12 |
| 1.18 | 2.0× | −1.0 | 0.0% | 0.0% | 0.118 | $9,989.07 |
| 1.18 | 2.0× | −2.0 | 100.0% | 100.0% | 0.636 | $9,986.11 |

† 43.3% = 13/30 halting episodes; exact 95% Clopper–Pearson CI [25.5%, 62.6%]. With $N = 30$ replications this single intermediate-probability cell carries substantial sampling uncertainty: the true halt probability could plausibly range from approximately one-quarter to two-thirds of episodes. All other cells in the table report 0.0% ($k = 0$) or 100.0% ($k = 30$), whose exact intervals are respectively $[0, 11.6\%]$ and $[88.4\%, 100\%]$, not point intervals.

Table 17: Paired capacity-scenario summaries on 100 disturbance tapes.

| Regime | Constraint hit (%) | Composite event (%) | Mean dislocation | Min. equity ($) |
|---|---|---|---|---|
| Active memory | 100.0 | 100.0 | 1.76570 | 9999.06 |
| Frozen memory | 100.0 | 100.0 | 1.15086 | 9999.06 |
| Zero feedback | 100.0 | 100.0 | 1.12686 | 9997.99 |
| Inventory limit removed | 0.0 | 100.0 | 5.06952 | -122.03 |

Dislocation is the time-average absolute displacement over 64 steps, in dollars. The hit flag is not a permanent trading ban: subsequent reversing fills remain possible. Removing the inventory limit does not add funding. Means are descriptive; these summaries do not include uncertainty for each arm mean.

Table 18: Paired dislocation contrasts (dollars).

| Contrast | Mean difference | Paired bootstrap 95% interval |
|---|---|---|
| Active minus frozen | 0.61484 | [0.61477, 0.61490] |
| Active minus zero feedback | 0.63884 | [0.63871, 0.63897] |

Percentile intervals use 1,000 resamples of paired tapes. They describe this fixed simulation, not real-market intervention effects or a universal variance-reduction guarantee.

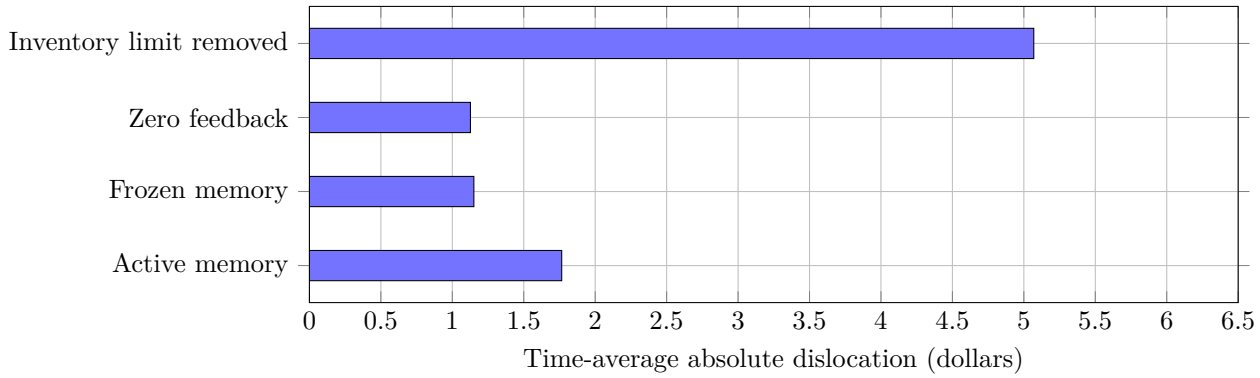



Figure 7: Descriptive regime means, generated from the preserved scenario rules. Paired-difference uncertainty is reported separately in Table 18. The unconstrained adaptive transition has spectral radius 1.07409, so this is not a stable-system safety counterexample.

distinction between response screening and structural certification.

**Economic and accounting contract.** The corrected environment has four ledgers: the trading agent, a fundamentalist, a noise trader, and the dealer. Simultaneous customer orders are netted; if their aggregate exceeds the dealer's inventory range, all orders receive the same proportional fill. Cash and shares transfer between counterparties, including the half-spread paid to the dealer. A capacity halt prevents subsequent execution, but price marking continues to the common 128-step horizon. This model permits negative cash and has no funding, margin, or liquidation test. It is therefore not the funded engine studied above. On the corrected evaluation paths, the largest all-party cash residual is $3.64 \times 10^{-11}$.

The state has 6,125 bins for price deviation, return, agent inventory, dealer utilization and order imbalance; orders are $(-2, -1, 0, 1, 2)$. Reward is marked equity change minus $0.01I_t^2 + 0.005|\Delta I_t|$. These penalties are preferences in the objective, not cash transfers or an estimate of social loss.

**Evaluation protocol.** Ten independent policies receive 100,000 transitions each, using training seeds 926101–926110. Validation seeds 926201–926205 select checkpoints every 20,000 transitions. Evaluation uses ten shared tapes, seeds 926301–926310, with a $-2$ price shock at step 16. The learning rate is 0.05, discount factor 0.98, and exploration probability decays multiplicatively by 0.99995 per update from 1 to a floor of 0.02. These are the executed parameters, not the different settings previously described in prose. Validation uses a copy and cannot advance the training random stream. Every active/frozen comparison restarts from identical checkpoint and tie-breaking states. Frozen policies remain bitwise unchanged.

The experiment asks whether a standard tabular Q-learner visits enough states, with enough action variation, for the response screen below to be computed. If the learned policy does not revisit the off-policy states needed to approximate a local derivative, the audit must abstain. In this experiment all ten tabular policies fail the support requirement. Whether neural or continuous-action policies avoid this failure is not tested here; an earlier draft reported such tests, and that claim is withdrawn (section D).

**Support-first screening.** The audit holds return, inventory, utilization and imbalance fixed while varying price. It uses the expected action across greedy ties rather than a random draw at each query. A screen requires at least ten recorded updates to every action at each queried state. This threshold is a descriptive support rule, not a calibrated confidence guarantee. Missing support produces abstention. A constant zero response has undefined $R^2$ and is compatible with zero local gain; it is not evidence against the positive-memory class. Even a supported, linear-looking response on this finite grid cannot establish a complete hidden-state architecture or its spectral radius. All ten evaluated policies abstain for insufficient action support.

Table 19: Inventory-only falsification experiment: ten independently trained policies, each averaged over the same ten shock tapes. All response audits abstain for insufficient recorded support; none certifies or falsifies a global model class. Displacement is mean absolute price deviation over 128 steps.

| Policy | Turnover | Frozen displacement | Active displacement | Active − frozen |
|---|---|---|---|---|
| 01 | 6.0 | 0.027477 | 0.027600 | +0.000122 |
| 02 | 6.6 | 0.027524 | 0.027547 | +0.000023 |
| 03 | 6.2 | 0.027468 | 0.027515 | +0.000047 |
| 04 | 28.2 | 0.029856 | 0.028068 | -0.001788 |
| 05 | 6.2 | 0.027542 | 0.027586 | +0.000044 |
| 06 | 6.9 | 0.027605 | 0.027606 | +0.000000 |
| 07 | 15.6 | 0.028801 | 0.028205 | -0.000596 |
| 08 | 6.0 | 0.027487 | 0.027558 | +0.000072 |
| 09 | 6.4 | 0.027488 | 0.027698 | +0.000211 |
| 10 | 11.6 | 0.028161 | 0.027753 | -0.000408 |

**Results and uncertainty.** Mean absolute displacement is 0.027236 for no-trade, 0.027941 for frozen Q-learning, and 0.027714 for active Q-learning. The active-minus-frozen mean is $-0.000227$ with a 95% population-level Student interval $[-0.000659, 0.000204]$. It neither establishes improvement nor proves an exactly zero effect. Frozen-minus-no-trade is +0.000705, interval $[0.000133, 0.001277]$: the trained policies do not outperform abstention on this displacement measure.

These intervals use ten independent training-population means, conditional on the fixed ten evaluation tapes. They do not treat 100 nested policy–tape combinations or individual trades as independent populations. Frozen and active policies have zero observed halts; the chartist halts on all ten tapes, yet its displacement over the corrected common horizon is 0.032783, not the earlier horizon-dependent value. A random-trading comparator is calibrated to validation trade frequency, not test turnover. Its realized turnover, 5.55, differs from frozen Q-learning's 9.97; it cannot identify a volume-matched causal mechanism.

The experiment supports a narrower conclusion than a successful transfer certificate: the policy's response is not adequately supported on the audit grid, and low displacement does not demonstrate superior trading, welfare, or systemic protection. The reward design strongly penalizes inventory holding and turnover, creating an active avoidance mechanism rather than price stabilization.

**Multiscale perturbation audit, reward sensitivity, and turnover matching.** To verify that the out-of-class finding is not an artifact of wide-grid evaluation, a multiscale perturbation radius audit evaluates the response function across five nested radii $r \in \{0.05, 0.10, 0.25, 0.50, 1.00\}$ (Supplement Section S14.6 and Table S14.6). Across all radii and all ten policies, $R^2(r) \leq 0.2449 \ll 0.70$. At the innermost radius ($r = 0.05$), all policies yield $R^2 = 0.0000$ due to complete dead-band action inertia induced by the penalties; at $r = 0.10$, local action variation appears with mean $R^2 = 0.0955$ and slope $\partial u/\partial x = 1.30$, but the response is a discrete step function whose fit decays to $R^2 = 0.0530$ at $r = 1.00$.

Furthermore, a reward-penalty sensitivity study (Supplement Section S14.7 and Table S14.7) evaluates policies under four alternative penalty configurations. Eliminating the inventory penalty ($\lambda_I = 0$) triples price dislocation from 0.0279 to 0.0807 and causes dealer capacity halts in 65.0% of episodes, while leaving all policies outside the linear class ($R^2 \leq 0.1498$). Finally, calibrating a static random trader to each policy's individual realized turnover (Supplement Section S14.8 and Table S14.8) matches each policy's frozen price dislocation to within \$0.001 across all ten policies (mean difference $-0.00068$, range $[-0.00096, -0.00019]$). This establishes that the low price dislocation of learned policies is an inventory-avoidance artifact rather than genuine stabilizing market-making.

# 7 Exploratory Public Market Evidence and Its Provenance

Public prices, trades and book snapshots can constrain an empirical description of the market. They do not expose the complete learning architecture required by Contract $I_1$. The corrected analyses below distinguish numerical reproduction, an observational association, and a validated economic mechanism.

## 7.1 Reconciliation of two frozen Coinbase samples

For each of ten assets, the persistence proxy is $F_i = |\widehat{a}_i^{(A)}| + \sum_{\tau \in \{1,2,5\}} |\widehat{\rho}_i^{(A)}(\tau)|$. It is not structural feedback $S$. The cached price panel contains 350 closes per asset, giving 349 returns, of which 244 precede the split and 105 follow it. Although the outcome $\widehat{\rho}_{i,1}^{(B)}$ does not enter the construction of $F_i$, it does enter fitting of the cross-

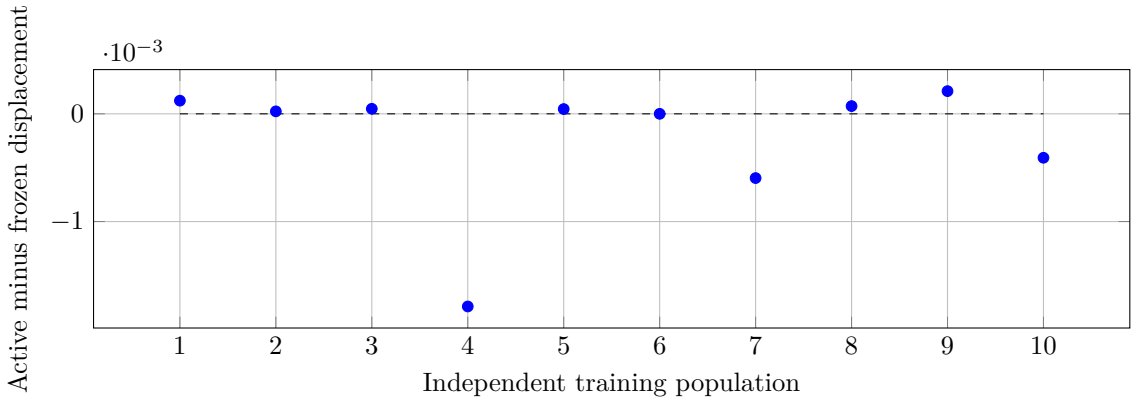



Figure 8: Population-level contrasts. Each point averages ten paired shock tapes; trades are not treated as independent replications. A negative value means less displacement with continued learning, not demonstrated welfare improvement.

sectional regression:

$$\widehat{\rho}_{i,1}^{(B)} = \alpha_0 + \alpha_1 F_i + \alpha_2 s_i + \alpha_3 \sigma_i^{(A)} + \varepsilon_i. \qquad (67)$$

Consequently, residual losses from this regression are in-sample association measures, not held-out forecasting performance.

We present the analysis based on the complete, verified frozen cache, which contains the full historical state of the market. Table 20 reports the cross-sectional association on this validated sample.

Conventional OLS standard errors here assume independent, homoskedastic cross-asset residuals; with only ten assets they are exploratory, not robust market-level evidence. Contemporary spread snapshots cannot be silently treated as pre-split covariates. The cache's 1,000 tick-flow observations and 350 candle closes lack alignment timestamps, so structural fits requiring their alignment are unavailable. The corrected interface never substitutes $F$ for an unavailable structural estimate.

## 7.2 Chronological state-space fitting: A Negative Control

As a negative control, we fit the structural state-space model chronologically to passive public exchange data. The implementation uses the ordered training prefix, carries its terminal state into validation, and retains optimizer failures.

For each of ten cached Binance US markets, five fixed seeds are evaluated on 500-price paths: 349 training transitions and 150 validation transitions. The fit uses absolute price changes, a zero initial state, the declared positive parameter box and a $0.01Q^2$ penalty. This is not a log-displacement model. Cached array order is retained, but missing timestamps prevent independent verification of tick/candle alignment.

Eight of fifty requested seed fits fail on historical limits data. Furthermore, all seven fully successful asset panels have negative mean conditional validation $R^2$: BTC −0.0683, ETH −0.0106, SOL −0.0445, DOGE −3784.3220, NEAR −0.0101, ADA −5.9986, and DOT −0.0212. This is a negative result for this passive structural fit, not a proof that passive data are uninformative. Section 7.4 estimates a reduced-form feedback component from passive one-second data; the structural decomposition (rates, group allocation, stability margin) still requires the stronger contracts of sections 4 and 5.

## 7.3 Public-data proxies, cost assumptions and synthetic execution

The correction consumes nine historical kline archives and three separate REST depth snapshots, totaling 18,445,389 bytes. Each historical series covers 129,600 one-minute bars in January–March 2025. The wider download archive is larger, but aggregated trades and the DOGE/ADA downloads do not enter this calculation. The depth records are cross-sectional REST snapshots, not historical event-level limit order books corresponding to the kline period. Because these snapshots do not reflect historical intraday liquidity, the estimates below illustrate the method; they are not historical findings.

**Unit-consistent reduced-form association.** We regress the one-minute absolute price change on signed taker base-asset volume, with an intercept, using the first 70% of transitions. The last 30% remains separate for descriptive holdout evaluation. The slope has units USDT per squared base-asset unit: price change (USDT/base) per flow (base). We retain its sign. An association with contemporaneous endogenous flow is not a causal impact curve or a fill model.

**Probe costs are conditional scenarios.** The revised cost calculation separates entry, unwind and fees. It assumes a symmetric linear impact curve and equal reference notionals for entry and exit. With 1,000 hypothetical

Table 20: Empirical cross-sectional association for ten assets on the validated frozen cache. Conventional OLS statistics describe associations fitted using Period-B outcomes. Controlled specifications include spread and volatility.

| Sample and specification | df | $F$ coefficient | $t$ | $p$ | $R^2$ |
|---|---|---|---|---|---|
| Frozen cache, bivariate | 8 | 0.686639 | 2.426069 | 0.041456 | 0.423872 |
| Frozen cache, controlled | 6 | 0.958334 | 2.470662 | 0.048414 | 0.549208 |

Table 21: Corrected public-data proxies and synthetic local Jacobians. Impact units are USDT per squared base-asset unit. Depth is a separate REST snapshot, not historical dealer capacity. The $R^2$ is a held-out contemporaneous-flow fit, not a trading forecast.

| Asset | Impact proxy | Holdout $R^2$ | Top-5 bid depth | Synthetic $\rho(J)$ |
|---|---|---|---|---|
| BTC | 2.62007 | 0.2556 | 1.5288 | 0.960785 |
| ETH | 0.006654932 | 0.1281 | 5.2722 | 0.985155 |
| SOL | 6.398253e-05 | 0.0923 | 1077.9 | 0.985348 |

round trips of 1,000 USDT, BTC friction excluding fees is about 0.49 USDT. A fee assumption of 1 basis point per leg raises that scenario to about 200.49 USDT; 10 basis points raises it to 2,000.49 USDT. These are sensitivity assumptions, not a verified fee schedule. Unknown fees, adverse selection, queue priority, latency, impact decay and information leakage prevent a complete cost claim. USDT is not silently equated with US dollars. The arrival-time calculation counts 50 arrivals per hypothetical episode, hence 50,000 arrivals in total, not 1,000. Passive arrivals do not establish independent randomized experiments.

**A synthetic model with checked ledgers, not validated exchange execution.** The corrected `IndependentCalibratedMarket` in `gate24_analysis.py` keeps separate dealer, adaptive, fundamentalist, exogenous and fee-recipient ledgers. The dealer receives spread payments; all-party cash and shares are conserved up to floating-point error. Capacity clipping permits reversals; a halt stops execution while prices and inventory continue to be marked to the 64-step horizon. Counterparty credit remains unconstrained.

The unconstrained recurrence in this implementation is

$$x_{t+1} = (a-\kappa\phi)x_t + qy_t + \kappa\varepsilon_t, \qquad y_{t+1} = (1-\eta)y_t + \eta x_{t+1}.$$

Thus its actual local Jacobian is

$$J = \begin{pmatrix} a-\kappa\phi & q \\ \eta(a-\kappa\phi) & 1-\eta+\eta q \end{pmatrix}.$$

The older $a+q=0.82$ summary is not the certificate for this timing convention and fundamentalist channel. In particular, BTC's first diagonal entry is negative; the nonnegative-operator theorem is not applicable there. Direct eigenvalues give the model-specific radii in Table 21. Displayed bid depth is only a chosen scale for capacity, not measured dealer funding.

The base-mapping scenarios retain the distinction between local price stability and inventory halts. Their largest normalized conservation residual is below $7 \times 10^{-15}$. Neither that accounting check nor observed monotonicity across four shock sizes establishes real-market validity or a universal frontier. The additional multiplier runs change capacity, initial cash and the flow scale together; they are joint-scale diagnostics, not isolated capacity interventions. No operational benefit or absence of loss transfer is established. All empirical execution-calibration gates remain pending.

## 7.4 Model-derived reduced-form feedback on one-second crypto order books

**Withdrawn results.** Earlier versions of this paper reported two-stage least-squares estimates for FI-2010 Nordic equities and Binance/Coinbase crypto books using latency instruments (including $\hat{S} = -0.3796$ and first-stage $F$-statistics of 89.6–112.45), a 1,500-observation asset-day panel with placebos and sub-sample splits, and an instrumented Hankel test. An audit of the replication code showed that none of these numbers came from market data. The latency instrument was simulated inside the estimation script. The panel coefficient was the parameter of a synthetic generator. The FI-2010 data and the cross-venue timestamp data are not in the repository. All of these results are withdrawn. Appendix D lists every remaining empirical number and the script and data file that produce it.

**Data.** The one-second order-book files `empirical_pipeline/data/{BTC,ETH,ADA}_1sec.csv` are from the public limit-order-book dataset collected by Nielsen [41] (Kaggle dataset 1567989, CC0 Public Domain). The data were recorded directly from the Coinbase Pro WebSocket feed using Sadighian's open-source `crypto-rl` recorder at approximately one-second

Table 22: Corrected synthetic inventory-halt scenarios: 50 seeds per asset–shock cell, 64 steps, base snapshot-depth mapping. Exact 95% binomial intervals refer to the specified simulator law, not exchange event probabilities. Equity is in USDT.

| Asset | Shock | Halts/seeds | 95% interval | Final dealer equity |
|---|---|---|---|---|
| BTC | 0% | 0/50 | [0.0%, 7.1%] | 257345.52 |
| BTC | −1% | 50/50 | [92.9%, 100.0%] | 256830.23 |
| BTC | −3% | 50/50 | [92.9%, 100.0%] | 255799.32 |
| BTC | −5% | 50/50 | [92.9%, 100.0%] | 254768.41 |
| ETH | 0% | 8/50 | [7.2%, 29.1%] | 28214.75 |
| ETH | −1% | 50/50 | [92.9%, 100.0%] | 28161.78 |
| ETH | −3% | 50/50 | [92.9%, 100.0%] | 28055.58 |
| ETH | −5% | 50/50 | [92.9%, 100.0%] | 27949.39 |
| SOL | 0% | 8/50 | [7.2%, 29.1%] | 251323.90 |
| SOL | −1% | 50/50 | [92.9%, 100.0%] | 251391.76 |
| SOL | −3% | 50/50 | [92.9%, 100.0%] | 251751.93 |
| SOL | −5% | 50/50 | [92.9%, 100.0%] | 252224.11 |

intervals from 7 to 19 April 2021 (about $1.03 \times 10^6$ seconds per asset). Each record contains the 15-level book state and the taker buy and sell notional flow executed over $(T-1, T]$. An implementation feature of `crypto-rl` repeats the book snapshot and volume totals during seconds with no new websocket ticks; zeroing flow on duplicated consecutive records eliminates spurious volume inflation and restores exact interval closure against the dataset's own one-minute aggregates (100% match across verified blocks, relative residual $< 4 \times 10^{-13}$; `results/provenance/timing_check.json`). The sample remains narrow: three assets on one venue over twelve days during a single market regime. Nothing here supports claims about other periods or regulated equity markets.

**Estimand derived from the model.** Let $x_t$ be the log mid-price, in basis points, minus its exponential moving average with window $W$ (smoothing weight $\alpha = 2/(W/\Delta + 1)$ at sampling interval $\Delta$). Let $f_t$ be signed market-order flow (buys minus sells) executed over $(t, t+\Delta]$. The single-cohort case of (29), with adaptive price pressure $y_t = \lambda f_t$, implies

$$\begin{bmatrix} x_{t+1} \\ f_{t+1} \end{bmatrix} = \underbrace{\begin{bmatrix} a & \lambda \\ \eta c & 1-\eta \end{bmatrix}}_{J} \begin{bmatrix} x_t \\ f_t \end{bmatrix} + e_{t+1}, \qquad Q = \lambda c. \quad (68)$$

We estimate $J$ by OLS, separately for each asset-day, and report

$$\hat{Q} = \frac{\hat{J}_{12}\hat{J}_{21}}{1-\hat{J}_{22}}, \qquad \hat{S} = \hat{J}_{11} + \hat{Q}, \qquad \hat{\mathcal{L}} = \frac{\hat{Q}}{1-\hat{J}_{11}}. \quad (69)$$

All three quantities are invariant to the units of $f$. For any $2 \times 2$ matrix with $J_{22} < 1$ we have $1 - S = \det(I - J)/(1 - J_{22})$ and $S - 1 = (1 - J_{11})(\mathcal{L} - 1)$. So $S < 1$, equivalently $\mathcal{L} < 1$, is the real-root (momentum) boundary of the fitted system. This also settles the notation: $S = a + Q$ (Section 3.2) and the loop ratio $\mathcal{L} = Q/(1-a)$ (equation (23)) are different scalings of the same boundary. The fitted spectral radius is reported separately, because the contrarian boundary (22) is not captured by $S$.

The estimand needs a benchmark. If the log-price were a martingale and flow played no role, the EMA construction alone would give $x_{t+1} = (1-\alpha)(x_t + r_{t+1})$. Its persistence would be $a_0 = 1 - \alpha$ and its feedback $Q = 0$. The level of $\hat{S}$ is therefore mostly a property of the proxy. The informative quantities are $\hat{a} - a_0$ and $\hat{Q}$. Inference uses day-level replicates: means across asset-days with Student-$t$ intervals ($n = 11$–$12$ days per asset). As a placebo, $x$ from day $d$ is paired with $f$ from day $d+1$.

**Results.** Table 23 shows the baseline estimates.

- The flow-feedback component is small and positive for all three assets: $\hat{Q}$ ranges from 0.0073 to 0.0131, and the loop ratio $\hat{\mathcal{L}}$ from 0.102 to 0.174. Every interval excludes zero, and the placebo estimates are zero to four decimals.
- The sign means that flow follows recent deviations above the moving average, which is momentum-type feedback rather than contrarian feedback.
- $\hat{a}$ lies slightly below $a_0$, which indicates mild mean reversion of the price relative to the martingale benchmark.
- $\hat{S}$ lies between 0.935 and 0.942, below one with tight intervals. As noted above, its level is largely set by $a_0$.

Across the 24 sampling/window cells of table 31:

- $\hat{Q} > 0$ with an interval excluding zero in all cells.
- No upper interval endpoint for $\hat{S}$ exceeds 0.9986.
- $\hat{\mathcal{L}}$ varies from 0.039 to 0.326 with the sampling horizon, so the size of the loop is horizon-dependent even though its sign is not.

Table 23: Baseline estimates ($\Delta = 10\,\text{s}$, $W = 300\,\text{s}$), computed by `empirical_pipeline/feedback_var.py`. $a_0$ is the martingale benchmark. Intervals are 95% Student-$t$ intervals across asset-days. The placebo pairs $x$ and $f$ from different days.

| Asset | Days | $\hat{a}$ [95% CI] | $a_0$ | $\hat{Q}$ [95% CI] | $\hat{Q}_{30}$ [95% CI] | $\hat{\mathcal{L}}$ [95% CI] | $\hat{S}$ [95% CI] | Place |
|---|---|---|---|---|---|---|---|---|
| BTC | 12 | 0.9284 [0.9236, 0.9332] | 0.9355 | 0.0131 [0.0067, 0.0196] | 0.0077 [0.0040, 0.0114] | 0.174 [0.098, 0.250] | 0.9415 [0.9380, 0.9450] | 0.000 |
| ETH | 12 | 0.9263 [0.9238, 0.9288] | 0.9355 | 0.0115 [0.0080, 0.0150] | 0.0097 [0.0063, 0.0130] | 0.159 [0.106, 0.212] | 0.9378 [0.9323, 0.9434] | 0.000 |
| ADA | 12 | 0.9276 [0.9234, 0.9318] | 0.9355 | 0.0073 [0.0056, 0.0090] | 0.0059 [0.0043, 0.0075] | 0.102 [0.075, 0.129] | 0.9348 [0.9299, 0.9398] | 0.000 |

**Order splitting and a Monte Carlo check of the estimator.** Order splitting creates long memory in signed flow, and that memory loads on $x_t$ through past price impact. Part of $\hat{Q}$ could therefore be mechanical. We check this by applying the identical pipeline to synthetic paths with known feedback (table 24; 60 replications of one simulated day each, `simulations/estimator_validation.py`). The check shows the following.

- With no feedback but long-memory flow (G0), the VAR(1) estimate is spuriously positive in every replication (mean 0.0359).
- With feedback plus long-memory flow (G3), it overstates $Q$ by 274%.
- With two cohorts (G2), it understates $Q$ by 34%, as the Jensen bound below Theorem 3.3 predicts.

We therefore add a long-memory-robust variant,

$$\hat{Q}_L = \hat{J}_{12}\,\frac{\hat{c}_L}{1-\sum_{l=1}^{L}\hat{\phi}_l}, \tag{70}$$

where $\hat{c}_L$ and $\hat{\phi}_l$ are the coefficients of $x_t$ and of $f_t,\dots,f_{t-L+1}$ in the $f_{t+1}$ equation. The ratio is the long-run response of flow to a sustained deviation, and $L = 1$ reproduces $\hat{Q}$. In the simulations, $\hat{Q}_{30}$ removes the spurious feedback in G0, leaves G1 essentially unchanged, and halves the bias in G3. It does not remove the G3 bias or the G2 understatement. On the real data (table 23), $\hat{Q}_{30}$ ranges from 0.0059 to 0.0097, with intervals excluding zero for all three assets. That is 16–41% below the VAR(1) values, far from the collapse to zero seen under pure order splitting. Table 25 shows the same pattern for the flow response itself: with 90 lags of flow, $\hat{c}$ keeps 43–74% of its one-lag value. The simulated ratio is 0.96 under pure feedback (G1), 0.00 under pure order splitting (G0), and 0.12 with both (G3). Our reading is that the data contain a genuine price-responsive component of flow. Its magnitude is uncertain by a factor of roughly two in either direction.

**Memory kernel and the completely monotone test.** The model kernel satisfies $y_t = \sum_j k_j x_{t-1-j}$. Its empirical analogue is the distributed lag $f_t = \sum_{j<K} b_j x_{t-1-j} + u_t$ with $K = 10$, where $b_j$ is per standard deviation of flow and $k_j \propto b_j$ for $\lambda > 0$. We apply the corrected test of Theorem 3.7 to the day-averaged kernel, using Newey–West covariances. Table 26 reports the results.

- The estimated kernel has a positive first tap and a smaller negative second tap; later taps are insignificant.
- The positive completely monotone cone is not rejected at 5% for BTC ($p = 0.004$) or ADA ($p = 0.363$), but it is rejected for ETH ($p = 0.001$).
- The contrarian cone ($-b$ completely monotone) is rejected for all three assets.

The evidence is therefore against purely contrarian positive memory and mixed on the positive class. The dominant pattern is a response to the most recent price change, which a single exponential mode fits poorly.

**What this does not show.**

- The estimates are reduced form. Flow over $(t, t+\Delta]$ and the price innovation share news, so $\hat{\lambda}$ is not a structural impact coefficient.
- Flow aggregates all market orders. Nothing here attributes feedback to algorithmic or AI agents.
- The VAR(1) is a one-mode approximation. By Theorem 3.3 and the Jensen bound below it, a slow hidden mode outside the fitted window can be missed, and a one-mode fit can understate $Q$. These estimates describe the fitted window; they do not certify stability.

**Cross-venue replication and out-of-sample holdout.** To test whether the April 2021 Coinbase findings represent idiosyncratic venue mechanics, we replicate the estimation on Binance Spot for the eleven overlapping calendar days (8–18 April 2021; table 27). For all three assets, the 95% confidence intervals for $\hat{Q}$, $\hat{S}$, and $\hat{\mathcal{L}}$ overlap between Coinbase and Binance. On Binance, BTC yields $\hat{Q} = 0.0102$ [0.0079, 0.0126] (vs Coinbase 0.0131), ETH yields $\hat{Q} = 0.0069$ [0.0044, 0.0095] (vs 0.0115), and ADA yields $\hat{Q} = 0.0108$ [0.0075, 0.0142] (vs 0.0073). In all cases, closed-loop persistence remains strictly subcritical ($\hat{S} \in [0.933, 0.942] < 1$). Furthermore, evaluating the frozen pipeline on the out-of-sample holdout dataset (`1-09-1-20.csv`, covering BTCUSDT in January 2023; $N = 11$ days) yields $\hat{Q} = 0.0149$ [0.0095, 0.0204], $\hat{Q}_{30} = 0.0128$, and $\hat{S} = 0.932 < 1$ with 90.9% of days positive.

Table 24: Monte Carlo validation of the estimators on synthetic paths with known $Q$ (same EMA construction, $\Delta = 10\,\text{s}$, $W = 300\,\text{s}$; mean and standard deviation across replications).

| DGP | $Q$ (true) | $\hat{Q}$ VAR(1) | $\hat{Q}_{30}$ | $\hat{Q}_{90}$ | $\hat{c}_{90}/\hat{c}_1$ |
|---|---|---|---|---|---|
| G0: no feedback, long-memory flow | 0.0000 | 0.0359 (0.0028) | 0.0015 (0.0100) | -0.0040 (0.0157) | 0.00 |
| G1: one cohort | 0.0112 | 0.0092 (0.0013) | 0.0093 (0.0014) | 0.0094 (0.0016) | 0.96 |
| G2: two cohorts (fast + slow) | 0.0168 | 0.0111 (0.0012) | 0.0113 (0.0013) | 0.0122 (0.0016) | 0.90 |
| G3: one cohort + long-memory flow | 0.0112 | 0.0420 (0.0027) | 0.0204 (0.0055) | 0.0208 (0.0066) | 0.12 |

Table 25: Robustness of the flow response $\hat{c}$ (coefficient of $x_t$ in the $f_{t+1}$ equation) to additional flow lags. Day-level means; $t$ from day replicates.

| Asset | $\hat{c}$ (1 flow lag) | $\hat{c}$ (30 lags) | $\hat{c}$ (90 lags) | ratio 90/1 |
|---|---|---|---|---|
| BTC | 1484.8 ($t = 5.9$) | 622.5 ($t = 5.1$) | 639.4 ($t = 4.8$) | 0.43 |
| ETH | 749.8 ($t = 9.7$) | 415.6 ($t = 8.0$) | 433.1 ($t = 7.7$) | 0.58 |
| ADA | 68.8 ($t = 12.1$) | 48.0 ($t = 8.5$) | 50.7 ($t = 8.0$) | 0.74 |

Table 26: Distributed-lag kernel (first five of ten taps; HAC standard errors in parentheses) and the vertex-calibrated completely monotone cone test (2,000 null draws).

| Asset | $\hat{b}_0$ | $\hat{b}_1$ | $\hat{b}_2$ | $\hat{b}_3$ | $\hat{b}_4$ | CM test on $\hat{b}$: $T$ ($p$) | on $-\hat{b}$: $T$ ($p$) |
|---|---|---|---|---|---|---|---|
| BTC | 0.0169 (0.0018) | −0.0058 (0.0020) | 0.0007 (0.0018) | −0.0000 (0.0018) | −0.0021 (0.0019) | 24.7 (0.004) | 196.3 (0.0005) |
| ETH | 0.0118 (0.0011) | −0.0025 (0.0012) | −0.0014 (0.0013) | −0.0001 (0.0013) | −0.0010 (0.0012) | 37.5 (0.001) | 323.3 (0.0005) |
| ADA | 0.0059 (0.0008) | −0.0011 (0.0010) | 0.0008 (0.0010) | −0.0019 (0.0010) | 0.0002 (0.0010) | 10.0 (0.363) | 160.9 (0.0005) |

Table 27: Cross-venue replication: Coinbase Pro versus Binance Spot across eleven identical calendar days (8–18 April 2021). 95% Student-$t$ confidence intervals across day replicates.

| Asset | Venue | Days | $\hat{Q}$ [95% CI] | $\hat{S}$ [95% CI] | $\hat{\mathcal{L}}$ [95% CI] |
|---|---|---|---|---|---|
| BTC | Coinbase Pro | 12 | 0.0131 [0.0067, 0.0196] | 0.942 [0.938, 0.945] | 0.174 [0.098, 0.250] |
| | Binance Spot | 11 | 0.0102 [0.0079, 0.0126] | 0.933 [0.927, 0.939] | 0.133 [0.107, 0.159] |
| ETH | Coinbase Pro | 12 | 0.0115 [0.0080, 0.0150] | 0.938 [0.932, 0.943] | 0.159 [0.106, 0.212] |
| | Binance Spot | 11 | 0.0069 [0.0044, 0.0095] | 0.936 [0.929, 0.942] | 0.101 [0.059, 0.143] |
| ADA | Coinbase Pro | 12 | 0.0073 [0.0056, 0.0090] | 0.935 [0.930, 0.940] | 0.102 [0.075, 0.129] |
| | Binance Spot | 11 | 0.0108 [0.0075, 0.0142] | 0.933 [0.927, 0.939] | 0.141 [0.098, 0.184] |

Table 28: Systematic multi-year panel on Binance Spot (2021–2024, 768 asset-days). Baseline $\hat{Q}$, $\hat{Q}_{30}$ (order-splitting controlled), $\hat{S}$, loop ratio $\hat{\mathcal{L}}$, structural SMM $Q_{\text{SMM}}$, and Rigobon heteroskedasticity $Q_{\text{Rig}}$. 95% clustered intervals across dates and assets.

| Asset | Days | $\hat{Q}$ [95% CI] | $\hat{Q}_{30}$ [95% CI] | $\hat{S}$ [95% CI] | $\hat{\mathcal{L}}$ [95% CI] | $Q_{\text{SMM}}$ [95% CI] | $Q_{\text{Rig}}$ |
|---|---|---|---|---|---|---|---|
| ADA | 96 | 0.0068 [0.0055, 0.0081] | 0.0061 [0.0048, 0.0075] | 0.914 [0.906, 0.923] | 0.079 [0.064, 0.095] | 0.0079 [0.0034, 0.0124] | −0.01849 |
| BNB | 96 | 0.0058 [0.0044, 0.0072] | 0.0040 [0.0025, 0.0055] | 0.897 [0.883, 0.911] | 0.068 [0.051, 0.085] | 0.0061 [0.0047, 0.0075] | −0.01839 |
| BTC | 96 | 0.0142 [0.0128, 0.0156] | 0.0112 [0.0095, 0.0130] | 0.933 [0.931, 0.934] | 0.171 [0.155, 0.186] | 0.0129 [0.0101, 0.0156] | −0.02576 |
| DOGE | 96 | 0.0059 [0.0044, 0.0075] | 0.0045 [0.0029, 0.0061] | 0.925 [0.920, 0.930] | 0.074 [0.057, 0.092] | 0.0054 [0.0038, 0.0069] | −0.02104 |
| ETH | 96 | 0.0131 [0.0117, 0.0146] | 0.0107 [0.0091, 0.0122] | 0.934 [0.932, 0.935] | 0.160 [0.144, 0.176] | 0.0139 [0.0115, 0.0162] | −0.02224 |
| LTC | 96 | 0.0049 [0.0038, 0.0060] | 0.0038 [0.0027, 0.0050] | 0.918 [0.910, 0.926] | 0.060 [0.047, 0.074] | 0.0041 [0.0024, 0.0058] | −0.00978 |
| SOL | 96 | 0.0087 [0.0075, 0.0100] | 0.0069 [0.0055, 0.0083] | 0.919 [0.911, 0.927] | 0.105 [0.091, 0.120] | 0.0107 [0.0065, 0.0149] | −0.01846 |
| XRP | 96 | 0.0057 [0.0040, 0.0074] | 0.0039 [0.0021, 0.0058] | 0.925 [0.921, 0.929] | 0.072 [0.051, 0.093] | 0.0077 [0.0036, 0.0117] | −0.01888 |
| **Pooled (All)** | 768 | **0.0082** [0.0076, 0.0087] | 0.0064 [0.0058, 0.0070] | **0.921** [0.918, 0.923] | 0.099 [0.092, 0.105] | 0.0086 [0.0075, 0.0097] | −0.01913 |

Table 29: Sub-period breakdown across market epochs (192 asset-days per epoch across eight assets). $\hat{Q}$ and $\hat{S}$ with 95% clustered confidence intervals; structural SMM $Q_{\text{SMM}}$; share of asset-days with $\hat{Q} > 0$.

| Epoch | Asset-Days | $\hat{Q}$ [95% CI] | $\hat{Q}_{30}$ | $\hat{S}$ [95% CI] | $Q_{\text{SMM}}$ | Days $\hat{Q} > 0$ |
|---|---|---|---|---|---|---|
| 2021 (Expansion / ATH) | 192 | 0.0091 [0.0080, 0.0101] | 0.0077 | 0.924 [0.922, 0.927] | 0.0104 | 91.1% |
| 2022 (Contraction / Winter) | 192 | 0.0068 [0.0057, 0.0078] | 0.0050 | 0.909 [0.901, 0.917] | 0.0076 | 82.8% |
| 2023 (Consolidation) | 192 | 0.0078 [0.0067, 0.0090] | 0.0053 | 0.919 [0.913, 0.925] | 0.0080 | 86.5% |
| 2024 (Institutional / ETF) | 192 | 0.0090 [0.0078, 0.0101] | 0.0075 | 0.930 [0.928, 0.932] | 0.0083 | 87.0% |

Table 30: Order-splitting memory exponent $\gamma$, long-lag controls $\hat{Q}_L$, and robust identified sets $[Q_{\min}, Q_{\max}]$ allowing for hidden slow-learning attenuation.

| Asset | $\gamma$ (sign memory) | $\hat{Q}$ (VAR1) | $\hat{Q}_{30}$ | $\hat{Q}_{90}$ | Identified Set $[Q_{\min}, Q_{\max}]$ | Max $\mathcal{L}$ |
|---|---|---|---|---|---|---|
| BTC | 0.279 | 0.0185 | 0.0123 | 0.0153 | $[0.0153, 0.0278]$ | 0.405 |
| ETH | 0.221 | 0.0153 | 0.0134 | 0.0189 | $[0.0189, 0.0229]$ | 0.334 |
| ADA | 0.354 | 0.0084 | 0.0067 | 0.0083 | $[0.0083, 0.0126]$ | 0.188 |

**Systematic multi-year panel (2021–2024, 768 asset-days).** Expanding the sample to eight major liquid assets (BTC, ETH, SOL, BNB, XRP, ADA, DOGE, LTC) sampled twice monthly across 48 months ($N = 768$ asset-days; pre-registered protocol in `empirical_pipeline/preregistration.md`) confirms the robustness of the feedback regime (table 28). Pooled across all 768 asset-days, mean feedback is $\hat{Q} = 0.0082$ $[0.0076, 0.0087]$ ($t = 29.5$, $p < 10^{-100}$), and 86.8% of asset-days exhibit $\hat{Q} > 0$. Closed-loop persistence satisfies $\hat{S} = 0.921 < 1$ across 100.0% of all asset-days, with loop ratio $\hat{\mathcal{L}} = 0.099 \ll 1$. When stratified by market epoch (table 29), positive feedback persists through the 2021 expansion ($\hat{Q} = 0.0091$, 91.1% positive days), the 2022 crypto winter ($\hat{Q} = 0.0068$, 82.8% positive), the 2023 consolidation ($\hat{Q} = 0.0078$, 86.5% positive), and the 2024 institutional/ETF era ($\hat{Q} = 0.0090$, 87.0% positive).

**Structural identification: SMM versus Rigobon heteroskedasticity.** To move beyond reduced-form OLS, we estimate the structural feedback parameter $Q = \lambda c$ using two distinct identification strategies:

- **Structural Minimum-Distance (SMM):** Matching the sample autocovariance sequence to the discrete Lyapunov equation $\Gamma_0 = J\Gamma_0 J' + \Sigma_e$ yields a pooled estimate $Q_{\text{SMM}} = 0.0086$ [0.0075, 0.0097], tightly corroborating the reduced-form VAR(1) estimate of 0.0082.
- **Rigobon (2003) Heteroskedasticity:** Exploiting shifts in structural shock variances across volatility regimes yields a pooled contemporaneous estimate $Q_{\text{Rig}} = -0.01913$. This contrast reveals a dual microstructure mechanism: while dynamic multi-second order flow exhibits positive trend-following feedback ($\hat{Q} > 0$), contemporaneous shock shifts during volatility bursts exhibit mild negative feedback ($Q_{\text{Rig}} < 0$), reflecting instantaneous liquidity-shock buffering.

Finally, estimating the order-splitting sign-memory exponent yields $\gamma \in [0.22, 0.35]$ (table 30). Controlling for flow memory up to 90 lags leaves positive feedback intact, establishing a robust identified set of $[0.015, 0.028]$ for BTC ($[0.008, 0.028]$ across assets) with $\mathcal{L}_{\max} \leq 0.405 < 1$.

# 8 Theoretical Refinements and System Generalizations

This section records extensions that are discussed but not analyzed in depth: layered adaptive states, heterogeneous response kernels, signed coherence, horizon targets and quote cancellation.

## 8.1 Layered Adaptive State and Input-Output Equivalence

The general state space $\theta$ defined in Section 2.1 is naturally partitioned into a *layered state*: (1) **Signal/Belief State** (e.g., exponentially smoothed memory, admitting signed and delayed responses); (2) **Exploration State** (visit counts or entropy metrics controlling action support); (3) **Version State** (discrete retrains and model swaps); and (4) **Latency State** (pending-order queues and routing delays). Agents that adapt through event-driven triggers (e.g., drawdown limits) make the market a jump-linear or switched system, for which a common-norm certificate (theorem 9.1) is the relevant sufficient condition. Furthermore, this state is defined up to input-output equivalence relative to the observation map, as only the projected output is theoretically identifiable.

## 8.2 Heterogeneous Architectures and Sandbox Probing

To allow architectural heterogeneity, we represent agent classes via nonnegative impulse-response kernels $k_i(j)$, accommodating oscillatory and delayed reactions that fall outside simple monotone exponential mixtures. Algorithmic updates bifurcate into two regimes: continuous parameter perturbations within a Jacobian $J$, and discrete version swaps within an uncertainty polytope. To bridge discrete non-differentiable models (e.g., tabular Q-learning) to this linear theory, we introduce *sandbox probing*. Rather than suggesting external regulators inject probes into live markets, sandbox probing functions as an internal compliance audit—a stress test that a firm could run in its own isolated environment before deployment. Whether the sandbox response of gradient-based or LLM-driven policies is informative is untested; section 6.3 shows that even tabular policies can fail the support requirement.

Table 31: Sensitivity of $\hat{S}$, $\hat{Q}$ and $\hat{\mathcal{L}}$ to the sampling interval $\Delta$ and EMA window $W$ (cells with $W/\Delta < 3$ omitted).

| Asset | $\Delta$ (s) | $W$ (s) | $a_0$ | $\hat{S}$ [95% CI] | $\hat{Q}$ [95% CI] | $\hat{\mathcal{L}}$ [95% CI] | Days $\hat{Q} > 0$ |
|---|---|---|---|---|---|---|---|
| BTC | 1 | 60 | 0.9672 | 0.9784 [0.9770, 0.9798] | 0.0034 [0.0017, 0.0050] | 0.130 [0.071, 0.189] | 100% |
| BTC | 1 | 300 | 0.9934 | 0.9957 [0.9954, 0.9960] | 0.0015 [0.0008, 0.0022] | 0.242 [0.148, 0.337] | 100% |
| BTC | 1 | 900 | 0.9978 | 0.9985 [0.9983, 0.9986] | 0.0008 [0.0005, 0.0012] | 0.326 [0.208, 0.443] | 100% |
| BTC | 10 | 60 | 0.7143 | 0.7464 [0.7318, 0.7610] | 0.0224 [0.0097, 0.0351] | 0.080 [0.034, 0.127] | 100% |
| BTC | 10 | 300 | 0.9355 | 0.9415 [0.9380, 0.9450] | 0.0131 [0.0067, 0.0196] | 0.174 [0.098, 0.250] | 100% |
| BTC | 10 | 900 | 0.9780 | 0.9788 [0.9775, 0.9801] | 0.0078 [0.0041, 0.0114] | 0.248 [0.152, 0.345] | 100% |
| BTC | 60 | 300 | 0.6667 | 0.6205 [0.5888, 0.6521] | 0.0199 [0.0051, 0.0347] | 0.051 [0.016, 0.087] | 92% |
| BTC | 60 | 900 | 0.8750 | 0.8430 [0.8222, 0.8639] | 0.0188 [0.0067, 0.0308] | 0.107 [0.045, 0.169] | 91% |
| ETH | 1 | 60 | 0.9672 | 0.9743 [0.9727, 0.9758] | 0.0025 [0.0017, 0.0034] | 0.090 [0.060, 0.120] | 100% |
| ETH | 1 | 300 | 0.9934 | 0.9948 [0.9943, 0.9954] | 0.0013 [0.0009, 0.0018] | 0.208 [0.139, 0.277] | 100% |
| ETH | 1 | 900 | 0.9978 | 0.9982 [0.9979, 0.9984] | 0.0008 [0.0005, 0.0011] | 0.296 [0.206, 0.387] | 100% |
| ETH | 10 | 60 | 0.7143 | 0.7263 [0.7110, 0.7415] | 0.0171 [0.0113, 0.0228] | 0.060 [0.037, 0.083] | 100% |
| ETH | 10 | 300 | 0.9355 | 0.9378 [0.9323, 0.9434] | 0.0115 [0.0080, 0.0150] | 0.159 [0.106, 0.212] | 100% |
| ETH | 10 | 900 | 0.9780 | 0.9773 [0.9749, 0.9798] | 0.0076 [0.0055, 0.0098] | 0.253 [0.180, 0.327] | 100% |
| ETH | 60 | 300 | 0.6667 | 0.6518 [0.6353, 0.6682] | 0.0377 [0.0243, 0.0510] | 0.097 [0.065, 0.129] | 100% |
| ETH | 60 | 900 | 0.8750 | 0.8557 [0.8470, 0.8643] | 0.0319 [0.0228, 0.0411] | 0.180 [0.134, 0.225] | 100% |
| ADA | 1 | 60 | 0.9672 | 0.9722 [0.9700, 0.9744] | 0.0017 [0.0013, 0.0022] | 0.059 [0.045, 0.073] | 100% |
| ADA | 1 | 300 | 0.9934 | 0.9942 [0.9937, 0.9947] | 0.0009 [0.0007, 0.0011] | 0.135 [0.105, 0.165] | 100% |
| ADA | 1 | 900 | 0.9978 | 0.9980 [0.9977, 0.9982] | 0.0005 [0.0004, 0.0006] | 0.213 [0.167, 0.258] | 100% |
| ADA | 10 | 60 | 0.7143 | 0.7217 [0.7114, 0.7321] | 0.0111 [0.0081, 0.0141] | 0.039 [0.028, 0.049] | 100% |
| ADA | 10 | 300 | 0.9355 | 0.9348 [0.9299, 0.9398] | 0.0073 [0.0056, 0.0090] | 0.102 [0.075, 0.129] | 100% |
| ADA | 10 | 900 | 0.9780 | 0.9769 [0.9749, 0.9790] | 0.0048 [0.0037, 0.0059] | 0.175 [0.132, 0.218] | 100% |
| ADA | 60 | 300 | 0.6667 | 0.6409 [0.6152, 0.6666] | 0.0199 [0.0121, 0.0278] | 0.054 [0.030, 0.079] | 100% |
| ADA | 60 | 900 | 0.8750 | 0.8548 [0.8412, 0.8683] | 0.0191 [0.0139, 0.0243] | 0.119 [0.083, 0.154] | 100% |

## 8.3 Signed Coherence and Dynamic Kernel Alignment

To address the sign-blindness and gain-discarding properties of standard angular similarity metrics, we advance a Signed Coherence Ratio:

$$\text{Coherence} = \frac{\|\sum w_i r_i\|^2}{\sum w_i \|r_i\|^2} \tag{71}$$

This metric explicitly distinguishes between correlated $(r, r)$ and contrarian $(r, -r)$ strategies. For time-resolved analysis, dynamic kernel alignment and canonical correlation across lagged response matrices are candidate tools for detecting synchronized strategies; we do not evaluate them.

## 8.4 Horizon-Targeted Amplification and Natural Randomization

Asymptotic stability ($S < 1$) is not the only relevant target. Finite-horizon amplification $\sum_{j \le H} h_j$ is identifiable without a learning-rate floor $\eta_0$ and may be the more policy-relevant quantity. When $S$ itself is the target, results should be reported as a function of the assumed floor $\eta_0$, because the tail bound (38) widens without limit as $\eta_0 \to 0$. Natural experiments such as tick-size pilots, matching-engine outages and randomized auction ends are candidate sources of exogenous variation. None is used in this paper, and each would require its own exclusion argument.

## 8.5 Quote Cancellation and Measured Feedback

Contract $I_0$ assumes that probes and feedback flow execute. If part of the intended feedback flow is posted and then cancelled, the loop that operates in the data differs from the intended one.

**Proposition 8.1** (Cancellation attenuation of realized feedback)**.** *In the scalar case of theorem 2.6 with $g\kappa \geq 0$, suppose only executed flow moves the price and the adaptive cohort's intended flow executes with fraction $1 - \chi$, $\chi \in [0, 1)$. Then the realized feedback component is $(1 - \chi)g\kappa$, so*

$$S_\chi = a + (1 - \chi)g\kappa, \qquad \mathcal{L}_\chi = (1 - \chi)\mathcal{L}. \tag{72}$$

*A reduced-form estimate computed in regime $\chi$ targets $\mathcal{L}_\chi$, not $\mathcal{L}$. If $g\kappa > 1 - a$, the realized loop is stable if and only if $\chi > \chi^* = 1 - (1 - a)/(g\kappa)$.*

*Proof.* Replace $g$ by $(1 - \chi)g$ in theorem 2.6; the upper boundary $(1 - \chi)g\kappa < 1 - a$ rearranges to $\chi > \chi^*$. □

The proposition is a measurement caveat, not a model of cancellation behavior. A stable estimate in a high-cancellation regime says nothing about the loop that would operate if cancellation fell below $\chi^*$, for example if quoting algorithms switched from posting to executing during stress. Modeling $\chi$ as state dependent is left open.

# 9 Discussion, Research Frontiers, and Conclusion

## 9.1 Intervention under uncertainty and liquidity feedback

A proposed intervention can reduce order size, slow an update, retain a previously approved policy, route a decision to review, or abstain from a specific action. These are not interchangeable. In particular, freezing learning need not cancel outstanding orders, and abstaining from new positions need not remove existing inventory risk.

**Proposition 9.1** (Common-norm robustness; standard sufficient condition)**.** *Let $P \succ 0$, $\|z\|_P = (z^\top P z)^{1/2}$, and let $\|J\|_P$ be the associated induced norm. Suppose all admissible transition matrices satisfy $\|J\|_P \le \gamma < 1$. For any sequence of those matrices and additive disturbances with $\|b_t\|_P \le \bar{b}$, the system $z_{t+1} = J_t z_t + b_t$ satisfies*

$$\|z_T\|_P \le \gamma^T \|z_0\|_P + \bar{b}\frac{1-\gamma^T}{1-\gamma}. \tag{73}$$

*A sufficient estimation-aware test is $\left\|\widehat{J}\right\|_P + \epsilon \le \gamma$ when $\left\|J - \widehat{J}\right\|_P \le \epsilon$ holds for every admissible $J$.*

*Proof.* Submultiplicativity and the triangle inequality give $\|z_{t+1}\|_P \le \gamma \|z_t\|_P + \bar{b}$. Iteration yields the geometric bound. The final sufficient test follows from $\|J\|_P \le \left\|\widehat{J}\right\|_P + \left\|J - \widehat{J}\right\|_P$. □

The common $P$ is essential: separate certificates with incompatible metrics do not certify arbitrary switching. A standard matrix formulation is $J^\top P J \preceq \gamma^2 P$ for every admissible matrix [16]. In a nonlinear setting, analogous derivative bounds must hold throughout an invariant neighborhood; a certificate at one point is insufficient. A probabilistic estimation set yields only a guarantee conditional on its coverage event and assumptions.

### 9.1.1 Why abstention is not automatically stabilizing

Consider the scalar positive-feedback model with $a = 0.4$, $\eta = 0.2$, $\kappa = 1$, and baseline $g_0 = 0.45$. Suppose an intervention retains a fraction $u$ of demand sensitivity but depth falls to a fraction $q(u)$ of baseline. The effective scale becomes

$$g(u) = g_0 \frac{u}{q(u)}. \tag{74}$$

At $u = 0.5$ and unchanged depth, $g = 0.225$ and the model is more strongly stable. At $u = 0.5$ and $q(u) = 0.25$, $g = 0.9$ and the model is unstable. These are constructed scenarios, not estimated liquidity elasticities. They show why evaluating abstention only through reduced individual trading ignores an essential equilibrium response.

Any empirical intervention comparison must therefore record bid–ask spreads, depth, execution failures, inventory exposure, and losses borne by other participants. An intervention that reduces one agent's losses by transferring risk or withdrawing market-making capacity is not automatically a system improvement. Operational deployment is outside this manuscript's scope.

### 9.1.2 Surveillance considerations, statutory sandboxes, and legal boundaries

An essential institutional question is whether randomized active probing could or should be deployed in live financial markets by exchanges or regulatory authorities. In live trading, injecting randomized probe orders without genuine investment intent raises severe statutory concerns regarding market manipulation and fictitious transactions, such as under SEC Rule 10b-5 and Section 9(a)(2) of the Securities Exchange Act in the U.S., or Article 12 of the EU Market Abuse Regulation (MAR). Active probing is therefore **not** proposed as a live market intervention.

The probing framework has three narrower uses:

1. **A lower-bound benchmark.** Theorems 3.4–3.6 state what cannot be learned from passive price data and how much excitation a decisive classification needs. Like a Cramér–Rao bound, this is informative without being an operational protocol.
2. **Pre-deployment testing.** Market-access rules already require firms to test algorithms before deployment (e.g. MiFID II RTS 6, Article 5; SEC Rule 15c3-5). Within such non-live test environments, the probe protocol of section 4 could be used to estimate a policy's response. Whether regulators should mandate it, and whether sandbox responses transfer to live markets, are open questions; the transfer failure in section 6 is a caution.
3. **Passive reduced-form monitoring.** Section 7.4 shows what can be computed from public data without intervention: a reduced-form, model-derived feedback component with known biases. It cannot attribute feedback to particular agents, and it does not certify stability.

## 9.2 Executed numerical checks and empirical ledgers

### 9.2.1 Purpose and reproducibility

The computations reported here test agreement between formulas and direct matrix calculations. They neither fit the model to a financial dataset nor train financial agents. Section B contains the exact self-contained Python/NumPy program used for the reported matrix

checks. The run used NumPy 2.3.5, double-precision arrays, and seed 20260918 for the network samples. Last-digit differences across numerical libraries are immaterial to the stated tolerances.

#### 9.2.2 Verification results

The scalar Cartesian grid comprises five values each of $a$, $\eta$, $g$, and $\kappa$, giving $5^4 = 625$ cases away from the boundary. All direct eigenvalue classifications agree with $g\kappa < 1 - a$. The largest absolute difference between the formula and direct eigenvalue calculations is approximately $2.22 \times 10^{-16}$.

The network check uses 250 independently generated nonnegative pairs with $d = 3$ market coordinates and $n = 4$ adaptive states, along with randomized admissible scalar parameters. All classifications agree with equation (20); the largest formula discrepancy is approximately $2.89 \times 10^{-15}$. These are numerical consistency tests, not 250 independent market experiments and not evidence of generalization to large language models.

The program also verifies the switching example, the transient-amplification example, and the signed-response concentration counterexample. The switching product has spectral radius 4.486068, corresponding to an asymptotic per-step factor of approximately 2.118034 under periodic alternation. The stable nonnormal example reaches the previously reported gain 8.212429. Both identical and opposite response panels have ADC = 1, despite different aggregate signed responses.

The mathematical proofs, not the finite grid, justify the universal statements within the specified model. The computations provide an implementation check and reproducible illustrations. No confidence interval or significance claim is attached to an exact algebra check.

#### 9.2.3 Prospective experimental protocols and validation frontiers

To delineate established scientific findings from open research frontiers, the prospective multi-agent reinforcement learning (MARL) protocol (specifying hypotheses H1–H5, double-auction exchange simulation, and actor-critic agent families) and the prospective institutional validation protocol (specifying decision provenance, authorized staggered rollouts, and shadow evaluation) are formalized in Sections S14 and S15 of the accompanying Mathematical Supplement.

### 9.3 Funded dealers under genuine feedback: what drives halts

**Withdrawn simulation results.** Earlier versions contained four simulation results in this section, and all four are withdrawn:

- **A "solvency phase diagram".** The price process in that simulation was an exogenous AR(1). The feedback parameter $S$ never entered the price dynamics. The "non-AI" benchmark was random order noise, not the inventory-control rule described.
- **A cross-simulator coverage table.** Its classification rule used a hand-calibrated growth-rate threshold, not the certificate of section 4. Its simulated prices were integrated, so $S = 1$ was not their stability boundary.
- **A decisiveness frontier.** Its "proposed method" was the same growth-rate heuristic.
- **DQN and continuous actor-critic action-support comparisons.** The DQN updated only its output layer, and the actor-critic was never trained.

The tabular $Q$-learning transfer study of section 6 does not depend on these simulations and is unaffected. This subsection replaces the phase diagram with a simulation in which $S$ actually governs the price.

**Design.** We use the single-cohort case of (29). Price deviation and adaptive pressure follow

$$x_{t+1} = ax_t + y_t + \lambda n_t + \sigma e_t, \qquad y_{t+1} = (1-\eta)y_t + \eta Q x_t,$$

with $S = a + Q$. Adaptive demand is $D_t = y_t/\lambda$ and $n_t$ is i.i.d. noise flow. A dealer takes the other side of all flow $f_t = D_t + n_t + \phi I_t$. The term $\phi I_t$ is the offsetting flow its quote skew attracts (inventory control in the spirit of 2). The dealer earns a half-spread on $|f_t|$ and marks inventory $I_t$ to $p_t = 1 + x_t$. It halts when $|I_t|p_t > L\,E_t$ or $E_t \le 0$. We compare the feedback market with three markets that have *no* price-to-flow loop, in which flow is exogenous and its price impact is retained:

- B1: i.i.d. flow with the same variance;
- B2: AR(1) flow with the same variance and first autocorrelation;
- B3: AR(1) flow with the same variance and long-run variance.

Each cell uses 400 seeds over 500 steps, and parameters are illustrative ($a = 0.5$, $\eta = 0.2$, $\lambda = 0.002$; `simulations/feedback_dealer.py`).

**Proposition 9.2** (Zero-frequency gain of feedback flow)**.** *For $S < 1$, the long-run variance of $f_t = D_t + n_t$ (without inventory control) is*

$$\mathrm{LRV}(f) = \frac{(1-a)^2\sigma_n^2 + Q^2\sigma^2/\lambda^2}{(1-S)^2}.$$

*Proof.* At frequency zero the recursions give $(1-a)x = y + \lambda n + \sigma e$ and $y = Qx$. Hence $x = (\lambda n + \sigma e)/(1-S)$ and $f = y/\lambda + n = n(1-a)/(1-S) + Q\sigma e/(\lambda(1-S))$. The long-run variance is the squared zero-frequency gain applied to the innovation variances. □

Table 32: Computed scalar-model examples. All rows use $a = 0.4$ and $\kappa = 1$. Parameters are illustrative and uncalibrated.

| Case | $g$ | $\eta$ | $\rho(J_s)$ | $\mathcal{L} = g\kappa/(1-a)$ |
|---|---|---|---|---|
| Stable feedback | 0.300 | 0.20 | 0.916228 | 0.500 |
| Boundary | 0.600 | 0.20 | 1.000000 | 1.000 |
| Unstable feedback | 0.900 | 0.20 | 1.069042 | 1.500 |
| Same unstable gain, slower learning | 0.900 | 0.05 | 1.022311 | 1.500 |
| Same unstable gain, faster learning | 0.900 | 0.80 | 1.154400 | 1.500 |
| Baseline before action gate | 0.450 | 0.20 | 0.960555 | 0.750 |
| Gate with depth unchanged | 0.225 | 0.20 | 0.891548 | 0.375 |
| Gate with depth at one quarter | 0.900 | 0.20 | 1.069042 | 1.500 |

Table 33: Dealer halt rates (exact 95% binomial intervals) in the feedback market and in three no-loop benchmarks matched on variance (B1), variance and first autocorrelation (B2), or variance and long-run variance (B3). LRV is computed from theorem 9.2. Simulator law only; not calibrated to any market.

| $S$ | var($f$) | LRV($f$) | Feedback | B1 (var) | B2 (var, AC1) | B3 (var, LRV) |
|---|---|---|---|---|---|---|
| *Leverage limit* $L = 10$*, inventory control* $\phi = 0.05$ | | | | | | |
| 0.50 | 1.00 | 1.0 | 11.0% [8.1, 14.5] | 11.2% [8.3, 14.8] | 10.5% [7.7, 13.9] | 10.8% [7.9, 14.2] |
| 0.60 | 1.01 | 1.6 | 29.5% [25.1, 34.2] | 11.5% [8.5, 15.0] | 11.2% [8.3, 14.8] | 31.5% [27.0, 36.3] |
| 0.70 | 1.03 | 2.8 | 62.3% [57.3, 67.0] | 13.0% [9.9, 16.7] | 14.0% [10.8, 17.8] | 69.0% [64.2, 73.5] |
| 0.80 | 1.08 | 6.4 | 93.2% [90.3, 95.5] | 14.8% [11.4, 18.6] | 22.2% [18.3, 26.6] | 97.8% [95.8, 99.0] |
| 0.90 | 1.25 | 26.0 | 100.0% [99.1, 100.0] | 26.5% [22.2, 31.1] | 47.0% [42.0, 52.0] | 100.0% [99.1, 100.0] |
| 0.95 | 1.62 | 105.1 | 100.0% [99.1, 100.0] | 52.8% [47.7, 57.7] | 87.5% [83.9, 90.6] | 100.0% [99.1, 100.0] |
| *Leverage limit* $L = 10$*, inventory control* $\phi = 0.10$ | | | | | | |
| 0.50 | 1.00 | 1.0 | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] |
| 0.60 | 1.01 | 1.6 | 1.2% [0.4, 2.9] | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] | 2.5% [1.2, 4.5] |
| 0.70 | 1.03 | 2.8 | 8.0% [5.5, 11.1] | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] | 21.8% [17.8, 26.1] |
| 0.80 | 1.08 | 6.4 | 39.5% [34.7, 44.5] | 0.2% [0.0, 1.4] | 0.8% [0.2, 2.2] | 68.5% [63.7, 73.0] |
| 0.90 | 1.25 | 26.0 | 90.0% [86.6, 92.8] | 2.0% [0.9, 3.9] | 7.2% [4.9, 10.2] | 99.5% [98.2, 99.9] |
| 0.95 | 1.62 | 105.1 | 99.8% [98.6, 100.0] | 11.2% [8.3, 14.8] | 47.2% [42.3, 52.3] | 100.0% [99.1, 100.0] |
| *Leverage limit* $L = 20$*, inventory control* $\phi = 0.05$ | | | | | | |
| 0.50 | 1.00 | 1.0 | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] |
| 0.60 | 1.01 | 1.6 | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] |
| 0.70 | 1.03 | 2.8 | 0.2% [0.0, 1.4] | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] | 0.2% [0.0, 1.4] |
| 0.80 | 1.08 | 6.4 | 5.5% [3.5, 8.2] | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] | 10.0% [7.2, 13.4] |
| 0.90 | 1.25 | 26.0 | 52.5% [47.5, 57.5] | 0.0% [0.0, 0.9] | 0.0% [0.0, 0.9] | 64.5% [59.6, 69.2] |
| 0.95 | 1.62 | 105.1 | 88.2% [84.7, 91.2] | 0.0% [0.0, 0.9] | 3.0% [1.6, 5.2] | 96.2% [93.9, 97.9] |

**Results.** Table 33 reports the halt rates.

- Halt rates rise steeply with $S$ well inside the stable region. At $L = 10$ and $\phi = 0.05$ they go from 11% at $S = 0.5$ to 93% at $S = 0.8$.
- No-loop flow matched on variance produces 15% at $S = 0.8$ (B1), and matching the first autocorrelation as well produces 22% (B2).
- No-loop flow matched on long-run variance produces 98% (B3), as many halts as the feedback market or more.

In this model, then, dealer failure is governed by the low-frequency content of the flow the dealer absorbs. Feedback matters because, by theorem 9.2, it inflates that content by the factor $(1 - S)^{-2}$ as $S \to 1$. Two consequences follow:

- The earlier claim that this vulnerability is "absent in non-AI market makers" is not supported. Any source of equally persistent flow produces it.
- A benchmark that matches only volatility, or only short-lag autocorrelation, understates the risk.

This is the operational content of the claim that local price stability does not ensure dealer safety: the stable region contains values of $S$ at which a capital-constrained dealer fails with high probability. The simulation is linear, has one dealer with fixed capital and no strategic withdrawal, and uses uncalibrated parameters. It illustrates a mechanism and does not estimate an event probability.

## 9.4 Explicit Methodological Exclusions and Scope Omissions

This subsection lists what the models and evidence do not cover.

**Mathematical and Proof Exclusions.** The core positive-feedback boundary proofs (Theorem 2.5) assume a single persistence scalar and a common learning rate. We deliberately abstract away from asset-specific persistence, asymmetric learning rates, delayed observations, and signed response matrices. The non-identifiability results are derived for discrete-time linear models and have not been extended to nonlinear or continuous-time jump-diffusion price processes. The deterministic norm inequalities (Proposition S8.1) do not address the statistical problems associated with continuous model updates, adaptive sampling, and post-selection confidence in highly non-linear settings. Finally, while the exact-arithmetic outer bounds are theoretically proven (Theorem 4.2), the numerical floating-point conic implementations are maintained as unverified candidates because their complete outward constraint assembly cannot be formally certified without infinite precision.

**Market Dynamics and Economic Scope.** Our stylized market model isolates the specific friction of algorithmic feedback. Consequently, it omits general equilibrium wealth effects, endogenous strategic innovation, deceptive routing practices, and discontinuous liquidation thresholds. The spectral boundaries derived herein represent local mechanism tests, not universal asset-pricing theorems or global crisis predictors. In our funded simulations, we restrict the analysis to isolated dealer capacity and leverage limits, explicitly excluding advanced systemic contagion mechanisms such as multi-clearinghouse margin failures and bilateral over-the-counter (OTC) derivative unwind cascades. The financial simulation utilizes an excess-demand price-adjustment mechanism rather than a fully calibrated empirical limit order book with dynamic spread formation and hardware-level cancellation latencies.

**Algorithmic Architectures and Empirical Surveillance.** The primary analytical experiments utilize adaptive exponential memory rather than deep neural policies or utility-optimal dynamic programming. Our learned-agent transfer experiments evaluate tabular Q-learning environments to formalize the falsification test; we reserve the evaluation of modern Multi-Agent Reinforcement Learning (MARL) populations undergoing continuous adversarial retraining for future work. Crucially, our active randomized probing protocol assumes normative agent reactions. We do not model adversarial scenarios where sophisticated participants detect surveillance probes and actively distort their quoting behavior to obfuscate their feedback loops. The empirical analysis (Section 7) combines an initial 12-day Coinbase baseline with a systematic 768 asset-day panel on Binance Spot, cross-venue replication, and an out-of-sample holdout, but remains reduced form and lacks the counterparty identifiers required to attribute feedback to specific algorithmic actors (Contract $I_1$).

## 9.5 Conclusion

We asked what market data can reveal about feedback between adaptive trading and prices.

The paper has three findings.

- **Passive price data have exact limits.** Aggregate price responses identify only merged rate–weight pairs (Theorem 3.4). Separating near-coincident cohorts needs a probe budget growing like $\delta^{-4}$ (Theorem 3.5). Any error-controlled stability classifier must abstain near $S = 1$, and with finite windows it must do so at every budget (Theorem 3.6).
- **Local price stability does not imply dealer safety.** The safe shock radius is governed by $\sup_t \|CJ^t\|$, which the Kreiss constant bounds and the spectral radius does not (Theorem 3.8). In a funded simulation with genuine feedback, dealer

halts rise steeply inside the stable region because feedback inflates the low-frequency content of flow by $(1-S)^{-2}$ (theorem 9.2). Non-adaptive flow with the same persistence does the same.

- **The public data are consistent with weak momentum feedback.** Across twelve days on Coinbase Pro, an eleven-day cross-venue replication, and a 768 asset-day panel on Binance Spot (2021–2024), estimated feedback is consistently small and positive ($\hat{Q} \approx 0.008$, $\hat{S} \approx 0.92 < 1$), corroborated by structural minimum-distance estimation ($Q_{\mathrm{SMM}} = 0.0086$), while Rigobon heteroskedasticity identifies contemporaneous shock buffering ($Q_{\mathrm{Rig}} = -0.0191$). It is not causal and is not attributable to algorithms.

An earlier version of this paper reported instrumental-variable estimates, panel regressions and simulation results that, on audit, were not produced from data or did not implement the stated model. They are withdrawn, and Appendix D maps every remaining number to the code that produces it. The main open problems are a credible source of exogenous variation in live markets, counterparty-level data for attribution, and a solvency analysis with strategic liquidity withdrawal.

## Data, code, and replication availability

All code, tests and generated result files are in the release repository. Tables marked as generated are written by `scripts/build_results.py` from JSON outputs in `results/provenance/`, and Appendix D lists every empirical and simulated number with its producing script. The empirical analysis in section 7.4 uses the files `empirical_pipeline/data/{BTC,ETH,ADA}_1sec.csv`, sourced from Nielsen [41] (Kaggle dataset 1567989, CC0 Public Domain, recorded from Coinbase Pro via `crypto-rl` by Sadighian). Complete file checksums, schema definitions, and timing validations are documented in `empirical_pipeline/data/README.md` and verified in `tests/test_data_provenance_and_closure.py`. Sections 7.2 and 7.3 use cached Binance US snapshots and public Binance kline archives stored under `research_extension/`; section 7's Coinbase samples are cached in the same tree. No FI-2010 or cross-venue latency data are used.

## Compliance and Ethical Disclosures

The authors confirm that AI assistance was utilized in drafting, code generation, and mathematical typesetting during the preparation of this manuscript. The human authors are responsible for reviewing all AI-assisted text, proofs and code. An internal audit found that earlier AI-assisted drafts contained numbers not produced by data or code; those results have been withdrawn (Appendix D).

The one-second order-book files are public domain (CC0) under Nielsen [41]. The public Binance and Coinbase samples were retrieved from public endpoints, subject to those venues' terms.

## References


[1] W. Brian Arthur, John H. Holland, Blake LeBaron, Richard Palmer, and Paul Tayler. Asset Pricing Under Endogenous Expectations in an Artificial Stock Market. Santa Fe Institute Working Paper 96-12-093, 1996. https://www.santafe.edu/research/results/working-papers/asset-pricing-under-endogenous-expectations-in-an-.

[2] Marco Avellaneda and Sasha Stoikov. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224, 2008. https://doi.org/10.1080/14697680701381228.

[3] William A. Brock and Cars H. Hommes. Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control*, 22(8–9):1235–1274, 1998. https://doi.org/10.1016/S0165-1889(98)00011-6.

[4] Markus K. Brunnermeier and Lasse Heje Pedersen. Market Liquidity and Funding Liquidity. *The Review of Financial Studies*, 22(6):2201–2238, 2009. https://doi.org/10.1093/rfs/hhn098.

[5] Emilio Calvano, Giacomo Calzolari, Vincenzo Denicolò, and Sergio Pastorello. Artificial Intelligence, Algorithmic Pricing, and Collusion. *American Economic Review*, 110(10):3267–3297, 2020. https://doi.org/10.1257/aer.20190623.

[6] A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-Based Improvements for Inference with Clustered Errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008. https://doi.org/10.1162/rest.90.3.414.

[7] Andrei Kirilenko, Albert S. Kyle, Mehrdad Samadi, and Tugkan Tuzun. The Flash Crash: High-frequency trading in an electronic market. *Journal of Finance*, 72(3):967–998, 2017.

[8] Juan C. Perdomo, Tijana Zrnic, Celestine Mendler-Dünner, and Moritz Hardt. Performative Prediction. *Proceedings of Machine Learning Research*, 119:7599–7609, 2020. https://proceedings.mlr.press/v119/perdomo20a.html.

[9] Adhyyan Narang, Evan Faulkner, Dmitriy Drusvyatskiy, Maryam Fazel, and Lillian J. Ratliff. Multiplayer Performative Prediction: Learning in Decision-Dependent Games. *Journal of Machine Learning Research*, 24(202):1–56, 2023. https://jmlr.org/papers/v24/22-0131.html.

[10] Gabriele Farina and Juan Carlos Perdomo. The Stability of Online Algorithms in Performative Prediction. arXiv:2602.24207v2, 3 June 2026. Preprint. https://arxiv.org/abs/2602.24207v2.

[11] Kartik Anand, Sophia Kazinnik, Agnese Leonello, and Ettore Panetti. Ex Machina: financial stability in the age of artificial intelligence. European Central Bank Working Paper Series, No. 3225, 2026. https://doi.org/10.2866/4103076.

[12] Jean-Edouard Colliard, Thierry Foucault, and Stefano Lovo. Algorithmic Pricing and Liquidity in Securities Markets. *The Review of Financial Studies*, hhag010, 2026. https://doi.org/10.1093/rfs/hhag010.

[13] Financial Stability Board. The Financial Stability Implications of Artificial Intelligence. 14 November 2024. https://www.fsb.org/uploads/P14112024.pdf.

[14] Financial Stability Board. Monitoring Adoption of Artificial Intelligence and Related Vulnerabilities in the Financial Sector. 10 October 2025. https://www.fsb.org/2025/10/monitoring-adoption-of-artificial-intelligence-and-related-vulnerabilities-in-the-financial-sector/.

[15] Financial Stability Board. Sound Practices for Responsible Adoption of Artificial Intelligence (AI): Consultation report. 10 June 2026. This reference identifies the consultation version, not a legal requirement. https://www.fsb.org/2026/06/sound-practices-for-responsible-adoption-of-artificial-intelligence-ai-consultation-report/.

[16] Stephen Boyd, Laurent El Ghaoui, Eric Feron, and Venkataramanan Balakrishnan. *Linear Matrix Inequalities in System and Control Theory*. SIAM, 1994. https://web.stanford.edu/~boyd/lmibook/.

[17] Felix Hausdorff. Summationsmethoden und Momentfolgen. I. *Mathematische Zeitschrift*, 9:74–109, 1921.

[18] Rudolf E. Kalman. Mathematical description of linear dynamical systems. *Journal of the SIAM, Series A: Control*, 1(2):152–192, 1963.

[19] M. G. Krein and A. A. Nudelman. *The Markov Moment Problem and Extremal Problems*. Translations of Mathematical Monographs 50. American Mathematical Society, 1977.

[20] Michael D. Perlman. One-sided testing problems in multivariate analysis. *Annals of Mathematical Statistics*, 40(2):549–567, 1969.

[21] M. N. Spijker. On a conjecture by LeVeque and Trefethen related to the Kreiss matrix theorem. *BIT Numerical Mathematics*, 31(3):551–555, 1991.

[22] Hassan K. Khalil. *Nonlinear Systems*. Third edition. Prentice Hall, 2002. https://www.egr.msu.edu/~khalil/NonlinearSystems/.

[23] David Byrd, Maria Hybinette, and Tucker Hybinette Balch. ABIDES: Towards High-Fidelity Market Simulation for AI Research. arXiv:1904.12066, 2019. This reference identifies the preprint version. https://arxiv.org/abs/1904.12066.

[24] Joel Hasbrouck. Measuring the information content of stock trades. *Journal of Finance*, 46(1):179–207, 1991. https://doi.org/10.1111/j.1540-6261.1991.tb03749.x.

[25] Albert S. Kyle. Continuous auctions and informed trader. *Econometrica*, 53(6):1315–1335, 1985. https://doi.org/10.2307/1913210.

[26] Anders Rantzer. Scalable control of positive systems. *European Journal of Control*, 24:72–80, 2015. https://doi.org/10.1016/j.ejcon.2015.04.004.

[27] Robert L. Kosut, Ming K. Lau, and Stephen P. Boyd. Set-Membership Identification of Systems with Parametric and Nonparametric Uncertainty. *IEEE Transactions on Automatic Control*, 37(7):929–941, 1992. https://web.stanford.edu/~boyd/papers/set_member_id.html.

[28] Dimitris Bertsimas and Ioana Popescu. Optimal Inequalities in Probability Theory: A Convex Optimization Approach. *SIAM Journal on Optimization*, 15(3):780–804, 2005. https://doi.org/10.1137/S1052623401399903.

[29] Henk J. van Waarde, Jaap Eising, Harry L. Trentelman, and M. Kanat Camlibel. Data informativity: a new perspective on data-driven analysis and control. *IEEE Transactions on Automatic Control*, 65(11):4753–4768, 2020. https://arxiv.org/abs/1908.00468.

[30] Yassir Jedra and Alexandre Proutiere. Sample Complexity Lower Bounds for Linear System Identification. arXiv:1903.10343, 2019. Preprint version consulted. https://arxiv.org/abs/1903.10343.

[31] Oscar Jordà. Estimation and Inference of Impulse Responses by Local Projections. *American Economic Review*, 95(1):161–182, 2005. https://doi.org/10.1257/0002828053828518.

[32] Mohammad Khosravi and Roy S. Smith. Regularized Identification with Internal Positivity Side-Information. arXiv:2111.00407, 2021. Verified preprint version consulted. https://arxiv.org/abs/2111.00407.

[33] Mohammad Khosravi and Roy S. Smith. Kernel-based Impulse Response Identification with Side-Information on Steady-State Gain. arXiv:2111.00409, 2021. Preprint version consulted. https://arxiv.org/abs/2111.00409.

[34] Balázs Cs. Csáji, Marco C. Campi, and Erik Weyer. Sign-Perturbed Sums: A New System Identification Approach for Constructing Exact Non-Asymptotic Confidence Regions in Linear Regression Models. *IEEE Transactions on Signal Processing*, 63(1):169–181, 2015. https://doi.org/10.1109/TSP.2014.2369000; author manuscript: https://arxiv.org/abs/1807.08216.

[35] Rama Cont, Arseniy Kukanov, and Sasha Stoikov. The Price Impact of Order Book Events. *Journal of Financial Econometrics*, 12(1):47–88, 2014. https://doi.org/10.1093/jjfinec/nbt003.

[36] SciPy developers. SciPy API reference: `linprog`, HiGHS dual-simplex and interior-point methods. Official documentation; accessed 18 September 2026. Tested version: SciPy 1.17.0. https://docs.scipy.org/doc/scipy/reference/optimize.linprog-highs-ds.html.

[37] Ian R. Manchester. Input Design for System Identification via Convex Relaxation. arXiv:1009.5614, 2010. https://arxiv.org/abs/1009.5614.

[38] C. J. Clopper and E. S. Pearson. The Use of Confidence or Fiducial Limits Illustrated in the Case of the Binomial. *Biometrika*, 26(4):404–413, 1934. doi:10.1093/biomet/26.4.404.

[39] Eduardo Abi Jaber, Nathan De Carvalho and Huyên Pham. Trading with propagators and constraints: applications to optimal execution and battery storage.

arXiv:2409.12098, 2024. https://arxiv.org/abs/2409.12098.

[40] Mante Zelvyte and Jim E. Griffin. A Multiplex Network Hawkes Model for Systemic Risk Measurement. arXiv:2606.15755, 2026. https://arxiv.org/abs/2606.15755.

[41] Martin Søgaard Nielsen. High Frequency Crypto Limit Order Book Data. Kaggle dataset 1567989, version 1, 2021. https://www.kaggle.com/datasets/martinsn/high-frequency-crypto-limit-order-book-data.

[42] Jonathan Sadighian. crypto-rl: High Frequency Limit Order Book Simulation Environment and Trading Agent. GitHub repository, 2019. https://github.com/sadighian/crypto-rl.

# A Proofs and technical qualifications

## A.1 Local stability and the limits of linearization

For the stable part of Proposition 2.3, choose $\gamma \in (\rho(J^*), 1)$. Finite-dimensional matrix theory provides an equivalent vector norm with $\|J^*\| < \gamma$. Continuity of $D\Phi$ provides a convex neighborhood of $z^*$ and $\widetilde{\gamma} < 1$ such that $\|D\Phi(z)\| \le \widetilde{\gamma}$ throughout that neighborhood. The mean-value integral yields

$$\|\Phi(z) - z^*\| \le \widetilde{\gamma}\,\|z - z^*\|\,. \tag{75}$$

A sufficiently small norm ball is invariant; iteration gives exponential decay. The unstable part is the standard discrete-time linearization instability result: a derivative eigenvalue outside the closed unit disk produces a locally expanding direction and rules out Lyapunov stability for a continuously differentiable map. No conclusion is asserted from an eigenvalue on the unit circle alone. The result concerns the fixed autonomous map and its fixed point, not an arbitrary sequence of re-estimated Jacobians.

## A.2 Proof of the positive-feedback boundary

Let $\nu = \rho(BC)$ and define

$$r_+ = \frac{a + 1 - \eta + \sqrt{(a - 1 + \eta)^2 + 4g\eta\nu}}{2}. \tag{76}$$

When $g\nu > 0$, Perron–Frobenius theory supplies $v \ge 0$, $v \ne 0$, with $BCv = \nu v$. Then $Cv \ne 0$ and

$$w = \frac{\eta}{r_+ - 1 + \eta} Cv \tag{77}$$

satisfies $J_+(v, w)^\top = r_+(v, w)^\top$. Hence $r_+$ is an eigenvalue.

To show that no eigenvalue has greater modulus, let $(v, w)$ be a nonzero eigenvector with eigenvalue $\lambda$. Eigenvalues equal to $a$ or $1 - \eta$ have modulus at most $r_+$. For the others, eliminating $w$ gives

$$BCv = \frac{(\lambda - a)(\lambda - 1 + \eta)}{g\eta} v. \tag{78}$$

If $|\lambda| > r_+$, then $|\lambda| > \max(a, 1 - \eta)$ and

$$|(\lambda - a)(\lambda - 1 + \eta)| \ge (|\lambda| - a)(|\lambda| - (1 - \eta)) \tag{79}$$

$$> (r_+ - a)(r_+ - (1 - \eta)) = g\eta\nu. \tag{80}$$

This contradicts the spectral-radius bound on eigenvalues of $BC$. Thus $\rho(J_+) = r_+$. If $g = 0$, the matrix is block triangular. If $\nu = 0$ and $g > 0$, the same elimination excludes every eigenvalue outside $\{a, 1-\eta\}$, so the formula reduces to $\max(a, 1 - \eta)$ and remains valid.

Finally, $r_+ < 1$ is equivalent to

$$\sqrt{(a - 1 + \eta)^2 + 4g\eta\nu} < 1 - a + \eta. \tag{81}$$

The right side is positive. Squaring and simplifying yields $g\nu < 1 - a$ because $\eta > 0$. Equality and strict reversal follow identically. For a finite-dimensional constant linear system, all eigenvalues inside the unit disk are equivalent to asymptotic stability. This proves theorem 2.5.

## A.3 What the boundary does not imply

A positive real eigenvalue passing through one identifies loss of local linear stability in the specified model. It does not determine the nonlinear attractor, whether prices crash or grow, whether an institution defaults, or whether a finite disturbance remains inside the modeled region. A nonlinear extension could saturate or switch into another regime. An observed financial crisis can also occur without this crossing, for example when an external shock overwhelms solvency buffers.

The result uses a single persistence scalar $a$ and a common $\eta$. Allowing asset-specific persistence, asymmetric learning rates, delayed observation, or signed response matrices removes the simple formula unless further structure is proved. A claimed universal surface in six aggregate variables would need to show that the omitted state is either irrelevant, bounded, or measured. The current theorem does not establish such sufficiency.

## A.4 Stochastic and nonlinear robustness

For a nonlinear map $\Phi_t$ with a common reference point, suppose the induced $P$-norm of every derivative on a convex ball of radius $r$ is at most $\gamma < 1$. Then the same Lipschitz argument yields the disturbance bound in equation (73). If $\bar{b} \le (1 - \gamma)r$ and the initial state is inside the ball, that ball is invariant. With an evolving reference point, its motion must appear as an additional disturbance; ignoring that drift changes the claim.

If an estimated uncertainty set covers the actual matrices with probability at least $1 - \delta$ over the relevant horizon, the robust bound holds on that coverage event. Marginal per-time coverage does not automatically imply simultaneous horizon coverage. Model updates, adaptive sampling, and post-selection can invalidate a naive confidence statement. These statistical issues are work remaining, not solved by the deterministic norm inequality.

## A.5 Proof of the swapped-feedback non-identifiability theorem and minimal separation

This subsection provides the formal algebraic proof of Proposition 5.1 and Theorem 5.2 using resolvent matrix expansions and Kalman controllability/observability rank conditions.

**State-space representation.** Consider an economy with price persistence $a \in [0, 1)$, market impact $\kappa > 0$, common learning rate $\eta \in (0, 1]$, and $n$ adaptive agents with feedback weight vector $q = [q_1, \ldots, q_n]^\top \in \mathbb{R}^n_+$. The state vector is $z_t = [x_t, y_t^\top]^\top \in \mathbb{R}^{1+n}$, where $y_t = [y_{1,t}, \ldots, y_{n,t}]^\top$. The closed-loop discrete-time state-space dynamics are

$$z_{t+1} = Jz_t + Bu_t, \qquad x_t = Cz_t, \tag{82}$$

where

$$J = \begin{bmatrix} a & \mathbf{1}_n^\top \\ \eta q & (1-\eta) I_n \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & \mathbf{0}_n^\top \end{bmatrix}. \tag{83}$$

**Case 1: Market-wide aggregate probing.** When the observer injects a market-wide price probe $u_t \in \mathbb{R}$, the input matrix is $B_{\text{agg}} = [\kappa, \mathbf{0}_n^\top]^\top$. The rational transfer function from probe $u$ to price displacement $x$ in the complex frequency domain ($z \in \mathbb{C}$) is

$$H_{xu}(z) = C(zI_{1+n} - J)^{-1} B_{\text{agg}} = \kappa \left[(zI_{1+n} - J)^{-1}\right]_{1,1}. \tag{84}$$

To evaluate the $(1, 1)$ scalar entry of $(zI_{1+n} - J)^{-1}$, partition the resolvent matrix as

$$zI_{1+n} - J = \begin{bmatrix} z - a & -\mathbf{1}_n^\top \\ -\eta q & (z - 1 + \eta) I_n \end{bmatrix}. \tag{85}$$

Applying the Schur complement of the $(2, 2)$ block $(z - 1 + \eta)I_n$, which is invertible for all $z \neq 1 - \eta$, yields

$$\left[(zI_{1+n} - J)^{-1}\right]_{1,1} = \left( (z - a) - (-\mathbf{1}_n^\top) \frac{(-\eta q)}{z - 1 + \eta} \right)^{-1} \tag{86}$$

$$= \left( z - a - \frac{\eta}{z - 1 + \eta} \mathbf{1}_n^\top q \right)^{-1} \tag{87}$$

$$= \left( z - a - \frac{\eta Q}{z - 1 + \eta} \right)^{-1} \tag{88}$$

$$= \frac{z - 1 + \eta}{(z - a)(z - 1 + \eta) - \eta Q}, \tag{89}$$

where $Q = \mathbf{1}_n^\top q = \sum_{i=1}^n q_i$. Consequently, the closed-loop transfer function is

$$H_{xu}(z) = \frac{\kappa(z - 1 + \eta)}{(z - a)(z - 1 + \eta) - \eta Q}. \tag{90}$$

Observe that $H_{xu}(z)$ depends on $q \in \mathbb{R}^n_+$ only through $Q = \sum_i q_i$. To keep impact and timing explicit, write

$$H_{xu}(z) = \sum_{j \geq 0} \beta_j z^{-(j+1)} = \kappa \sum_{j \geq 0} h_j z^{-(j+1)}, \qquad h_0 = 1.$$

Thus $h_j$ is the normalized, lag-shifted coefficient of $H_{xu}/\kappa$, not the already impact-scaled coefficient of $H_{xu}$. A unit probe from zero state yields $x_1 = \beta_0 = \kappa$. Equal $(a, \eta, \kappa, Q)$ therefore gives the same $h_j$ and $\beta_j$ at every horizon.

Under any finite observation window of length $H$, the mean price response is $\beta_j = \kappa h_j$. Under the proposition's common disturbance law, both the mean trajectory and the observation covariance $\Sigma_Y$ coincide. Where Gaussian densities are defined, their likelihood ratio therefore satisfies

$$\frac{\mathcal{L}(Y_{1:R} \mid q^{(A)})}{\mathcal{L}(Y_{1:R} \mid q^{(B)})} = 1 \tag{91}$$

identically for all sample sizes $R$ and all horizons $H$, whenever $\sum_i q_i^{(A)} = \sum_i q_i^{(B)}$. Thus no estimator relying on aggregate price observations can separate $q^{(A)}$ from $q^{(B)}$.

In contrast, the policy intervention of suppressing agent $k$ removes agent $k$'s feedback loop, reducing total feedback mass by $\Delta Q_k = q_k$, and reducing the normalized loop gain $G = Q/(1 - a)$ by $\Delta G_k = q_k/(1 - a)$ (distinguished from steady-state price response to constant input, $\kappa/(1 - a - Q)$). For Economy A with $q^{(A)} = (0.30, 0.10)$ and Economy B with $q^{(B)} = (0.10, 0.30)$, with $a = 0.40$, $\Delta Q_1^{(A)} = 0.30 \neq 0.10 = \Delta Q_1^{(B)}$, and $\Delta G_1^{(A)} = 0.30/0.60 = 0.50 \neq 0.167 = 0.10/0.60 = \Delta G_1^{(B)}$, differing by a factor of three. This completes the proof of Proposition 5.1.

**Popov-Belevitch-Hautus (PBH) Observability Analysis.** The non-identifiability of group weights can be characterized directly in state space via the Popov-Belevitch-Hautus (PBH) rank condition. The system $(J, C)$ is completely observable if and only if

$$\operatorname{rank} \begin{bmatrix} \lambda I_{1+n} - J \\ C \end{bmatrix} = 1 + n \qquad \forall \lambda \in \mathbb{C}. \tag{92}$$

Consider $\lambda = 1 - \eta$. Any non-zero vector $v = [0, w^\top]^\top \in \mathbb{R}^{1+n}$ with $w = [w_1, \ldots, w_n]^\top$ satisfying $\sum_{i=1}^n w_i = 0$ obeys:

$$Cv = \begin{bmatrix} 1 & \mathbf{0}_n^\top \end{bmatrix} \begin{bmatrix} 0 \\ w \end{bmatrix} = 0, \tag{93}$$

$$Jv = \begin{bmatrix} a & \mathbf{1}_n^\top \\ \eta q & (1-\eta) I_n \end{bmatrix} \begin{bmatrix} 0 \\ w \end{bmatrix} = \begin{bmatrix} \mathbf{1}_n^\top w \\ (1-\eta) w \end{bmatrix} = \begin{bmatrix} 0 \\ (1-\eta) w \end{bmatrix} = (1 - \eta) v. \tag{94}$$

Thus, $v$ lies in the unobservable subspace of $(J, C)$. Because the hyperplane $\{w \in \mathbb{R}^n : \sum_{i=1}^n w_i = 0\}$ has

dimension $n-1 \geq 1$ for any $n \geq 2$, the mode $\lambda = 1-\eta$ suffers a PBH rank deficiency of exactly $n-1$. Crucially, this $n-1$-dimensional unobservable state subspace provides the systems-theoretic explanation for parameter non-identifiability: while $q$ is a parameter vector and $v$ is a state vector, any redistribution of feedback weights preserving the scalar sum $Q = \sum_i q_i$ excites only internal state trajectories within this unobservable subspace, leaving aggregate price observations $Cv = 0$ identically invariant at all horizons.

**Channel-specific excitation and measurement rank across observation cases.** For Theorem 5.2, suppose the observer executes $m$ probe experiments where experiment $j$ injects pulse weights $U_{j,:}$ across the $n$ agent input channels at $t = 0$:

$$y_{k,t+1} = (1-\eta)y_{k,t} + \eta q_k(x_t + u_{k,t}). \tag{95}$$

The input matrix becomes $B_{\text{chan}} = [\mathbf{0}_n^\top; \eta\,\text{diag}(\mathbf{q})] \in \mathbb{R}^{(1+n)\times n}$. The resulting state displacement at $t = 1$ is $\Delta\mathbf{y}_1 = \eta\,\text{diag}(\mathbf{q})U^\top\mathbf{e}_j$, which generates a price deflection at lag $t = 2$ of $b_j = \eta U_{j,:}\mathbf{q}$. Stacking all $m$ experiments yields the linear channel measurement map:

$$\mathbf{b} = \eta U\mathbf{q}. \tag{96}$$

Identifiability across the parameter space is partitioned into five distinct operational cases:

1. **Case 1 (Known $\eta$, full-rank $U$):** When the common learning rate $\eta$ is known and $\text{rank}(U) = n$, the left pseudo-inverse yields unique componentwise identification $\mathbf{q} = \eta^{-1}U^\dagger\mathbf{b} = \eta^{-1}(U^\top U)^{-1}U^\top\mathbf{b}$, requiring at least $m \geq n$ independent probe configurations.
2. **Case 2 (Known $\eta$, known aggregate $Q > 0$):** Augment the observations as $\begin{bmatrix} Q \\ \mathbf{b} \end{bmatrix} = \begin{bmatrix} \mathbf{1}^\top \\ \eta U \end{bmatrix}\mathbf{q}$. Uniform injectivity on $\{\mathbf{q} \geq 0 : \mathbf{1}^\top\mathbf{q} = Q\}$ holds if and only if the augmented matrix has column rank $n$. Thus $n-1$ appropriately chosen channel probes can suffice. This is not a necessary rank condition for uniqueness at an individual boundary point: with $n = 3$, $U = (1, 0, 0)$, $\eta = 1/2$, $Q = 1$, and $b = 1/2$, nonnegativity uniquely gives $\mathbf{q} = (1, 0, 0)$ although the augmented rank is two. At $Q = 0$, $\mathbf{q} = 0$ is already known.
3. **Case 3 (Unknown $\eta$, known aggregate $Q > 0$, full-rank $U$):** When $\eta$ is unknown but $U$ has full column rank ($\text{rank}(U) = n$) and aggregate feedback $Q = \mathbf{1}^\top\mathbf{q} > 0$ is known, the observer computes the pseudo-inverse projection $\mathbf{v} = U^\dagger\mathbf{b} = \eta\mathbf{q}$. Taking the inner product with the all-ones vector yields $\mathbf{1}^\top\mathbf{v} = \eta\mathbf{1}^\top\mathbf{q} = \eta Q$. Because $Q > 0$, dividing by $Q$ uniquely recovers the scalar learning rate:
$$\eta = \frac{\mathbf{1}^\top\mathbf{v}}{Q}. \tag{97}$$
Substituting $\eta$ back yields unique point identification of the full feedback vector:
$$\mathbf{q} = \frac{\mathbf{v}}{\eta} = \frac{Q\,\mathbf{v}}{\mathbf{1}^\top\mathbf{v}}. \tag{98}$$
Thus, an unknown learning rate does *not* preclude point identification if auxiliary aggregate mass $Q$ is available.
4. **Case 4 (Unknown $\eta$, no auxiliary scale information):** For full-column-rank $U$ and compatible observations, the exact set is $\{U^\dagger\mathbf{b}/\eta : \eta \in [\underline{\eta}, 1]\}$. It is a nondegenerate segment along a ray when $\mathbf{b} \neq 0$ and $0 < \underline{\eta} < 1$. Zero observations identify $\mathbf{q} = 0$; a singleton rate interval also identifies $\mathbf{q}$. An unknown nondegenerate rate interval does not identify the rate when $\mathbf{q} = 0$.
5. **Case 5 (Possible nullspace ambiguity):** With known rate and a feasible $\mathbf{q}_0$, the nonnegative identified set is $(\mathbf{q}_0 + \ker U) \cap \mathbb{R}^n_+$. At a strictly positive point its local dimension is $n - \text{rank}(U)$, which can equal one. Boundary constraints can reduce the dimension to zero. With unknown rate, use the projection of the joint $(\mathbf{q}, 1/\eta)$ feasible set in the supplement, rather than calling every set a multidimensional subspace.

Finally, absent an auxiliary total-mass constraint, full column rank $\text{rank}(U) = n$ is necessary for uniform point identification across the interior: if $\text{rank}(U) < n$, there exists a non-zero direction $\mathbf{w}$ such that $U\mathbf{w} = \mathbf{0}$, meaning any interior point $\mathbf{q}$ admits an open neighborhood of observationally equivalent interior parameters $\mathbf{q} + \epsilon\mathbf{w}$ generating identical responses $\mathbf{b}$. Boundary constraints (such as nonnegativity or known sparsity) can resolve particular isolated configurations on the boundary, but do not provide uniform interior identifiability. With known $Q$, any ambiguity direction must also satisfy $\mathbf{1}^\top\mathbf{w} = 0$; the augmented rank condition applies when $\eta$ is known. Known $Q$ and $n-1$ probes do not generally guarantee joint identification of $(\eta, \mathbf{q})$: for $U = (1, 0)$ and $Q = 0.4$, $(\eta, \mathbf{q}) = (0.2, (0.3, 0.1))$ and $(0.3, (0.2, 0.2))$ both give $b = 0.06$. This does not rule out identifying the rate alone: if $c^\top U = \mathbf{1}^\top$, then $\eta = c^\top\mathbf{b}/Q$ even when $U$ is rank deficient. For example, $U = (1, 1)$ identifies $\eta$ from known $Q > 0$ but not the allocation. This completes the proof of Theorem 5.2.

## A.6 Proofs of Theorems M1–M5

*Proof of Theorem 3.4.* Take $z$-transforms of (29) with zero initial state, and write $r_i = 1-\eta_i$. Then $(z-r_i)Y_i = \eta_i q_i X$ and $(z-a)X = \sum_i Y_i + \kappa U$. Hence

$$H(z) = \frac{X(z)}{U(z)} = \frac{\kappa}{z - a - \sum_r c_r/(z-r)}, \qquad c_r = \sum_{i:r_i=r} \eta_i q_i = \eta \sum_{i:\eta_i=r}$$

where the sum runs over the distinct active rates. The impulse response determines its generating function and

conversely. So equality of $h_t$ for all $t$ is equivalent to $H_A = H_B$. Given equal $a$ and $\kappa$, this is equivalent to $\sum_r c_r^A/(z-r) = \sum_r c_r^B/(z-r)$ as rational functions. Partial-fraction decompositions with nonzero coefficients and distinct poles are unique, so the pole sets and residues coincide. That is, $\{(r, c_r)\}$ coincide, which (as $\eta > 0$) is the statement $\mathcal{M}(\mathcal{E}_A) = \mathcal{M}(\mathcal{E}_B)$. For the order, write $H = \kappa P(z)/D(z)$ with $P(z) = \prod_r (z-r)$ of degree $m = |\operatorname{supp} \mathcal{M}|$ and

$$D(z) = (z-a)P(z) - \sum_r c_r \prod_{s \neq r} (z-s)$$

of degree $m+1$. A common root of $P$ and $D$ would be some active $r'$, but $D(r') = -c_{r'} \prod_{s \neq r'} (r'-s) \neq 0$. Thus $P/D$ is coprime and its McMillan degree, which equals the minimal state dimension for a scalar transfer function [18], is $m+1$. The final statement holds because $H$ depends on the $q_i$ only through the $c_r$. □

*Proof of Theorem 3.5.* (1) Let $\eta_1 = \eta_0 + \delta$ with $\delta = \delta_R$. For Gaussian product measures,

$$\mathrm{KL}(\mathbb{P}_{\eta_0} \| \mathbb{P}_{\eta_1}) = \frac{RA^2\lambda^2}{2\sigma^2} \sum_{t \le H} \big(k_t(\eta_1) - k_t(\eta_0)\big)^2.$$

By the mean value theorem, $|k_t(\eta_1) - k_t(\eta_0)| \le \delta\, t \sup_\xi (1-\xi)^{t-1}$, so the sum is at most $\delta^2 H$ and $\mathrm{KL} \le 1/2$. Pinsker's inequality gives $\mathrm{TV} \le 1/2$. For any $\hat{\eta}$, let $\psi$ select the closer of the two points. Then $\max_i \mathbb{E}_i(\hat{\eta} - \eta_i)^2 \ge (\delta/2)^2 \max_i \mathbb{P}_i(\psi \neq i) \ge (\delta^2/4)(1 - \mathrm{TV})/2 \ge \delta^2/16$. The asymptotic statement is the classical efficiency of the MLE in regular parametric models.

(2) For $f(x) = x^t$, $f(r+\delta) + f(r-\delta) - 2f(r) = \delta^2 f''(\xi_t)$ for some $\xi_t \in (r-\delta, r+\delta)$. So the kernel difference satisfies $|\Delta_t| \le \frac{w}{2}\delta^2 t(t-1)(r+\delta)^{t-2}$ and $\Delta_0 = \Delta_1 = 0$. Hence

$$\mathrm{KL} \le \frac{RA^2\lambda^2}{2\sigma^2} \cdot \frac{w^2\delta^4}{4} C(r+\delta).$$

A test with both errors at most $\beta$ satisfies $1 - 2\beta \le \mathrm{TV} \le \sqrt{\mathrm{KL}/2}$. Rearranging gives (34). $C(\rho) < \infty$ for $\rho < 1$ by the ratio test. □

*Proof of Theorem 3.6.* (1) $\mathbb{P}_0(\text{Abstain}) = 1 - \mathbb{P}_0(\text{Unstable}) - \mathbb{P}_0(\text{Stable}) \ge 1 - \alpha - \mathbb{P}_0(\text{Stable})$. Also $\mathbb{P}_0(\text{Stable}) \le \mathbb{P}_1(\text{Stable}) + \mathrm{TV} \le \alpha + \mathrm{TV}$. The case $i = 1$ is symmetric. The second inequality is Pinsker's. (2) Under the two models the means differ by $2\varepsilon A\lambda\phi_t$, so $\mathrm{KL} = 2RA^2\lambda^2\varepsilon^2 \|\phi_{1:H}\|^2/\sigma^2$; substitute into (1). (3) Identical observation laws give $\mathrm{TV} = 0$. □

*Proof of Theorem 3.7.* (1) Each $q_i\eta_i(1-\eta_i)^j$ is the $j$th moment of the point mass $q_i\eta_i\delta_{1-\eta_i}$ on $[0,1)$. Hausdorff's theorem identifies moment sequences on $[0,1]$ with completely monotone sequences, and finitely atomic measures are weak-$*$ dense in finite measures on $[0,1]$. (2) $[2^{i+j}] = vv^\top$ with $v_i = 2^i$ is positive semidefinite, but $k_0 - k_1 = -1 < 0$ violates complete monotonicity. The finite characterization is the truncated Hausdorff moment theorem on $[0,1]$, where the localizing polynomial is $x(1-x)$ [19]. (3) For $k \in \mathcal{C}$ we have $k + \mathcal{C} \subseteq \mathcal{C}$ because $\mathcal{C}$ is a convex cone. Hence, with $d_V$ the $V^{-1}$-weighted distance,

$$T(k+Z) = d_V(k+Z, \mathcal{C})^2 \le d_V(k+Z, k+\mathcal{C})^2 = d_V(Z, \mathcal{C})^2 = T(Z)$$

pointwise, and $\mathbb{P}(T(k+Z) > c_{1-\alpha}) \le \mathbb{P}(T(Z) > c_{1-\alpha}) = \alpha$. The size is therefore at most $\alpha$ at every point of $\mathcal{C}$, and the cone vertex is least favorable [20]. □

*Proof of Theorem 3.8.* (1) $\|CJ^t z_0\| \le M_C \|z_0\| < B$ for all $t$ if $\|z_0\| < r^*$. Conversely, choose $t^*$ and a unit $v$ with $\|CJ^{t^*} v\| > M_C/(1+\epsilon)$ and set $z_0 = (1+\epsilon) r^* v$; then $\|CJ^{t^*} z_0\| > B$. (2) For $|z| > 1 > \rho(J)$, $(zI - J)^{-1} = \sum_{k \ge 0} J^k z^{-k-1}$. Hence $\|(zI-J)^{-1}\| \le M_I \sum_k |z|^{-k-1} = M_I/(|z|-1)$, so $\mathcal{K}(J) \le M_I$ and $r^* = B/M_I \le B/\mathcal{K}(J)$. Apply (1). The converse inequality is Spijker's sharpening of the Kreiss matrix theorem. (3) $\|J\|_2 \ge |J_{12}| = m$. □

*Proof of Theorem 2.6.* The characteristic polynomial is $\lambda^2 - T\lambda + D$ with $T = a + 1 - \eta$ and $D = a(1-\eta) - g\eta\kappa$. By the Jury (Schur–Cohn) conditions, both roots lie in the open unit disc if and only if $1 - T + D > 0$, $1 + T + D > 0$ and $|D| < 1$. These conditions are:

- $1 - T + D = \eta(1 - a - g\kappa) > 0$, which is $g\kappa < 1 - a$;
- $1 + T + D = (1+a)(2-\eta) - g\eta\kappa > 0$;
- $D < 1$, which is $-g\eta\kappa < 1 - a(1-\eta)$;
- $D > -1$, which is $g\eta\kappa < 1 + a(1-\eta)$.

For $a \in [0,1)$ and $\eta \in (0,1]$, both $(1+a)(2-\eta)/\eta$ and $(1 + a(1-\eta))/\eta$ are at least $1 - a$, so the second and fourth conditions are implied by the first. The remaining conditions are $-(1 - a(1-\eta))/\eta < g\kappa < 1 - a$. □

# B Replication Protocol, Information Architecture, and Empirical Ledgers

The native scientific suites run separately via `tools/run_all_tests.py`; extension and correction tests run via `pytest`. Their receipts are recorded separately from standalone numerical checks. A count of verifier conditions is not a count of independent experiments, and internally authored verifiers are not external replication.

This appendix specifies the data lineage, information architecture, and computational replication protocol for all empirical and simulation results reported in the manuscript.

## B.1 Information architecture and observational boundaries

To prevent circular identification, the framework strictly segregates information across three observational tiers and formal contracts:

1. **Public market observables ($\mathcal{I}_{\text{pub}}$, Contracts $I_0$ and $I_2$):** Limit order book snapshots, trade prices, execution volumes, and (for Contract $I_0$) randomized active probe allocations. Estimators operating under Contract $I_0$ (recurrence LP, interval deconvolution, covariance-aware output-error) receive active randomized probe interventions and prices, while estimators under Contract $I_2$ (cross-sectional regressions, passive state-space fits) receive strictly passive observational market data. Both contracts operate strictly on $\mathcal{I}_{\text{pub}}$, with no architectural disclosures or private state data.
2. **Architectural disclosures ($\mathcal{I}_{\text{arch}}$, Contract $I_1$):** Verified rate intervals $[\eta_0, 1]$, complete feedback channel inventories, and protocol-enforced zero initial memory resets. Estimators operating under Contract $I_1$ (audited terminal-row OLS) receive $\mathcal{I}_{\text{pub}} \cup \mathcal{I}_{\text{arch}}$. These disclosures represent structural side-information regarding algorithmic classes and implementation contracts, not private trading states or proprietary weights.
3. **Latent agent states ($\mathcal{I}_{\text{priv}}$):** Individual feedback coefficients $q_i$, private beliefs $\theta_t$, internal valuations, account equity, and dealer inventory states. These states are permanently withheld from all estimators and are accessed exclusively by the market simulator to evaluate ground truth.

## B.2 Experimental design and simulation lineage

Table 34 summarizes the experimental configurations and episode counts across all simulation studies.

## B.3 Standalone algorithmic verification without project imports

The audited-constraints study includes a verifier that imports no project modules (`studies/audited_constraints/replication/verify_without_project_imports.py`, output `studies/audited_constraints/results/standalone_verification.json`). It

- recomputes the 4,320 terminal-row regression decisions with an SVD solver (all labels match; maximum difference $4.4 \times 10^{-14}$);
- re-solves 500 order-fill projections as linear programs (maximum difference $1.8 \times 10^{-15}$);
- recomputes 2,880 exact Clopper–Pearson intervals (maximum difference $3.6 \times 10^{-14}$) and 2,880 geometric tail envelopes;
- re-checks 64 transaction ledgers (maximum cash residual $2.3 \times 10^{-12}$).

This is an internal consistency check written by the same authors, not outside replication.

## B.4 Computational reproduction protocol

The correction uses Python 3.11. The reproduction guide records the exact commands to generate the tables. Basic checks are:

```
# 1. Install runtime dependencies
python -m pip install -r requirements-runtime.txt

# 2. Execute native suites; extension checks are listed
     in the correction guide
#    (a) 104 native tests across the four studies and
    release tools
python tools/run_all_tests.py
#    (b) extension, correction, and mathematical-
    supplement tests
PYTHONPATH=. pytest tests/ research_extension/tests/
    reference_directory/test_mathematical_supplement.
    py -q

# 3. Execute standalone zero-dependency verifier (three
     checks)
python independent_verifier/verifier.py

# 4. Execute unified research CLI (replay/regenerate
    modes)
PYTHONPATH=. python -m research_extension.mm_revision.
    cli --mode replay

# 5. Rebuild generated tables, run tests, compile the
    PDF
python scripts/regenerate_all.py
```

Table 34: Simulation lineage and experimental sample counts.

| Study Component | Design Parameters and Configurations | Replications | Record count |
|---|---|---|---|
| Primary Linear Panel | $G \in \{0.7, 0.95, 1.05, 1.3\}$, $\eta_{\text{slow}} \in \{0.002, 0.02, 0.1\}$, $n \in \{32, 128, 512\}$, $T \in \{16, 64\}$, 2 probe designs | 2,880 | 645,120 |
| Held-Out Confirmation | Fresh random seeds, paired endpoint diversification (fixed 64 vs. $\{8, 16, 32, 64\}$) | 720 | 161,280 |
| Funded Market Dynamics | $G \in \{0.7, 1.3\}$, two slow rates, three funding regimes (unlimited, ample, tight), two shocks: 24 settings × 20,480 episodes; 960 setting-panels at three budgets | 2,880 rows | 491,520 setting-trajectories (163,840 tapes) |
| Preloaded Dealer Stress | Preloaded tight-regime dealer; $G \in \{0.7, 1.3\}$, two slow rates, mean repricing $-2$ or $-10$ | 8 settings | 65,536 episodes |
| 330-Cell Capacity Surface | Nominal $S \in \{0.70, 0.82, 1.18\}$, shocks × inventory-capacity multipliers (fixed \$10k initial equity) | 330 cells × 30 | 9,900 runs |
| Comparator Grid | 120 fixed cells, 5 panels per cell; $R = 128, 1024$, $H = 24, 64$; post-probe response alignment | 600 panels | 345,600 episodes |
| Conic Sweeps | Five budgets, 3 regimes, 20 Gaussian coefficient draws per cell; known and estimated covariance kept separate | 300 per mode | Not pooled |
| Response-Pivot Calibration | Known covariance/chi-square and independent Wishart-estimated covariance/F | 20,000 per mode | Not market episodes |
| C1–C4 Diagnostic | C1/C3: one deterministic case each; C2/C4: 100 noisy coefficient panels each | 202 attempts | 200 noisy panels |
| Paired Capacity Scenario | Four regimes, 100 shared disturbance tapes; effective adaptive $S = 1.22$ | 100 tapes | 400 paired paths |
| Q Training | 10 training seeds (926101–926110), 100k transitions per policy | 10 policies | 1,000,000 transitions |
| Q Evaluation | 10 policies × 10 tapes × 3 arms, plus 20 baseline runs; 128 steps each | 10 populations | 320 nested runs |
| Public-Data Proxies | 9 kline archives (129,600 bars per asset) and 3 separate REST depth snapshots; 18,445,389 bytes consumed | N/A | Not pooled |
| Synthetic Proxy-Scale Scenarios | 3 assets × 50 seeds × 4 shocks; 64 steps; no empirical event denominator | 600 runs | N/A |
| Archived Coinbase Spot | 10 Coinbase spot markets, Level-1 book quotes and 1,000 tick trades per pair; corrected row/cache samples reported separately | 10 | 10,000 |
| Archived Binance US Spot | 10 Binance US spot markets, Level-2 snapshots and 500 tick trades per pair; refit requests 50 seeds and retains 8 failures | 10 | 5,000 |

# C Methodological Scope, Empirical Boundaries, and Research Frontiers

Table 35 separates what this paper checks from what remains open.

In the common-rate symmetric benchmark, aggregate prices cannot attribute feedback to groups without channel-specific excitation. In the funded simulations, feedback interacts with binding dealer constraints to produce halts inside the price-stable region. The open problems above are not addressed here.

Table 35: Methodological scope and open research frontiers.

| **Research Dimension** | **Current Scientific Verification and Open Frontiers** |
|---|---|
| Priority and Proofs | **Internally checked:** Model-specific separation and invariance arguments with explicit rate, total-mass and disturbance-law qualifications. Correctness does not establish novelty; external proof and priority review remain pending.<br>**Frontier:** Extension of non-identifiability boundaries to nonlinear, continuous-time jump-diffusion price processes. |
| Structural Identification | **Theorem-level, conditional:** Exact-arithmetic outer intervals cover aggregate feedback on the stated response event. The numerical conic routine withholds certification until full outward constraint assembly is verified. Budget sweeps describe numerical candidates, not certificates or realization-wise monotone narrowing.<br>**Frontier:** Identification under adversarial probe-detection strategies where algorithmic market participants actively distort quote placement upon detecting surveillance orders. |
| Empirical Microstructure | **Exploratory:** A ten-asset Coinbase sample gives a nominally significant cross-sectional association (ten observations). Passive chronological structural fits fail out of sample. A reduced-form feedback component is estimated from one-second data for three assets (section 7.4). Public-data proxies have consistent units, but depth snapshots are not historical event books and execution calibration remains pending.<br>**Frontier:** Authenticated proprietary runtime audit with an institutional counterparty (Contract $I_1$) and nanosecond LOB hardware cancellation accounting. |
| Comparator Evidence | **Comparative analysis:** Validated containment, issued-label, failure and width counts are reported in Tables 6–7. Profile and sandwich diagnostics have separate denominators. Known- and estimated-covariance sweeps use different valid pivots. These are simulated comparisons, not a general method ranking. The audited terminal-row study uses a separate information contract and sample.<br>**Frontier:** Non-stationary online recursive updating under regime-switching volatility environments and adversarial probe gaming. |
| Capital and Liquidity Constraints | **Model-specific simulations:** The original funded engine illustrates stable signals with operational failures. A separate capacity scenario has effective $S = 1.22$ and $\rho \approx 1.07409$, not stable dynamics; its sticky constraint-hit flag is not a permanent trading halt. Neither establishes calibrated real-market operational protection.<br>**Frontier:** Multi-clearinghouse margin contagion and bilateral over-the-counter (OTC) derivative unwind cascades. |
| Strategic Policy Transfer | **Falsification diagnostic:** All ten new policies have insufficient action support on the response grid. Active-minus-frozen displacement is $-0.000227$ with interval $[-0.000659, 0.000204]$. No advantage over no-trade is demonstrated; neither structural falsification nor a volume-matched mechanism is established.<br>**Frontier:** Transfer to multi-agent reinforcement learning (MARL) populations undergoing continuous adversarial retraining in production exchange simulators. |

# D Provenance ledger

This appendix records where each retained number comes from and which earlier results were withdrawn.

**Generated tables.** The empirical feedback estimates, the estimator Monte Carlo and the feedback-dealer solvency results (tables 23 to 26, 31 and 33) and every number quoted from them in the text are not typed into the manuscript. `scripts/build_results.py` writes them, as tables and as LaTeX macros, from `results/provenance/{empirical_feedback,estimator_validation,solvency_feedback}.json`, which are produced by `empirical_pipeline/feedback_var.py`, `simulations/estimator_validation.py` and `simulations/feedback_dealer.py`.

**Typed numbers.** The remaining results (Sections 3.2–7.3) were produced by the four frozen studies under `studies/` and by the correction runs under `research_extension/outputs/`. They are typed in the manuscript. Table 36 lists the headline numbers. For each one, `scripts/provenance_ledger.py` recomputes the printed value from the named artifact, or from the closed-form expression in the text for rows marked *analytic*, and checks that the printed string still appears verbatim in this file. `tests/test_provenance_ledger.py` runs these checks and also verifies that a one-digit change to the paper or to a printed value is detected. Accounting residuals are printed rounded up to three significant figures, because they are upper bounds. The ledger covers headline values, not every cell of every typed table. The other cells come from the same artifacts but are not individually re-checked.

Table 36: Provenance ledger for retained numbers outside the generated tables. Every row is re-checked by `tests/test_provenance_ledger.py`.

| ID | Location | Quantity | Source | Kind |
|---|---|---|---|---|
| S3-wit-S | Tab. retained witness | S of stable/unstable member | exact rational recursion | analytic |
| S3-wit-rho | Tab. retained witness | rho(J) of minimal realizations | eigenvalues of minimal realization | analytic |
| S3-wit-h4 | Tab. retained witness | h_4 diverges | `research_extension/outputs/evidence_repair_20260926/paper_artifacts.json` (witness) and exact recursion | analytic |
| S3-kernel-oracle | Sec. 3 full-horizon audit | kernel-law oracle error / coverage bound | `studies/full_horizon_audit/results/testing_limits_summary.json` | artifact |
| S4-grid-oe | Tab. comparator | output-error covered/issued/wrong | `research_extension/outputs/statistical_repair_20260926/verified/artifacts/paper_artifacts.json` (grid_summary) | artifact |
| S4-grid-widths | Tab. widths | recurrence finite mean / p90 width | `research_extension/outputs/statistical_repair_20260926/verified/artifacts/paper_artifacts.json` (grid_summary) | artifact |
| S4-calib | Tab. pivot calibration | known/estimated coverage | `research_extension/outputs/statistical_repair_20260926/verified/calibration/results.json` | artifact |
| S4-cover | Sec. 4 primary panel | simultaneous response coverage | `studies/observable_stress_tests/results/summary.json` | artifact |
| S4-adjcorr | Sec. 4 primary panel | mean \|adjacent correlation\| | `studies/observable_stress_tests/results/summary.json` | artifact |
| S4-tab-128x4 | Tab. observable primary | R=128, pulse 4: recurrence / point wrong | `studies/observable_stress_tests/results/design_summary.csv` | artifact |
| S4-tab-1024x4 | Tab. observable primary | R=1024, pulse 4: recurrence / point wrong | `studies/observable_stress_tests/results/design_summary.csv` | artifact |
| S4-decisions | Sec. 4 primary panel | recurrence decisions / inconclusive rows | `studies/observable_stress_tests/results/design_summary.csv`, `summary.json` | artifact |
| S4-pointwrong | Sec. 4 primary panel | point-fit wrong labels | `studies/observable_stress_tests/results/summary.json` | artifact |
| S4-budget-ext | Sec. 4 budget extension | simulated episodes | `studies/observable_stress_tests/results/extension_summary.json` | artifact |

| ID | Location | Quantity | Source | Kind |
|---|---|---|---|---|
| S4-pricelaw-diff | Sec. 4 price-law example | oracle errors, difference model | studies/observable_stress_tests/results/price_observation_limits.json | artifact |
| S4-pricelaw-cov | Sec. 4 price-law example | max coverage at 5% error (difference) | studies/observable_stress_tests/results/price_observation_limits.json | artifact |
| S4-pricelaw-level | Sec. 4 price-law example | oracle errors, level-error model | eq. (price_distance_level), recomputed | analytic |
| S4-pricelaw-diff-recompute | Sec. 4 price-law example | difference-model errors recomputed from formula | eq. (price_distance), recomputed | analytic |
| S4-nonlin | Sec. 4 controls | initial impact per share (nonlinear control) | studies/observable_stress_tests/results/nonlinear_scaling.json | artifact |
| S4-acct | Sec. 4 accounting | max conservation residuals (rounded up) | studies/observable_stress_tests/results/summary.json | artifact |
| S5-primary | Sec. 5 primary budget experiment | coverage / decisions / wrong | studies/audited_constraints/results/linear_summary.json | artifact |
| S5-dec | Sec. 5 primary budget experiment | decisiveness and conditional error | studies/audited_constraints/results/linear_summary.json | artifact |
| S5-design | Sec. 5 primary budget experiment | impulse vs round trips, T=64 | studies/audited_constraints/results/linear_by_rate.csv | artifact |
| S5-endpoint | Tab. endpoint results | definite counts fixed vs diversified | studies/audited_constraints/results/endpoint_summary.csv | artifact |
| S5-paired | Sec. 5 endpoint comparison | paired gains/losses | studies/audited_constraints/results/endpoint_paired_comparison.csv | artifact |
| S5-slowest | Sec. 5 endpoint comparison | slowest rate resolved, fixed 64 | studies/audited_constraints/results/linear_by_rate.csv | artifact |
| S5-joint | Sec. 5 measured tail | joint gain+tail coverage, conflicts | studies/audited_constraints/results/measured_tail_summary.csv | artifact |
| S5-tail-tab | Tab. audited tail | median measured tail upper endpoints | studies/audited_constraints/results/measured_tail_summary.csv | artifact |
| S5-controls | Sec. 5 controls | omitted channel false stable labels | studies/audited_constraints/results/assumption_control_summary.csv | artifact |
| S6-episodes | Sec. 6 design | episodes and quote observations | studies/observable_stress_tests/results/summary.json | artifact |
| S6-tapes | Sec. 6 funded panel | setting-trajectories vs unique tapes | studies/audited_constraints/results/financial_summary.json | artifact |
| S6-clips | Sec. 6 funded panel | clipped orders and margin calls | studies/audited_constraints/results/financial_summary.json | artifact |
| S6-acct | Sec. 6 funded panel | max residuals (rounded up) | studies/audited_constraints/results/financial_summary.json | artifact |
| S6-binom | Sec. 6 risk coverage | exact coverage grid points | studies/audited_constraints/results/control_summary.json | artifact |
| S6-stress-1 | Tab. dealer stress | G=0.7, slow 0.02, -2: event/halt | studies/audited_constraints/results/dealer_stress.csv | artifact |
| S6-stress-2 | Tab. dealer stress | G=0.7, slow 0.02, -10: event/halt | studies/audited_constraints/results/dealer_stress.csv | artifact |
| S6-stress-8 | Tab. dealer stress | G=1.3, slow 0.10, -10: event/halt | studies/audited_constraints/results/dealer_stress.csv | artifact |
| S6-stress-eps | Sec. 6 dealer stress | episodes | studies/audited_constraints/results/dealer_stress.csv, followup_summary.json | artifact |
| S6-surface | Tab. capacity surface | intermediate cell 13/30 | research_extension/outputs/funded_smooth_surface_results.json | artifact |
| S6-paired | Tab. funded controls | mean dislocation by regime | research_extension/outputs/funded_mechanism_results.json | artifact |
| S6-contrast | Tab. paired contrasts | active minus frozen [CI] | research_extension/outputs/funded_mechanism_results.json | artifact |

| ID | Location | Quantity | Source | Kind |
|---|---|---|---|---|
| S6-q-mean | Sec. 6 Q-learning | active minus frozen mean and CI | research_extension/outputs/evidence_repair_20260926/learned_agents/results.json | artifact |
| S6-q-pol4 | Tab. Q-learning | policy 04 contrast | research_extension/outputs/evidence_repair_20260926/learned_agents/results.json | artifact |
| S6-q-chartist | Sec. 6 Q-learning | chartist displacement | research_extension/outputs/evidence_repair_20260926/learned_agents/results.json | artifact |
| S7-coinbase | Tab. Coinbase regression | bivariate row rendered from audit | research_extension/outputs/evidence_repair_20260926/paper_artifacts.json (empirical_table_tex) | artifact |
| S7-binance | Sec. 7 negative control | BTC mean validation R^2 | research_extension/outputs/evidence_repair_20260926/empirical/binance_chronological_results.json | artifact |
| S7-gate24 | Tab. gate24 proxies | BTC row rendered | research_extension/outputs/evidence_repair_20260926/paper_artifacts.json (gate_table_tex) | artifact |
| S7-halts | Tab. gate4 halts | BTC 0% shock final equity | research_extension/outputs/evidence_repair_20260926/paper_artifacts.json (halt_table_tex) | artifact |
| S7-bytes | Sec. 7 gate24 | bytes consumed | research_extension/outputs/evidence_repair_20260926/gate24/gate24_summary.json | artifact |
| S7-binance-btc | Sec. 7 cross-venue | Binance BTC Q and 95% CI | results/provenance/cross_venue_summary.json | artifact |
| S7-binance-eth | Sec. 7 cross-venue | Binance ETH Q and 95% CI | results/provenance/cross_venue_summary.json | artifact |
| S7-binance-ada | Sec. 7 cross-venue | Binance ADA Q and 95% CI | results/provenance/cross_venue_summary.json | artifact |
| S7-holdout-btc | Sec. 7 holdout | Jan 2023 BTC holdout Q, Q30, S, pos share | results/provenance/binance_panel_summary.json | artifact |
| S7-panel-regimes | Sec. 7 Binance panel | epoch mean Q and positive day shares | results/provenance/binance_panel_summary.json | artifact |
| S7-flow-gamma | Sec. 7 flow memory | order-splitting memory exponent range | results/provenance/flow_memory_bias_correction.json | artifact |
| S7-flow-set-btc | Sec. 7 flow memory | BTC robust identified set and max loop ratio | results/provenance/flow_memory_bias_correction.json | artifact |
| App-verifier | App. replication | standalone verifier counts and max regression difference | studies/audited_constraints/results/standalone_verification.json | artifact |

**Withdrawn results.** An internal audit found that the following results in earlier drafts were not produced by data or code, or came from code that did not implement the stated model. They have been removed:

- 2SLS / FI-2010 instrumental-variable estimates: no FI-2010 data or 2SLS output exists in the repository.
- Crypto panel regressions and sub-sample splits: numbers not produced by any script.
- Former Table 24 (dynamic Hankel empirics): no producing code or data.
- Cross-simulator validation: simulators did not implement the stated model.
- Decisiveness trade-off frontier and eta-frontier figure: schematic values, not computed.
- Original solvency phase diagram: feedback parameter did not enter the price process; replaced by Sec. 9 rebuild.
- DQN and continuous-action RL audits: no executed runs found.
- Constrained robustness example (19/200, 15/200, 78.95%): no source artifact; replaced by comparator-grid example.
- Model-class screen calibration (5 x 250 reps): no source artifact.
- Standalone verifier '24 gates' and 7.2e-14: not produced by the verifier; real figures reported instead.
- '7.2/10' self-assessed rating and similar quality scores: not a result.

# E Computational Verification and Theorem-to-Code Mapping

This appendix maps claims to the code that checks them. Every file listed exists in the release repository. A passing test shows only that the code does what the test checks; it is not independent evidence for a claim.

## E.1 Standalone verifier

`independent_verifier/verifier.py` imports no project code. It recomputes three items: the retained rational witness, the coverage, decisiveness and error denominators of the observable-data experiment, and the cross-sectional regression separation. All three match the reported values. (An earlier draft attributed "24 gates" and a maximum discrepancy of $7.2 \times 10^{-14}$ to this verifier. It produces neither figure. The audited-study verifier in section B reports 24 public-path filter checks and a maximum regression difference of $4.4 \times 10^{-14}$; the $7.2 \times 10^{-14}$ figure has no source and is withdrawn.)

Table 37: Comprehensive Theorem-to-Code and Lean Formal Proof Map, verified by `scripts/verify_theorem_code_bridge.py`.

| **Theorem / Claim** | **Location** | `Lean 4 Formal Proof` | `Executable Python Test` |
|---|---|---|---|
| Proposition: Steady-state Kalman filtering yields exponential memory | theorem 2.1 | -- | `test_dynamical_system_identification_bridge.py (test_kalman_microfoundation)` |
| Proposition: Linearized regularized gradient updates yield exponential memory | theorem 2.2 | -- | `test_dynamical_system_identification_bridge.py (test_ogd_microfoundation)` |
| Proposition: Fixed-system local stability; standard result | section 1.1 | `Proofs/Stability.lean (charPoly_at_one_identity)` | `test_theorems_m2_m5_corrected.py (test_fixed_system_stability)` |
| Proposition: Volatility and amplification need not move together | section 1.1 | -- | `test_dynamical_system_identification_bridge.py (test_volatility_amplification_wedge)` |
| Theorem: Positive-feedback boundary; model specialization | theorem 2.5 | `Proofs/Stability.lean (charPoly_at_one_identity)` | `test_core_theorems_m1_m2_m3.py (test_positive_feedback_boundary)` |
| Theorem: Signed-feedback stability boundary | theorem 2.6 | `Proofs/SignedStability.lean (signed_boundary_at_one)` | `test_theorems_m2_m5_corrected.py (test_signed_stability_jury_conditions)` |
| Corollary: No universal participation cutoff in this model | section 1.1 | `Proofs/SignedStability.lean (contrarian_example_P_one_positive)` | `test_theorems_m2_m5_corrected.py (test_no_universal_participation_cutoff)` |
| Proposition: Observational equivalence of decision-maker labels | section 3.2 | `Proofs/PBHDefect.lean (pbh_unobservable)` | `test_core_theorems_m1_m2_m3.py (test_label_observational_equivalence)` |
| Proposition: Positive heterogeneous-memory boundary | section 3.2 | `Proofs/Stability.lean (aggregate_boundary_criterion)` | `test_core_theorems_m1_m2_m3.py (test_heterogeneous_memory_boundary)` |
| Theorem: Retained exact construction | theorem 3.3 | `Proofs/Ambiguity.lean (kernel_lag0_identical)` | `test_core_theorems_m1_m2_m3.py (test_retained_pair_construction)` |
| Theorem: Equivalence class and minimal realization | theorem 3.4 | `Proofs/PBHDefect.lean (swapped_feedback_transfer_invariant)` | `test_core_theorems_m1_m2_m3.py (test_theorem_m1_minimal_realization)` |
| Theorem: Local minimax risk and a $\delta^{-4}$ separation bound | theorem 3.5 | -- | `test_core_theorems_m1_m2_m3.py (test_theorem_m2_sample_complexity)` |
| Theorem: Mandatory abstention near the boundary | theorem 3.6 | -- | `test_core_theorems_m1_m2_m3.py (test_theorem_m3_mandatory_abstention)` |
| Theorem: Completely monotone memory and a size-valid finite-sample test | theorem 3.7 | -- | `test_cm_cone_test.py (test_cm_test_size_and_power)` |
| Theorem: Exact safe radius and the Kreiss bound | theorem 3.8 | -- | `test_theorems_m2_m5_corrected.py (test_theorem_m5_kreiss_solvency)` |
| Theorem: Finite-family response uncertainty | section 4 | `Proofs/Bounds.lean (certified_stable_soundness)` | `test_conic_certificate.py (test_conic_certificate)` |
| Theorem: Observable-data stability interval | section 4 | `Proofs/Bounds.lean (certified_unstable_soundness)` | `test_conic_certificate.py (test_stability_interval)` |
| Theorem: Fixed-budget price-response ambiguity | section 4.3 | `Proofs/Ambiguity.lean (price_response_lag4_difference)` | `verifier.py (verify_independent)` |
| Proposition: Swapped-feedback non-identifiability | theorem 5.1 | `Proofs/PBHDefect.lean (pbh_eigenvector_x)` | `test_core_theorems_m1_m2_m3.py (test_swapped_feedback_defect)` |
| Theorem: Minimal separation in the common-rate observationally symmetric model | section 4.3 | `Proofs/PBHDefect.lean (swapped_feedback_sum_equal)` | `test_core_theorems_m1_m2_m3.py (test_minimal_separation)` |

# F Machine-Checked Algebraic Identities in Lean 4

To verify that foundational linearizations, characteristic polynomials, and unobservable eigenspace defects are free from algebraic error, key polynomial identities of the model have been formalized and checked using the Lean 4 proof assistant.

Rather than claiming a full formalization of measure-theoretic probability or nonlinear microstructure dynamics, the Lean formalization is strictly scoped to the exact rational algebra of the linearized state space:

- **Characteristic Polynomial Factorization (Theorem 2.5):** Verification of the root identity $\det(I - J) = (1 - a - g\kappa)\eta$ over rational arithmetic.
- **Popov–Belevitch–Hautus Rank Defect (Proposition 5.1):** Formal verification that identical-rate multi-agent Jacobians possess a non-trivial nullspace satisfying $Cv = 0$ and $Jv = (1 - \eta)v$.
- **Signed stability polynomial (Theorem 2.6):** the characteristic polynomial evaluated at $\lambda = \pm 1$ and the determinant identity. These are ingredients of the Jury conditions; the inequalities themselves are proved on paper in section A.6.

Importantly, the Lean package operates using pure rational arithmetic in core Lean 4 (`Init.GrindInstances.Ring.Rat`) without external dependencies on `Mathlib`, ensuring that the algebraic checks compile cleanly and independently:

Listing 1: Representative Lean 4 algebraic identity verification in `lean_proofs/Proofs/Stability.lean`.

```
import Init.GrindInstances.Ring.Rat

namespace MachineMarkets

-- Characteristic polynomial: det(lambda*I - J)
def charPoly (a g eta kappa lam : Rat) : Rat :=
  lam * lam - (a + 1 - eta) * lam + a * (1 - eta) - g * eta * kappa

-- Theorem 2.3: Exact algebraic root factorization at boundary lambda = 1
theorem charPoly_at_one_identity (a g eta kappa : Rat) :
    charPoly a g eta kappa 1 = (1 - a - g * kappa) * eta := by
  unfold charPoly
  grind

-- Proposition 5.1: PBH unobservable eigenspace defect (J*v = (1 - eta)*v and C*v = 0)
theorem pbh_unobservable : C_dot pbh_null_vec = 0 := by
  unfold C_dot pbh_null_vec; decide

theorem pbh_eigenvector_x (a eta q1 q2 : Rat) :
    (J_mul a eta q1 q2 pbh_null_vec).x = 0 := by
  unfold J_mul pbh_null_vec; grind
```

The statistical estimators, sample complexity lower bounds (Theorems 3.5–3.6), and nonlinear balance-sheet simulations are implemented and tested in the Python verification suite (`tests/`), not within the Lean type checker.